Flooding the filter: a game-theoretic model of AI, submission pressure, and quality collapse

~~Yuhui Fang~~

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Abstract

Generative AI is transforming ~~the nature of academic publishing . By September 2024, more than one-fifth of computer-science preprints showed evidence of large-language-model modification, while monthly submissions to arXiv reached a record 24, 226, roughly 50% higher than in late 2022. Verification~~ academic publishing by dramatically reducing the cost of manuscript production while review capacity grows far more slowly. Submission volumes have surged to record levels, a rising share of peer reviews contain AI-generated text, and verification infrastructure has not kept pace~~: reviewer acceptance rates have declined steadily over two decades, and between 6.5% and 16.9% of conference peer reviews are now themselves AI-modified~~. We study this asymmetry—the fact that AI scales paper production faster than it scales screening—using a game-theoretic model in which AI adoption raises submission volume and lowers drafting cost faster than it raises review capacity, while also increasing the rate at which low-quality manuscripts pass screening. We find three results. First, extreme slot scarcity can stabilize honest submission norms even under degraded screening, because the publication lottery becomes too hard to win. Second, gradual AI adoption can trigger a discrete tipping point into a low-quality long-run outcome; once tipped, the system is self-sustaining, so recovery requires a large coordinated intervention, not incremental reform. Third, expanding publication capacity without scaling verification accelerates the tipping. These results point to a concrete policy agenda of verification-centric institutions (structured audits, submission quotas, and decoupled certification) rather than simply more publication slots.

1 Introduction

We ask what happens to a publishing *system* when it becomes substantially cheaper and faster to produce a manuscript than to review one. By September 2024, more than one-fifth of computer-science preprints on arXiv showed measurable evidence of large-language-model modification (**?**), and at least 13.5% of all biomedical abstracts published that year contained AI-assisted text (**?**). Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% higher than in late 2022 (**?**). Verification has not kept pace. ~~Reviewer acceptance rates have~~ The rate at which researchers accept reviewer invitations has declined steadily over two decades (**?**), and between 6.5% and 16.9% of conference peer reviews now contain substantial AI-generated text (**?**). More than half of active researchers surveyed in 2025 reported using AI tools to write reviews, often against explicit journal policy (**?**). The question is not whether individual researchers behave ethically, but whether the system as a whole can absorb an asymmetric shock to production and verification.

While the conventional response to AI in science focuses on disclosure requirements, detection tools, and editorial guidelines, these measures address individual conduct and leave untouched a structural problem: even if every author is honest, reviewers can scrutinize only a finite number of manuscripts carefully per cycle. When submission volume rises faster than review capacity, screening degrades—not because of misconduct, but because of congestion. A reviewer who faces twice as many papers in the same time cannot give each submission the same attention. Congested screening lowers the effective quality threshold, and a lower threshold raises the expected payoff from submitting weak or low-quality work. We focus on this feedback between volume and screening quality as the central mechanism.

We distinguish two bottlenecks in the publishing system, and we argue that AI affects them asymmetrically. The first is verification capacity: the number of papers that reviewers can evaluate carefully

in a given period, denoted K_{rev}. The second is allocation capacity: the number of papers the journal can publish per period, denoted K. AI raises submission volume V by reducing the marginal cost of drafting and resubmitting papers, lowering the effort cost c of each submission. It may also increase the false-positive rate λ—the probability that a low-quality manuscript passes peer review—because AI-assisted writing produces polished text that is harder to distinguish from genuine contributions on surface features alone. Review capacity, by contrast, does not scale symmetrically. Deep expertise, data and code verification, and replication checking remain human-intensive tasks, and journal policies in many fields explicitly restrict AI use in reviews (**?**). Production, in short, scales faster with AI adoption than verification does.

We model this asymmetry in a game-theoretic framework, drawing on strategic models of scientific communication (**??**). We consider a population of authors who choose between two strategies: submitting only genuine work (the cooperative strategy) or submitting everything including low-quality manuscripts (the defecting strategy). A journal applies peer review, subject to both a review capacity K_{rev} and a publication slot constraint K. We introduce an AI adoption intensity parameter $a \in [0, 1]$. Higher a raises submission volume V with elasticity γ, raises review capacity K_{rev} with elasticity $\delta < \gamma$, lowers submission cost c, and raises the false-positive rate λ. The production elasticity exceeds the verification elasticity by assumption, formalizing the asymmetry observed empirically. The net effect of a on the profitability of submitting a low-quality paper, $\Pi_{bad}(x, a) = r \cdot \bar{\rho}(x, a) \cdot \pi_0(x, a) - c(a)$, is positive: AI makes spamming more attractive.

We find three results. First, extreme ~~publication scarcity stabilizes~~ slot scarcity can stabilize honest submission norms even ~~when screeningis imperfect. When~~ under degraded screening, because the publication lottery ~~is rare enough to win, the expected payoff from submitting a low-quality paper falls below the submission cost, and defection is unprofitable~~becomes too hard to win. Scarcity, paradoxically, acts as a ~~deterrent~~deterrent—but AI gradually erodes this protection by lowering the submission cost $c(a)$, so the deterrent requires progressively tighter slot constraints as adoption deepens. Second, ~~as a rises, the system can cross a tipping threshold: the long-run outcome shifts abruptly from a high-quality equilibrium to a~~ gradual AI adoption can trigger a discrete tipping point into a low-quality ~~one. This tipping occurs because AI simultaneously lowers the break-even point for spam and erodes the screening mechanism that would otherwise deter it. Third~~long-run outcome; once tipped, the system ~~exhibits persistent path dependence. The~~ is self-sustaining. Recovery requires a large coordinated intervention, not incremental reform: the value of a at which quality collapses is strictly lower than the value at which quality recovers~~. Once the system has tipped into the low-quality long-run outcome, incremental reforms cannot restore it; a large coordinated intervention is needed to cross the recovery threshold.~~

~~While~~ , so the system exhibits hysteresis. Third, expanding publication capacity (raising K) ~~seems like the obvious remedy, we find that doing so without raising review capacity~~ without scaling verification makes the situation worse, not better. More slots ~~increase the expected reward~~ raise the expected payoff from submitting marginal work~~, accelerating the drift toward the defecting strategy.~~ ; under AI adoption, where $c(a)$ is already low and $\lambda(a)$ already elevated, even a modest improvement in the publication lottery is enough to push the system across the tipping threshold.

These results point to a concrete policy agenda of verification-centric institutions—structured audits, submission quotas, and decoupled certification—rather than simply more publication slots. Effective intervention must ~~instead~~ target the two primary levers: raising genuine review capacity K_{rev} and reducing the false-positive rate λ. ~~Practical instruments that move these levers include structured code and data audit requirements, submission quotas or tokens, cascading triage systems, and layered certification schemes that decouple preprint dissemination from editorial endorsement.~~ These are not incremental reforms; they constitute a different institutional design, one oriented toward verification rather than volume.

We proceed as follows~~: Section~~ . Section 2 documents the empirical asymmetry between AI-assisted production and verification. Section 3 reviews related literature. Section 4 presents the ~~game-theoretic model; Section ?? derives the replicator dynamics and characterizes the equilibrium regimes; Section 6 examines parameter sensitivity, hysteresis, and strategic levers; and~~ baseline model. Section 5 introduces the AI adoption parameter and derives the extended spam payoff. Section 6 analyses how AI adoption shifts equilibria, erodes the stability window, and amplifies the slot expansion paradox. Section 7 reports agent-based simulation results. Section 8 translates the analytical results into a five-principle institutional agenda and evaluates the mechanisms using Figure 11. Section 9 provides a welfare analysis of the

model's regimes, identifies the two externalities that cause private incentives to diverge from the social optimum, and derives normative policy Corollaries from the welfare comparison. Section 10 derives testable empirical predictions, proposes proxies for the key model channels, and outlines an identification strategy. Section 11 discusses model limitations and extensions, including multi-journal competition and endogenous rewards. Section 12 concludes.

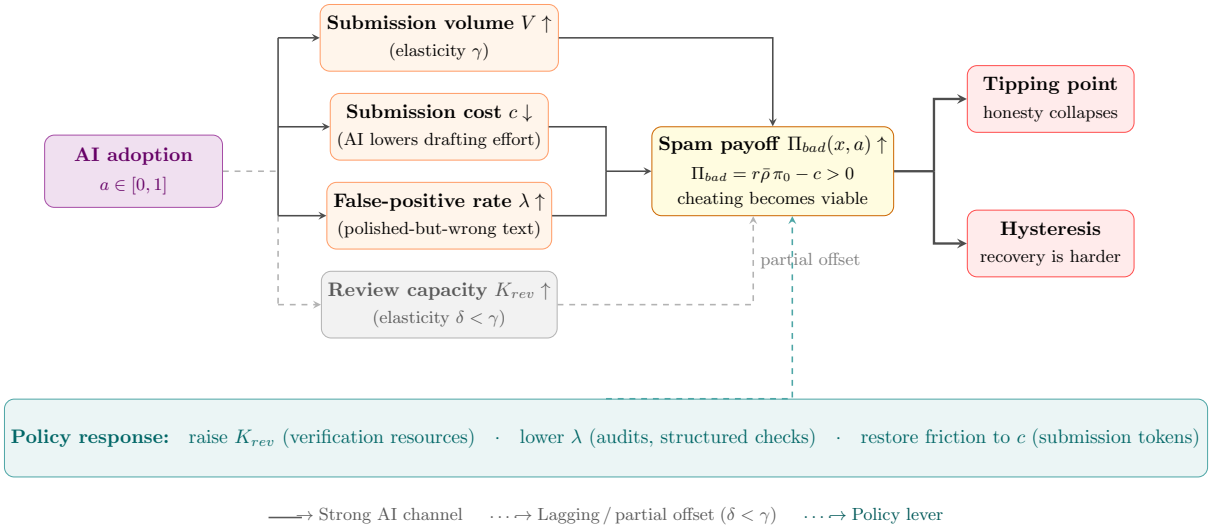


Figure 1: **Mechanism diagram: AI as an asymmetric shock to the publication system.** AI adoption (a) raises submission volume (V, elasticity γ) and lowers drafting cost (c), while polished AI text increases the false-positive rate (λ). Review capacity (K_{rev}) rises more slowly (elasticity $\delta < \gamma$), so the partial offset from AI-assisted reviewing is insufficient to prevent congestion. Together, these shifts raise the profitability of submitting low-quality work ($\Pi_{bad} = r\bar{\rho}\,\pi_0 - c$), pushing the system toward a tipping point and widening the gap between the collapse and recovery thresholds (hysteresis). Effective policy must raise K_{rev}, reduce λ, and restore friction to c (dashed teal arrow).

2 Stylized Facts: AI as an Asymmetric Shock

We draw on evidence that AI dramatically lowers the cost of producing a manuscript. Large language models have cut the time required to draft, revise, and resubmit academic text by a documented margin. In a randomised experiment with 453 college-educated professionals, **?** found that access to ChatGPT reduced the time to complete an incentivised writing task by approximately 40% while simultaneously raising assessed output quality by 18 percentage points, with the largest gains accruing to weaker writers. **?** report that researchers can revise a full manuscript in under five minutes at an API cost below \$0.50. These micro-level reductions compound at the submission level: iterative resubmission across journals was once a significant labour cost, and AI converts that friction into a near-zero marginal activity, giving every author—regardless of the underlying quality of their work—a cost-effective reason to submit or resubmit. In our model, this channel corresponds to a downward shift in the submission cost parameter $c(a)$.

Aggregate data confirm a sharp rise in submission pressure. Macro-level data are consistent with the micro-level cost story. Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% above the late-2022 baseline; the first month to cross 20,000 submissions was May 2023, shortly after the public release of ChatGPT (**?**). More than one-fifth of computer-science preprints sampled in September 2024 showed measurable evidence of large-language-model modification (**?**), and a vocabulary-based analysis of 15 million biomedical abstracts found that at least 13.5% of all 2024 *PubMed* papers contain AI-assisted text, with rates exceeding 40% in some sub-disciplines (**?**). Alongside this volume growth, annual retraction counts exceeded 10,000 for the first time in 2023, with paper-mill activity implicated in mass retraction events at major publishers. These data show that submission volume V has shifted materially upward as AI adoption has spread, independently of any change in

the fraction of authors who genuinely produce high-quality work. In our model, this corresponds to an upward shift in $V(x, a)$ with elasticity γ with respect to a.

While text production scales with AI, verification does not. Rigorous peer review requires reading carefully for logical consistency, checking data against claimed analyses, running or inspecting code, and evaluating genuine novelty—tasks that remain human-intensive in ways that text generation is not. Confidentiality requirements for submitted manuscripts prevent reviewers from uploading papers to third-party AI services, and many journals explicitly forbid AI use in the review process (**?**). Even where AI-assisted reviewing is nominally permitted, the marginal reviewer who can detect a subtle methodological flaw is a scarce specialist whose attention is not fungible with a language model's output. Reviewer acceptance rates have declined continuously for two decades (**?**), and the existing burden is already extreme: reviewers contribute more than 100 million hours annually to the peer review system worldwide, with the load concentrated among a small fraction of active researchers (**?**). We therefore expect that a system already under strain before the AI era cannot absorb a near-doubling of submission volume through a commensurate increase in review capacity. In our model, this channel is captured by $K_{rev}(a) = K_{rev}(0) \cdot (1 + \delta a)$, where the elasticity δ is strictly smaller than the production elasticity γ, reflecting the structural and policy constraints described above.

AI also polishes manuscripts without improving their reliability, which may raise the false-positive rate. A third channel operates through the quality of the screening signal rather than through volume. AI writing tools produce fluent, well-formatted, and stylistically consistent text (**?**), and this surface polish reduces the ability of reviewers to distinguish a low-quality submission from a genuine contribution on textual features alone, raising the probability that a flawed manuscript passes the review filter. The screening process degrades further when reviewers themselves turn to AI: between 6.5% and 16.9% of AI-conference peer reviews now contain substantial AI-generated text (**?**), and an experimental audit found that AI-assisted reviews award systematically higher scores, leading to elevated acceptance rates (**?**). Automated detection offers limited recourse—leading AI detectors exhibit false-negative rates above 70%, and adversarial paraphrasing can reduce recall to near zero (**?**). In our model, this channel corresponds to an upward shift in the false-positive rate $\lambda(a)$: as AI adoption rises, the probability that a low-quality submission passes review increases even holding the underlying composition of the author population constant.

We summarize these channels as a single asymmetry hypothesis. The four channels described above—rising submission volume, falling drafting cost, lagging review capacity, and rising false-positive risk—reduce to a single structural claim:

> *AI raises the elasticity of manuscript production above the elasticity of manuscript verification.*

We formalise this as the **AI asymmetry assumption**: with $a \in [0, 1]$ denoting AI adoption intensity,

$$\gamma > \delta \geq 0, \tag{1}$$

where γ is the elasticity of submission volume with respect to a and δ is the elasticity of review capacity with respect to a. This is not an axiom but a testable empirical statement about relative scaling rates. Table 1 translates each channel into a specific model parameter and states the direction of the shift. In Section 6, we examine how our qualitative results depend on the magnitude of the gap $\gamma - \delta$ and on the slope of $\lambda(a)$.

Table 1: **AI adoption channels and their model counterparts.** Each row maps an empirical mechanism to the corresponding model parameter, the direction of the shift, and the evidence base.

Empirical channel	**Mechanism**	**Parameter**	**Shift**	**Key evidence**
Lower drafting cost	AI reduces time and effort per manuscript; resubmission friction falls	$c(a)$	$\downarrow$	**?**; **?**
Higher submission volume	More authors submit more frequently; low-quality papers enter the pool	V (elasticity γ)	$\uparrow$	**?**; **?**; **?**
Lagging review capacity	Deep verification remains human-intensive; policy restricts AI in review; reviewer shortage pre-dates AI	K_{rev} (elasticity $\delta < \gamma$)	$\uparrow$ slowly	**?**; **?**; **?**
Rising false-positive rate	Polished AI text harder to distinguish; AI-assisted reviews inflate acceptance scores; detectors unreliable	$\lambda(a)$	$\uparrow$	**?**; **?**; **?**
Possible quality uplift	AI may also help authors improve genuine contributions (partial offset)	α	$\uparrow$ (offset)	**?**

3 Related Literature

We draw on research showing that peer review functions as a shared resource under increasing strain. Each submission consumes a fixed amount of reviewer time and attention, and when submission volume grows faster than the pool of willing reviewers, that resource degrades. **?** show that scientific contests become inefficient at high participation rates: total effort rises, but expected quality per participant falls. **?** reaches a parallel conclusion for journals: a small entry cost can deter low-quality submissions and relieve pressure on reviewers. **?** estimate that the global peer review system already consumed over 100 million hours of researcher time each year in the mid-2010s, and **?** value the US contribution alone at more than 1.5 billion USD in 2020. **?** model these pressures and find that screening quality declines even when reviewers act in good faith, simply because competing demands leave too little time per paper. The reviewing system, in short, was already near its limits before generative artificial intelligence entered the picture.

While congestion models treat reviewers as passive, a related body of work models authors as strategic agents—and this is where the feedback loop becomes visible. **?** show that when the review filter is imprecise, submitting everything regardless of quality can become individually rational, pushing the long-run outcome toward low average quality. **?** extend this to a multi-journal setting and find that prestige-maximising incentives can reward strategic over honest behaviour. **?** study honest signalling and show that the good outcome—where only strong work is submitted—is fragile: it collapses when the cost of mimicking quality falls even slightly. **?** document this logic empirically, arguing that "publish or perish" incentives act as a selection force that spreads high-volume, low-quality strategies through academic communities. At the aggregate level, **?** shows that most published findings fail to replicate, not because of fraud, but because structural incentives reward output over reproducibility. **?** and **?** trace this to career and grant structures that treat paper counts as a proxy for scientific value. Paper mills—commercial services producing fabricated manuscripts to order—represent the extreme end of this logic: annual retraction counts have reached record highs and single incidents have implicated thousands of papers at major publishers. Our model formalises the mechanism underlying these trends: when the

review filter is weak and slots are available, submitting low-quality work yields a positive expected payoff, and selection pressure then favours volume over quality.

In contrast to the strategic submission literature, recent empirical work documents how generative artificial intelligence reshapes both sides of the system simultaneously. On the production side, **?** find that access to a large language model cuts professional writing time by 40% and raises rated output quality by 18 percentage points. **?** detect language-model modification in more than one-fifth of computer-science preprints by late 2024, and **?** find AI-assisted text in at least 13.5% of biomedical abstracts from the same period. On the evaluation side, however, the picture is more troubling. **?** conclude that deep verification—checking data, code, and logical consistency—remains beyond current automated systems. **?** find that between 6.5% and 16.9% of reviews at major conferences already contain AI-generated text, and **?** show experimentally that AI-assisted reviews award higher scores, raising acceptance rates. **?** reports that more than half of surveyed researchers now use AI tools for reviewing, often against journal policy. These findings supply the empirical foundation for our asymmetry assumption: artificial intelligence raises submission productivity faster than it raises verification capacity.

A fourth strand addresses the economics of gatekeeping and institutional design. **?** shows that the equilibrium publication standard depends on how strongly readers value quality signals. **?** finds that journals maintain strict standards to protect their certification value: accepting too freely devalues their own signal. **?** argue that effective reform must target the false-positive rate directly—through data deposition, structured reporting, and replication requirements—rather than adjusting slot counts alone. **?** shows that dissemination and certification can be separated: preprint servers make manuscripts available immediately while editorial endorsement follows independently. **?** provides the sharpest economic case for submission fees: a small cost of entry restores friction, filters marginal submissions, and reduces reviewer overload.

What we contribute

While these literatures each address part of the problem, none unifies them. Models of strategic submission do not treat artificial intelligence as a structural shock that shifts multiple inputs at once. The empirical AI-in-science literature documents faster production, slower review growth, and higher false-positive risk, but does not connect these to a model with clear predictions about the long-run state of the system. Neither strand addresses the possibility that submission pressure can grow gradually until it crosses a threshold, after which the system shifts suddenly to a low-quality state and resists recovery. The institutional design literature does not ask how its proposals must change when drafting cost approaches zero and the false-positive rate rises continuously with adoption. We close these gaps by connecting all four channels—$c(a)$ falling, V rising, K_{rev} lagging, and $\lambda(a)$ rising—to the long-run state of the publishing system, characterising conditions under which quality collapses suddenly and is hard to restore, and deriving which institutional levers can prevent or reverse that shift.

4 Baseline Model

4.1 ~~Setting~~

Players and strategies. We consider a ~~publication game involving a finite~~ population of N authors ~~and a single academic journal . Our framework aligns with established game-theoretic models of scientific publishing ?, while specifically building on the incentive structures analyzed by Zollman et al. ?. Unlike previous static models, we extend the analysis to account for science volume. The fundamental unit of the game is a research paper, which possesses a binary intrinsic quality~~$q \in \{0, 1\}$~~. A paper is defined as "good "~~ submitting to a single journal over repeated periods. Each manuscript has binary quality: good ($q = 1$) ~~or "bad "~~ with probability α, and bad ($q = 0$) with probability $1 - \alpha$. Under replicator dynamics, intermediate submission strategies are evolutionarily unstable and are eliminated in the mean-field limit, reducing the effective strategy space to two options (**??**). Under **selective submission** (OG), an author submits only when $q = 1$. Under **universal submission** (AS), an author submits regardless of quality, hoping that imperfect screening allows bad papers through. We write $x \in [0, 1]$ for the share of authors

playing OG; the remaining $1 - x$ play AS. ~~In each period, authors make submission decisions, and the journal enforces quality control standards subject to a physical capacity constraint.~~

~~The publication process unfolds in a strictly sequential timeline, as illustrated in Figure ??:~~

~~Submission (Generation of~~

Two bottlenecks. The publication process has two distinct chokepoints; each creates a different strategic force. The first is **review capacity** K_{rev}: the number of papers that can receive careful scrutiny per period. Submission volume is $V(x) = N[1 - (1 - \alpha)x]$, which rises as the AS share grows. When V ~~)~~exceeds K_{rev}, the available reviewer time must be spread across more submissions: each paper receives less careful attention, and the effective depth of scrutiny per manuscript falls (**??**). The **screening intensity** is:

$$\eta(x) = \min\left(1,\ \frac{K_{rev}}{V(x)}\right). \tag{2}$$

The second bottleneck is **publication capacity** K: the number of slots the journal can fill per period. Papers that pass review form a peer-accepted pool $M(x)$; at most K are published. The probability that a peer-accepted paper is ultimately published—the **rationing probability**—is: ~~Based on their strategies, authors decide whether to submit their work. The aggregate collection of all submitted manuscripts forms the *Submission Pool*, denoted by $V \leq N$.~~

$$\bar{\rho}(x) = \min\left(1,\ \frac{K}{M(x)}\right). \tag{3}$$

~~**Peer Review (Generation of M):** The journal applies a peer review process to filter the submission pool. Acknowledging the inherent unreliability of assessment documented in empirical studies ?, we model reviewers as imperfect classifiers. They are subject to Type I errors (False Positives, rate λ) and Type II errors (False Negatives, rate~~

Screening errors and the spam payoff. Peer review is an imperfect filter (**??**). A good paper is falsely rejected with probability ϵ~~). Papers that successfully pass this screening form the set of *Peer-accepted papers*, denoted by M. It is important to note that membership in M signifies that a paper is "scientifically acceptable," but not necessarily "published."~~

~~**Capacity Check and Storage (K):** The journal has a fixed resource limit or *Journal Storage*, denoted by K. This constraint reflects the physical limits on publication slots or the administrative burden of processing manuscripts ?.~~

~~**Final Acceptance (ρ):** The final publication decision depends on the relationship between the volume of~~; a bad paper wrongly passes with probability λ. Both error rates apply at full scrutiny. When $\eta < 1$, reviewer time per paper is diluted: each submission receives only a fraction η of the careful attention required to apply the full filter, so the effective pass rates rise above their full-scrutiny values (**?**):

$$\pi_0(x) = 1 - \eta(x)(1 - \lambda), \qquad \pi_1(x) = 1 - \eta(x)\,\epsilon. \tag{4}$$

As $\eta \to 0$, both rates converge to 1: the journal loses all discriminatory power because reviewers cannot give any paper the attention the filter requires. The peer-accepted ~~papers (M)and the journal storage (K) . We define ρ as the *probability of being accepted conditional on being peer-accepted.* This variable captures the congestion effect:~~ pool is $M(x) = N\alpha\,\pi_1(x) + N(1-\alpha)(1-x)\,\pi_0(x)$. Because OG and AS authors submit good papers on identical terms, the strategic difference reduces to the net value of submitting a bad paper:

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c, \tag{5}$$

where r is the publication reward and c the submission cost. The analysis considers two parameter regions. In the primary region $\lambda r < c < r(1 - \varepsilon)$—which governs Set A and the GH and BS regimes—spamming is unprofitable without congestion but the incentive reactivates once screening

degrades: the adverse-selection logic of Akerlof's market for lemons (**?**). Set B explores the complementary region $\lambda r > c$, where spamming would be profitable even at full screening intensity ($\eta = 1$), but tight slot rationing ($\bar{\rho} \ll 1$) keeps the corrupt boundary stable by making the publication lottery too hard to win—the defence-by-congestion mechanism operating through $\bar{\rho}$ rather than through η.

Dynamics. The expected per-period fitness of each strategy differs only in the treatment of bad papers. An OG author submits only when $q = 1$ (probability α), earning expected fitness $f_{OG} = \alpha[r\bar{\rho}\,\pi_1 - c]$. An AS author submits regardless, earning $f_{AS} = \alpha[r\bar{\rho}\,\pi_1 - c] + (1-\alpha)[r\bar{\rho}\,\pi_0 - c]$. The fitness advantage of AS over OG is therefore $f_{AS} - f_{OG} = (1-\alpha)\,\Pi_{bad}(x)$. Substituting into the standard two-strategy replicator equation $\dot{x} = x(1-x)[f_{OG} - f_{AS}]$ gives:

$$\cancel{\rho}\dot{x} = -x(1-x)(1-\alpha)\,\Pi_{bad}(x). \tag{6}$$

When $\Pi_{bad} > 0$, AS spreads and x falls; when $\Pi_{bad} < 0$, OG spreads and x rises. Two forces compete as the AS share grows. First, rising volume degrades screening (η falls, π_0 rises): submitting bad papers becomes more attractive. Second, rising volume floods the publication queue ($\bar{\rho}$ falls): the publication lottery becomes harder to win. When the screening-collapse force dominates, the system tips toward full corruption. When the rationing force dominates, extreme slot scarcity deters spam even after screening has failed—a mechanism we call *defence by congestion*.

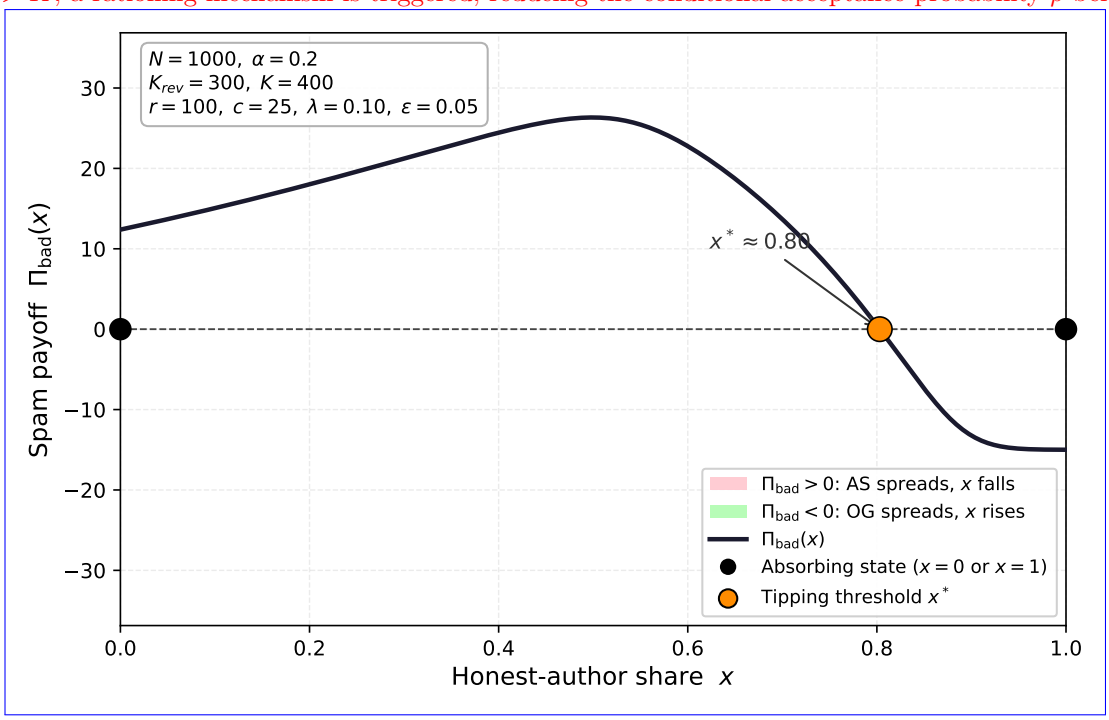
~~In this framework, a paper is published if and only if it passes two hurdles: the quality hurdle (peer review)and the congestion hurdle (storage constraint)~~

Four long-run outcomes. The signs of Π_{bad} at the two boundary states determine which long-run outcome the system reaches. Table 8 summarises the four cases. A key structural observation is that the two boundaries are controlled by different parameters: the honest boundary ($x = 1$) is secured by keeping λ low, while the corrupt boundary ($x = 0$) is held by keeping K tight. This means the two bottlenecks can be targeted separately as policy instruments—a point we return to in Section 6. Figure 2 illustrates the bistable case. Π_{bad} is positive near $x = 0$ (corruption reinforces itself) and negative near $x = 1$ (honesty reinforces itself), with an unstable tipping threshold at $x^* \approx 0.80$ separating the two regions. Which long-run outcome prevails depends entirely on where the system starts relative to this threshold.

Table 2: **Four long-run outcomes determined by boundary payoffs.** $\Pi_{bad}(1)$ is the spam payoff when almost all authors are honest (low-volume limit); $\Pi_{bad}(0)$ is the spam payoff when almost no authors are honest (maximum-load limit).

Regime	$\Pi_{\text{bad}}(1)$	$\Pi_{\text{bad}}(0)$	**Long-run outcome**
GH	< 0	< 0	**Global honesty.** Spamming is unprofitable everywhere. The system reaches $x = 1$ from any starting point. Primary lever: keep $\lambda < c/r$.
GC	> 0	> 0	**Global corruption.** Spamming is profitable everywhere. The system collapses to $x = 0$ from any starting point. Seen in predatory journals with weak review and ample slots.
BS	< 0	> 0	**Bistability.** Both $x = 1$ and $x = 0$ are long-run outcomes. An unstable tipping threshold x^* separates the regions; history determines which outcome prevails. Recovery from corruption requires a large intervention.
SC	> 0	< 0	**Stable coexistence.** Neither extreme is reached. A stable interior outcome x^* emerges, sustained by rationing. Seen in top-tier journals where the publication lottery is very hard to win.

~~**Schematic representation of the Threshold Competition Model.** The process unfolds in stages: (1) Authors from population N submit papers into pool V; (2) Peer review filters submissions into a set of peer-accepted papers M; (3) The volume M is compared against the journal storage capacity K. If $M > K$, a rationing mechanism is triggered, reducing the conditional acceptance probability ρ below~~



~~1.~~

Figure 2: **Baseline phase portrait (bistability).** The curve plots $\Pi_{bad}(x)$, the net payoff from submitting a bad paper, against the honest-author share x. Parameters: $N = 1000$, $K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$. Red shading: $\Pi_{bad} > 0$, so AS spreads and x falls. Green shading: $\Pi_{bad} < 0$, so AS shrinks and x rises. Two long-run outcomes coexist—full honesty ($x = 1$) and full corruption ($x = 0$)—separated by an unstable tipping threshold at $x^* \approx 0.80$ (orange marker). Which outcome prevails depends entirely on the starting position relative to x^*.

4.1 ~~Players, strategies, and quality generation~~

~~The publication game involves a single journal (J) and a finite population of authors $\mathcal{A} = \{1, 2, \ldots, N\}$. We treat~~

Box 1 The Baseline Model in Brief

Parameters. N~~as a finite exogenous parameter. This finite-population approach allows us to capture the congestion dynamics and "tragedy of the commons" effects inherent in resource-constrained peer review systems, as analyzed in the economic literature on academic publishing (??). Throughout, we use V and M to denote the realised *counts* of submitted and~~: author population; α: probability a paper is genuinely good; x: share of selective (OG) authors; K_{rev}: review capacity; K: publication slots; r: publication reward; c: submission cost; λ: false-positive rate (bad paper passes review); ε: false-negative rate (good paper rejected).

Two bottlenecks. Submission volume $V(x) = N[1 - (1 - \alpha)x]$ rises as the always-submit (AS) share grows.

$\eta(x) = \min\left(1, \frac{K_{rev}}{V(x)}\right)$ screening intensity — falls below 1 when volume exceeds capacity, diluting scrutiny per paper

$\bar{\rho}(x) = \min\left(1, \frac{K}{M(x)}\right)$ rationing probability — falls below 1 when accepted papers exceed slots

where $M(x) = N\alpha\,\pi_1(x) + N(1 - \alpha)(1 - x)\,\pi_0(x)$ is the peer-accepted ~~papers, respectively.~~ pool.

~~Quality generation.~~ ~~In each round, each author generates a manuscript with binary intrinsic quality $q \in \{0,1\}$, where $q = 1$ denotes a "good" paper and $q = 0$ a "bad" paper. Quality is generated i. i. d. across authors with~~

$$\cancel{\Pr(q = 1) = \alpha, \qquad \Pr(q = 0) = 1 - \alpha,}$$

~~where $\alpha \in (0,1)$ is an exogenous productivity parameter reflecting the baseline scientific capability of the community (?).~~ **Pass rates under congestion.**

$$\pi_0(x) = 1 - \eta(x)(1 - \lambda) \qquad \text{bad paper passes; rises as scrutiny is diluted}$$

$$\pi_1(x) = 1 - \eta(x)\,\varepsilon \qquad \text{good paper passes; rises as scrutiny is diluted}$$

~~Author strategies (S_A).~~ ~~Each author chooses a submission rule $s_A \in S_A$ that conditions on realised quality. Consistent with signaling models in scientific communication (?), we consider four pure strategies:~~

- **~~Only Good (OG):~~** ~~submit iff $q = 1$;~~
- **~~Always Submit (AS):~~** ~~submit regardless of q (strategic spamming) ;~~
- **~~Only Bad (OB):~~** ~~submit iff $q = 0$ (included for completeness);~~
- **~~Never Submit (NS):~~** ~~never submit.~~

~~Equivalently, s_A can be represented by (p_g, p_b), the submission probabilities for good and bad papers:~~

$$\cancel{(p_g, p_b) = \begin{cases} (1,0) & \text{if } s_A = \text{OG} \\ (1,1) & \text{if } s_A = \text{AS} \\ (0,1) & \text{if } s_A = \text{OB} \\ (0,0) & \text{if } s_A = \text{NS.} \end{cases}}$$

Spam payoff and dynamics.

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c \qquad \Longrightarrow \qquad \dot{x} = -x(1-x)(1-\alpha)\,\Pi_{bad}(x)$$

$\Pi_{bad} > 0$: AS spreads, x falls. $\Pi_{bad} < 0$: OG spreads, x rises.

~~Journal strategies (S_J).~~ ~~The journal commits to an editorial policy $s_J \in S_J$ that maps the perceived quality signal $\hat{q} \in \{0,1\}$ into an accept/reject decision:~~

- **~~Only Good (OG):~~** ~~accept iff $\hat{q} = 1$;~~
- **~~Always Accept (AA):~~** ~~accept all submissions (ignores $\hat{q}$);~~
- **~~Only Bad (OB):~~** ~~accept iff $\hat{q} = 0$;~~
- **~~Reject All (AR):~~** ~~reject all submissions (ignores $\hat{q}$).~~

Four long-run outcomes (determined by the signs of Π_{bad} at the two boundaries):

~~Exogenous parameters and endogenous variables.~~ ~~To clarify the mechanics of congestion, we distinguish between fixed environmental constraints and emergent outcomes: **Exogenous parameters:** Population size (N) , productivity (α), journal size (K) , review noise parameters (ϵ, λ) ,~~

Regime	$\Pi_{bad}(1)$	$\Pi_{bad}(0)$	Long-run outcome
GH	< 0	< 0	$x = 1$: all authors submit only good work
GC	> 0	> 0	$x = 0$: all authors submit regardless of quality
BS	< 0	> 0	Both $x = 1$ and ~~payoff primitives, discussed below: c, cost of submission for authors; r, the re~~
SC	> 0	< 0	Interior x^*; slot rationing deters spam

4.1 ~~Review noise and acceptance probabilities~~

Key insight: The honest boundary ($x = 1$) is secured by keeping λ low. The corrupt boundary ($x = 0$) is held by keeping K tight. The two bottlenecks can therefore be targeted *separately* as policy instruments.

~~Peer review acts as a stochastic filter between the submission volume$V$~~

5 The AI Extension

AI adoption as a parameter shift. Generative AI affects the publication system through multiple channels simultaneously. We capture this with a single scalar $a \in [0, 1]$, where $a = 0$ denotes the pre-AI baseline and $a = 1$ denotes full adoption across the author population. The parameter can be interpreted as the field-level intensity of AI use—a composite of the fraction of authors employing AI tools and the depth of their integration into the writing and submission workflow. We treat a as a comparative-statics parameter: we hold a fixed and trace how the equilibrium structure from Section 4 changes as a deepens. As established in Section 2 and illustrated in Figure 1, AI acts through four distinct channels, and the ~~peer-accepted volume M. Throughout, *pass* refers to passing the review filter and entering the peer-accepted pool M (not final publication, which is also affected by the journal size~~channels are not symmetric.

Channel 1—Production: more submissions at every composition. The most direct effect of AI is to lower the marginal cost of drafting and preparing a manuscript, which raises total submission volume at any given composition of the author population. Randomised experiments find that LLM access cuts professional writing time by roughly 40% (**??**). We formalise this as a multiplicative shift in total submission volume:

$$V(x, a) \;=\; N\big[1 - (1 - \alpha)x\big] \cdot (1 + \gamma a), \tag{7}$$

where $\gamma > 0$ is the **production elasticity**. System-level data confirm this: monthly arXiv submissions rose roughly 50% in two years, with over one-fifth of preprints showing measurable LLM modification (**??**). V rises with a regardless of x: AS authors, whose constraint was drafting effort rather than ideas, now prepare and submit low-quality manuscripts they would not previously have bothered producing; OG authors face a lower marginal cost of bringing a manuscript to submission-ready quality, converting preliminary work into formal submissions faster. The underlying quality probability α of any individual manuscript remains fixed; what changes is the number of submission-ready manuscripts each author produces per period.

~~**Noise parameters (ϵ, λ)**~~**Channel 2—Friction: lower cost of submitting bad papers.** ~~Consistent with empirical meta-analyses on review reliability (**??**), we assume reviewers are imperfect classifiers. Let $\hat{q} \in \{0, 1\}$ denote the perceived quality signal for a paper with true quality $q \in \{0, 1\}$. We define:~~ Beyond raising volume, AI directly lowers the cost of each submission attempt. We write:

$$\cancel{\epsilon = \Pr}c(\cancel{q}a) \;= \cancel{0 \mid q =}\; c_0(1 - \phi\, a), \qquad \cancel{\lambda = \Pr}\phi \in (\cancel{q = 1 \mid q =}0, 1), \tag{8}$$

where ~~ϵ is the false negative rate (acting as a tax on honesty) and λ the false positive rate (acting as an incentive for opportunism), a structure supported by empirical studies on scientific gatekeeping (**?**).~~ c_0 is the baseline cost and ϕ governs the rate at which AI erodes submission friction. The cost reduction enters Π_{bad} directly: a lower c raises the net payoff from submitting a bad paper even if acceptance probabilities are unchanged. The effect is also asymmetric across strategies, though the asymmetry is not per-paper: c enters both Π_{bad} and Π_{good} symmetrically as a per-submission cost. The asymmetry arises from submission frequency. OG authors submit only when they have genuine work ready, so the total cost reduction they realise each period is small. AS authors submit everything—good papers and bad—so the same per-paper reduction compounds across a much larger submission stream, directly amplifying the flood incentive.

~~**Conditional acceptance probabilities.** Given a journal policy s_J, define the probability of acceptance conditional on true quality as:~~

$$\cancel{\pi_q(s_J) \equiv \Pr(\text{pass} \mid q, s_J), \qquad q \in \{0, 1\}.}$$

~~Under the journal strategiesdefined in Section **??**, and aligned with theoretical models of gatekeeper~~

~~selection (?), the acceptance vector (π_1, π_0) is given by:~~

$$\cancel{(\pi_1(s_J), \pi_0(s_J))} = \begin{cases} (1-\epsilon,\ \lambda) & \text{if } s_J = \text{OG} \\ (1,\ 1) & \text{if } s_J = \text{AA} \\ \hline (\epsilon,\ 1-\lambda) & \text{if } s_J = \text{OB} \\ (0,\ 0) & \text{if } s_J = \text{AR}. \end{cases}$$

~~**Per-author pass-review probability ($\mathcal{P}$).** Let $(p_g(s_A), p_b(s_A))$ denote the submission probabilities conditional on realised quality. We define $\mathcal{P}(s_A, s_J)$ as the *ex-ante* probability that an author playing s_A contributes a paper to the peer-accepted pool M under journal policy s_J:~~

Channel 3—Vulnerability: weaker screening signal. AI-generated text is harder to distinguish on surface features (**?**), AI-assisted reviews award systematically higher scores (**??**), and automated detection remains unreliable—false-negative rates exceed 70% (**?**). Each mechanism raises the probability that a bad paper passes review. We write:

$$\cancel{\mathcal{P}}\lambda(\cancel{s_A, s_J}a) \ = \cancel{\alpha\, p_g(s_A)\, \pi_1(s_J)}\ \lambda_0 + \cancel{(1-\alpha)}\mu\, \cancel{p_b(s_A)\, \pi_0(s_J)}.a, \qquad \mu \geq 0, \tag{9}$$

~~Below we enumerate the effective acceptance probabilities for all author strategies under each journal regime.~~

~~**Case 1: Standard Review (Journal plays Only Good).** This~~ where λ_0 is the baseline ~~scenario where $\pi_1 = 1-\epsilon$ and $\pi_0 = \lambda$.~~

$$\begin{aligned} &\cancel{\mathcal{P}(\text{OG}, \text{OG}) = \alpha(1-\epsilon),} \\ &\cancel{\mathcal{P}(\text{AS}, \text{OG}) = \alpha(1-\epsilon) + (1-\alpha)\lambda,} \\ &\cancel{\mathcal{P}(\text{OB}, \text{OG}) = (1-\alpha)\lambda,} \\ &\cancel{\mathcal{P}(\text{NS}, \text{OG}) = 0.} \end{aligned}$$

false-positive rate and $\mu \geq 0$ is the sensitivity of screening quality to AI adoption. $\lambda(a)$ is a reduced-form parameter: it captures the net effect of AI adoption on the false-positive rate without modelling each mechanism separately. Rising λ enters $\pi_0 = 1 - \eta(1-\lambda)$ directly: even a carefully reviewed paper is more likely to pass.

~~**Case 2: Permissive (Journal plays Always Accept).** The screening is ignored $(\pi_1 = 1, \pi_0 = 1)$, making acceptance depend solely on submission.~~

$$\begin{aligned} &\cancel{\mathcal{P}(\text{OG}, \text{AA}) = \alpha,} \\ &\cancel{\mathcal{P}(\text{AS}, \text{AA}) = \alpha + (1-\alpha) = 1,} \\ &\cancel{\mathcal{P}(\text{OB}, \text{AA}) = 1-\alpha,} \\ &\cancel{\mathcal{P}(\text{NS}, \text{AA}) = 0.} \end{aligned}$$

Channel 4—Review assistance: capacity grows, but more slowly. AI can also assist reviewers—summarising papers, flagging inconsistencies, checking statistical reporting—so review capacity is not fixed. We write:

$$K_{rev}(a) \ = \ K_{rev}(0) \cdot (1 + \delta\, a), \tag{10}$$

where $\delta \geq 0$ is the **verification elasticity**. However, deep verification remains human-intensive: confidentiality rules restrict AI use in reviews (**??**), and reviewer capacity was already strained before the AI era (**?**). These constraints enforce the asymmetry $\gamma > \delta \geq 0$ (Equation 1).

~~**Case 3: Inverse Selection (Journal plays Only Bad).** The journal accepts papersperceived as bad $(\hat{q} = 0)$. Good papers pass only via false negatives $(\pi_1 = \epsilon)$, and bad papers pass via true negatives~~

~~$(\pi_0 = 1 - \lambda)$.~~

$$\cancel{\mathcal{P}(\mathrm{OG}, \mathrm{OB}) = \alpha\epsilon,}$$
$$\cancel{\mathcal{P}(\mathrm{AS}, \mathrm{OB}) = \alpha\epsilon + (1-\alpha)(1-\lambda),}$$
$$\cancel{\mathcal{P}(\mathrm{OB}, \mathrm{OB}) = (1-\alpha)(1-\lambda),}$$
$$\cancel{\mathcal{P}(\mathrm{NS}, \mathrm{OB}) = 0.}$$

Figure 3 summarises the four channels and their functional forms.

~~**Case 4: Closed (Journal plays Reject All).** No papers enter the peer-accepted pool $(\pi_1 = 0, \pi_0 = 0)$.~~

$$\cancel{\mathcal{P}(s_A, \mathrm{AR}) = 0 \quad \forall s_A \in S_A.}$$

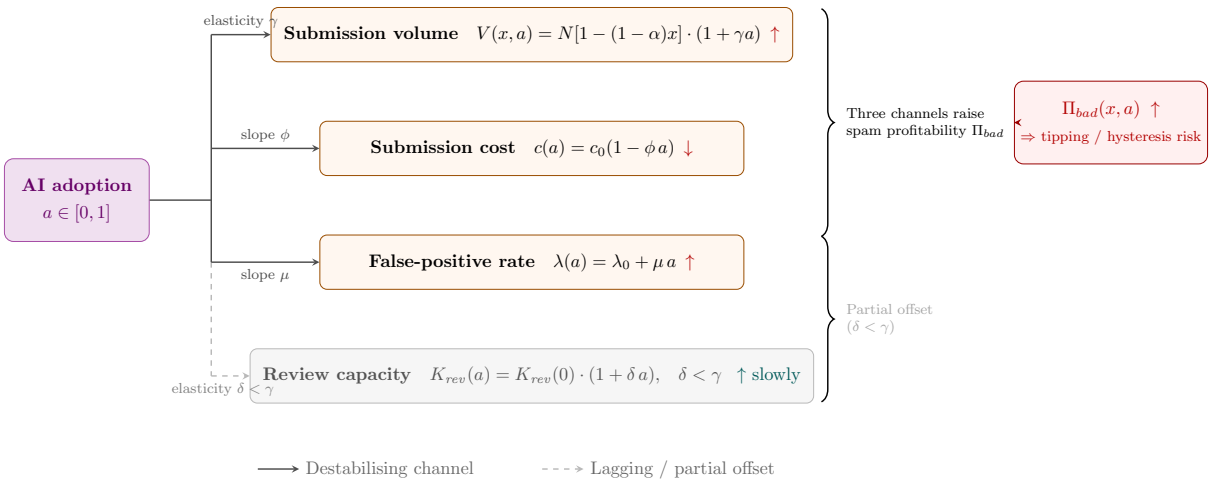


Figure 3: **AI adoption shifts four model parameters (Figure 3).** Solid arrows indicate channels that directly raise spam profitability Π_{bad}: rising submission volume (V, elasticity γ), falling submission cost (c, slope ϕ), and rising false-positive rate (λ, slope μ). The dashed arrow indicates the only offsetting channel—review capacity (K_{rev}, elasticity δ)—which grows more slowly than submission volume under the asymmetry assumption $\gamma > \delta$. All four channel equations are formal specifications of the informal channels described in Section 2.

~~**Aggregate peer-accepted volume ($E[M]$)**~~**The combined effect: spam becomes more profitable.** ~~To determine the system-wide congestion, we aggregate these individual passing probabilities across the entire population, a method consistent with burden estimation in peer review (?). Let $\mathbf{x} = \{x_{s_A} \mid s_A \in S_A\}$ denote the **population state**, where $x_{s_A} \in [0,1]$ represents the proportion of authors adopting strategy s_A, satisfying $\sum_{s_A \in S_A} x_{s_A} = 1$.~~

~~The expected total volume of peer-accepted papers~~ Substituting the four channels into the baseline model yields the AI-extended spam payoff:

$$\Pi_{bad}(x,a) \;=\; r \cdot \bar{\rho}(x,a) \cdot \pi_0(x,a) \;-\; c(a), \tag{11}$$

where $\eta(x,a) = \min(1, K_{rev}(a)/V(x,a))$, $\pi_0(x,a) = 1 - \eta(x,a)(1-\lambda(a))$, ~~$E[M]$, is the sum of contributions from all strategy groups:~~

$$\cancel{E[M] = \sum_{i=1}^{N} \mathcal{P}(s_{A,i}, s_J) = N \sum_{s_A \in S_A} x_{s_A} \mathcal{P}(s_A, s_J).}$$

~~This aggregate volume $E[M]$ is the critical endogenous variable that interacts with the journal capacity K to determine the rationing probability ρ defined in Eq. (1).~~

5.1 ~~Capacity constraint and rationing probability (ρ)~~

~~A key feature of~~ $\bar{\rho}(x,a) = \min(1, K/M(x,a))$, and

$$M(x,a) \;=\; (1+\gamma a)\big[N\alpha\,\pi_1(x,a) + N(1-\alpha)(1-x)\,\pi_0(x,a)\big]. \tag{12}$$

The factor $(1+\gamma a)$ reflects that AI inflates the total pool of papers reaching the publication queue proportionally to the same volume multiplier that governs $V(x,a)$: higher submission rates translate into a larger peer-accepted pool for any given pass rates π_0, π_1. This introduces a countervailing stabilising force. A larger M tightens slot rationing ($\bar{\rho} = K/M$ falls), reducing each spammer's probability of publication. Formally, $V\uparrow \Rightarrow M\uparrow \Rightarrow \bar{\rho}\downarrow \Rightarrow \Pi_{bad}\downarrow$: the ~~model is the separation between *scientific merit* (passing peer review and entering the set of peer-accepted papers) and *publication slots* (capacity-constrained publication). This distinction captures the reality that top-tier journals often reject scientifically sound papers solely due to space limitations (**??**).~~

~~**Journal capacity (K).** Let $K \in \mathbb{N}^+$ denote the journal's exogenous publication capacity: at most~~ very flood of submissions that degrades screening also clogs the publication queue. Note that this congestion effect compresses the *revenue* term $r\bar{\rho}\pi_0$ rather than raising the submission cost $c(a)$, which falls with a. Whether the destabilising channels ($\pi_0\uparrow$, $c\downarrow$) or the stabilising channel ($\bar{\rho}\downarrow$) dominates depends on the relative magnitudes of K ~~papers can be published per cycle due to limited pages and editorial resources (**??**).~~

~~**Uniform rationing and the rationing probability (ρ).** Let~~ and M ~~be the realised number of peer-accepted papers in a given round. If $M \le K$, all peer-accepted papers are published. If $M > K$, the journal applies *uniform rationing*: it selects exactly K papers uniformly at random from the M peer-accepted papers. Hence, the conditional publication probability for a peer-accepted paper is determined by the realised congestion:~~

$$\rho(M) \equiv \Pr(\text{published} \mid \text{pass}) = \min\!\left(1, \tfrac{K}{M}\right) = \begin{cases} 1 & \text{if } M \le K, \\ \frac{K}{M} & \text{if } M > K. \end{cases}$$

: when $K \gg M$ rationing is slack and the destabilising forces prevail; when $K \approx M$ the congestion defence is active. To isolate the mechanism most cleanly, consider the honest boundary $x = 1$, where only good papers are submitted and slot rationing is typically absent ($\bar{\rho}(1,a) \approx 1$ when publication slots exceed honest-author output):

$$\Pi_{bad}(1,a) \;\approx\; r \cdot \pi_0(1,a) \;-\; c(a). \tag{13}$$

~~**Ex-post vs. ex-ante congestion.** Across rounds, the peer-accepted count M is stochastic because it aggregates individual submission and review outcomes. Authors therefore form expectations over congestion given the population state $\mathbf{x}$ and journal policy s_J, implying an expected rationing factor:~~

$$\cancel{\mathbb{E}[\rho] = \mathbb{E}\big[\min\!\left(1, \tfrac{K}{M}\right)\big]}.$$

At $x = 1$, only good papers are submitted, so submission volume equals $V(1,a) = N\alpha(1+\gamma a)$. For both calibration sets this remains strictly below $K_{rev}(a)$ for all $a \in [0,1]$, so $\eta(1,a) = 1$: review capacity is not the binding constraint at the honest boundary.[1] The bad-paper pass rate therefore reduces to $\pi_0(1,a) = \lambda(a)$, and the GH condition at $x = 1$ simplifies to $\lambda(a) < c(a)/r$. The operative mechanism at the honest boundary is not review congestion but the erosion of the screening signal relative to the cost of submission: $\lambda(a)$ rises while $c(a)/r$ falls simultaneously. Provided $\mu \ge 0$ and $\phi > 0$:

$$\frac{\partial \Pi_{bad}(1,a)}{\partial a} \;=\; r \underbrace{\mu}_{>0 \text{ (rising } \lambda)} \underbrace{+\, c_0\phi}_{>0 \text{ (falling } c)} \;>\; 0. \tag{14}$$

[1] For Set A: $K_{rev}(a) = 300(1+0.1a) > 200(1+0.5a) = V(1,a)$ requires $a < 10/7 \approx 1.43$, well above the maximum adoption level.

~~Crucially, increases in low-quality or opportunistic submissions raise the expected volume $\mathbb{E}[M]$ and mechanically reduce $\mathbb{E}[\rho]$, generating a negative congestion externality on all authors. This mirrors the inefficiencies found in scientific funding competitions, where excessive participation degrades the system's overall value (??).~~ Setting $\Pi_{bad}(1, a^*) = 0$ gives a closed-form tipping point $a^* = (c_0 - r\lambda_0)/(r\mu + c_0\phi)$; for Set A, $a^* = (25 - 10)/(20 + 10) = 0.50$. Above a^*, $\Pi_{bad}(1, a) > 0$: the honest long-run outcome at $x = 1$ is no longer stable, and the system transitions from bistability into Global Corruption (or from GH into bistability if the corrupt boundary was previously deterred).

5.1 ~~Payoff analysis~~

~~In this subsection, we derive the expected payoffs for both authors and~~

Creation versus verification: why AI is structurally asymmetric. The four-channel decomposition shows that AI exerts pressure from both the supply and quality dimensions simultaneously. Channels 1 and 2 affect the supply side: they make it cheaper and faster to produce manuscripts. Channel 3 affects the screening side: it makes it harder to distinguish good submissions from bad ones at the review stage. Channel 4 is the only force working against the other three, but it is structurally constrained to grow more slowly. Writing is largely a syntactic task that language models perform well; evaluation is a semantic and domain-specific task requiring human expertise, data access, and contextual judgment (**?**). Institutions designed for the ~~journal.The introduction of the capacity constraint K fundamentally couples the payoffs: an individual's utility depends not only on their own strategy but also on the aggregate congestion generated by the population. This structure formally aligns with the class of *congestion games*, where individual payoffs are negatively correlated with the total utilization of a shared resource (**?**).~~ pre-AI era assumed rough parity between production and verification rates. That assumption no longer holds, and the asymmetry is not a temporary calibration issue—it reflects a fundamental difference in what AI can automate.[2]

~~**Notation and general framework** Table 3 summarizes the key parameters and variables. We employ a *mean-field approximation* where players base their decisions on the **expected** state of the system . This approach is standard in evolutionary game theory for modeling interactions in large finite populations, allowing deterministic analysis of stochastic processes (**??**)~~

Special cases: when stability can be preserved. Not every AI adoption trajectory leads to quality collapse. Equation (13) shows that the Global Honesty condition can persist even at high a if two circumstances hold jointly. First, if $\delta \approx \gamma$—review capacity scales nearly as fast as submission volume—then $\eta(1, a)$ does not fall and the congestion channel is neutralised. This could arise if structured AI-assisted verification pipelines (automated code checks, data-consistency audits) allow reviewers to handle more submissions without sacrificing depth. Second, if $\mu \approx 0$—AI tools do not degrade screening reliability, perhaps because structured reporting requirements close the gap that polished AI writing opens—then $\lambda(a)$ does not rise. If both conditions hold simultaneously, the cost reduction alone is typically insufficient to cross the tipping threshold, because $c(a)$ falls gradually while the honesty condition $c(a) > r\lambda(a)$ can remain satisfied. Table 4 maps the four combinations of these two conditions onto qualitative predictions about regime direction. The critical implication is that the institutional response determines which cell the system occupies: a is largely exogenous, but δ and μ are policy choices.

Testable predictions. The AI extension generates four empirically distinguishable predictions. First, quality degradation should be *nonlinear* in AI adoption: the system holds near the honest long-run

[2] We hold α (the probability a manuscript is genuinely good) fixed throughout the formal analysis. In principle, AI may also raise α via a quality-uplift channel: writing $\alpha(a) = \alpha_0(1 + \xi a)$ would add a partially offsetting force. Higher α increases the share of genuine work in the submission pool, which raises $M(x, a)$ and tightens slot rationing $\bar{\rho}$, thereby reducing Π_{bad}. However, the GH stability condition at $x = 1$ reduces to $\lambda(a) < c(a)/r$ when $\eta = 1$—a condition that does not involve α—so quality uplift does not shift the closed-form tipping threshold a^*. Evidence that AI raises genuine research quality is mixed and modest in magnitude (**?**); we therefore treat α as fixed to focus on the four identified destabilising channels.

outcome while a remains below the tipping threshold, then shifts abruptly rather than gradually. Smooth, continuous quality decline would be inconsistent with the bistable structure. Second, fields with higher AI adoption should show review-congestion signatures—longer turnaround times, rising desk-rejection rates, declining reviewer acceptance rates—as proxies for falling η (**?**). Third, acceptance rates should rise in fields with widespread AI-assisted reviewing, independently of any improvement in underlying research quality, as a signature of rising λ (**?**). Fourth, expanding publication slots should have *asymmetric* effects: before tipping, more slots increase access; after tipping, more slots accelerate quality collapse by raising the spam payoff Π_{bad}. We examine the regime consequences of each lever in Section 6.

Box 2 What AI Changes: Parameter Shifts and the Asymmetry

AI adoption parameter. $a \in [0, 1]$: field-level AI adoption intensity ($a = 0$ = pre-AI baseline; $a = 1$ = full adoption).

Four channels and their functional forms.

Channel	Functional form	Direction	Key parameter
Submission volume	$V(x, a) = N[1 - (1 - \alpha)x] \cdot (1 + \gamma a)$	$\uparrow$	elasticity γ
Submission cost	$c(a) = c_0(1 - \phi a)$	$\downarrow$	slope $\phi \in (0, 1)$
False-positive rate	$\lambda(a) = \lambda_0 + \mu a$	$\uparrow$	slope $\mu \geq 0$
Review capacity	$K_{rev}(a) = K_{rev}(0) \cdot (1 + \delta a)$	$\uparrow$ slowly	elasticity $\delta < \gamma$

~~**Submission and publication probabilities.** For an author playing strategy $s_A \in S_A$, we define the unconditional probability of submitting a paper as $P_{\text{sub}}(s_A)$. Based on the submission vectors (p_g, p_b) defined in Section **??**, this is:~~

$$\cancel{P_{\text{sub}}(s_A) = \alpha p_g(s_A) + (1 - \alpha) p_b(s_A).}$$

The asymmetry assumption: $\gamma > \delta \geq 0$ — production scales faster than verification. This is a testable empirical claim, not an axiom (Equation 1).

Net effect on spam payoff.

$$\frac{\partial \Pi_{bad}(x, a)}{\partial a} = \underbrace{r\mu}_{\lambda\uparrow} + \underbrace{c_0\phi}_{c\downarrow} + \underbrace{[\text{congestion terms}]}_{>0 \text{ when } \gamma > \delta} > 0 \qquad \text{under the calibrations of Table 9.}$$

~~Recall from Section **??** that $\mathcal{P}(s_A, s_J)$ denotes the probability of *entering the peer-accepted pool* M. However, entering M does not guarantee publication. Let $\bar{\rho}$ denote the **expected rationing probability**:~~

$$\cancel{\bar{\rho} = \mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \tfrac{K}{M}\right)\right].}$$

~~The joint probability of a paper being finally published is the product of passing review and passing the capacity check: $\bar{\rho} \cdot \mathcal{P}(s_A, s_J)$.~~ AI makes submitting low-quality work more attractive at every composition of the author population, provided the volume elasticity γ is moderate relative to the screening signal $1 - \lambda - \varepsilon$ (Proposition 1).

~~**Volumes: Submitted (V), Passed (M), and Published (Y).** It is crucial to distinguish between papers that pass peer review (M)and those that are finally published (Y). Let V_G and V_B be the expected volumes of good and bad submissions generated by the population. Let M_G and M_B be the expected volumes of papers entering the peer-accepted pool. We define Y_G and Y_B as the expected volumes of **finally published** papers :~~ **Closed-form tipping threshold** (honest boundary $x = 1$, uncongested case):

$$a^* = \frac{c_0 - r\lambda_0}{r\mu + c_0\phi}.$$

Above a^*, $\Pi_{bad}(1, a) > 0$: the honest long-run outcome is no longer self-sustaining. For Set A parameters: $a^* = (25 - 10)/(20 + 10) = 0.50$.

Key insight: a is largely exogenous, but δ (verification investment) and μ (screening vulnerability) are policy choices. Which cell of Table 4 a field occupies is an institutional decision, not a technological fate.

6 Equilibria, Tipping, and Hysteresis under AI Adoption

6.1 Equilibrium Conditions and the AI Shift

The long-run outcome of the publishing system is determined by the signs of the spam payoff Π_{bad} at two boundary states. At the *honest boundary* ($x = 1$), almost all authors are selective; the question is whether a single deviating author can profitably flood the system. At the *corrupt boundary* ($x = 0$), almost all authors submit everything; the question is whether the system can self-correct. These two boundaries are governed by different parameters. As established in Section 4, the honest boundary is secured by keeping λ low; the corrupt boundary is held by keeping K tight enough that the publication lottery deters spam even under degraded screening.

We define publication quality as the share of genuinely good papers among those accepted:

$$\cancel{Y_G} Q(x) \;= \cancel{M_G,\quad Y_B}\; \frac{N\alpha\,\pi_1(x)}{M(x)} \;= \cancel{M_B}\; \frac{\alpha\,\pi_1(x)}{\alpha\,\pi_1(x) + (1-\alpha)(1-x)\,\pi_0(x)}. \tag{15}$$

Since slot rationing applies uniformly to the peer-accepted pool $M(x)$, it does not alter the ratio of good to bad papers; quality is therefore determined entirely by the composition x at the long-run outcome. The four possible long-run outcomes—GH, GC, BS, SC—and their quality levels are summarised in Table 8.

~~**General payoff functions**~~

~~**Author's utility (U_A).** Let r be the reward for a published paper and c be the cost of submission. The expected payoff for an author playing s_A against a journal s_J is the expected reward from publication minus the reliable cost of submission:~~

$$\cancel{U_A(s_A, s_J) = r \cdot \underbrace{\bar{\rho} \cdot \mathcal{P}(s_A, s_J)}_{\text{Prob(Published)}} - c \cdot \underbrace{P_{\text{sub}}(s_A)}_{\text{Prob(Submitted)}}}$$

~~**Journal's utility (U_J).** Let B be the benefit of publishing a good paper, D be the penalty for a bad paper,~~ Under AI adoption $a > 0$, four simultaneous shifts push $\Pi_{bad}(x, a)$ upward at every interior x. Rising submission volume (elasticity γ) and ~~C_J be the unit cost of reviewing a submission. The journal's utility depends on the *published* volumes (Y_G, Y_B) and the *submitted* volume $(V = V_G + V_B)$. This formulation captures the trade-off between quality assurance and processing costs (?):~~

$$\cancel{U_J(s_J) = B \cdot Y_G - D \cdot Y_B - C_J \cdot V}$$

~~Substituting $Y = \bar{\rho} M$, we obtain the working form of the journal's payoff:~~

$$\cancel{U_J(s_J) = \bar{\rho}\,[B \cdot M_G(s_J) - D \cdot M_B(s_J)] - C_J \cdot V}$$

~~*Remark:* Eq. (6.1) relies on the approximation $\mathbb{E}[\rho M] \approx \mathbb{E}[\rho]\mathbb{E}[M]$. As N grows large,~~ lagging review capacity (elasticity $\delta < \gamma$) lower the screening intensity $\eta(x, a)$, raising the bad-paper pass rate $\pi_0(x, a)$. Falling submission cost $c(a)$ directly raises the net payoff from each submission attempt. Rising false-positive rate $\lambda(a)$ further degrades the review filter even for papers that receive full scrutiny. The combined derivative (Equation 14) is strictly positive under the asymmetry $\gamma > \delta$: AI makes spamming more attractive at every composition of the ~~relative fluctuations in M diminish, making this mean-field limit exact (?).~~

~~**Derivation of strategy-specific payoffs** We now explicitly list the payoff functions for the 8 primary scenarios (4 Author strategies and 4 Journal strategies).~~

~~**Author payoffs (under Journal Strategy $s_J = \text{OG}$).** In the standard academic context, the journal plays **Only Good (OG)**. The review filter applies $(\pi_1, \pi_0) = (1-\epsilon, \lambda)$. Substituting the specific $\mathcal{P}$ and P_{sub} values into Eq. (6.1) , we derive:~~

$$\begin{aligned}
U_A(\text{OG}, \text{OG}) &= \alpha\,[r\bar{\rho}(1-\epsilon) - c] \\
U_A(\text{AS}, \text{OG}) &= r\bar{\rho}\,[\alpha(1-\epsilon) + (1-\alpha)\lambda] - c \\
U_A(\text{OB}, \text{OG}) &= (1-\alpha)\,[r\bar{\rho}\lambda - c] \\
U_A(\text{NS}, \text{OG}) &= 0
\end{aligned}$$

~~**Journal payoffs (under Population State x).** The journal chooses a strategy s_J to maximize utility given the incoming flow of submissions. Let $V_G(\mathbf{x})$ and $V_B(\mathbf{x})$ be the volumes of good and bad submissions determined by the populationstate.~~

~~**1. Only Good (OG):** The journal filters papers. The passed volumes are $M_G = V_G(1-\epsilon)$ and $M_B = V_B\lambda$.~~

$$U_J(\text{OG}) = \bar{\rho}\,[BV_G(1-\epsilon) - DV_B\lambda] - C_J(V_G + V_B)$$

~~**2. Always Accept (AA):** The journal accepts everything $(M_G = V_G, M_B = V_B)$. Congestion is maximized here, so $\bar{\rho}$ will be minimal.~~

$$U_J(\text{AA}) = \bar{\rho}\,[BV_G - DV_B] - C_J(V_G + V_B)$$

~~**3. Only Bad (OB):** The journal applies inverse selection $(M_G = V_G\epsilon, M_B = V_B(1-\lambda))$~~ author population.

$$U_J(\text{OB}) = \bar{\rho}\,[BV_G\epsilon - DV_B(1-\lambda)] - C_J(V_G + V_B)$$

The analysis in this section traces the structural consequences of this shift for regime boundaries, stability windows, and recovery conditions.

~~**4. Reject All (AR):** The journal publishes nothing $(Y = 0)$. Review costs are still incurred for processed submissions.~~

$$U_J(\text{AR}) = -C_J(V_G + V_B) \quad \text{(normalized to 0 in dominance analysis)}$$

We use two representative parameter sets throughout (Table 9). Set A ($K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$) represents a typical journal prone to bistability, where the review bottleneck is the binding constraint. Set B ($K_{rev} = 1200$, $K = 180$, $r = 150$, $c = 30$, $\lambda = 0.4$) represents a top-tier journal where extreme slot scarcity is the defining constraint. AI elasticity parameters are fixed at $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$ throughout.

~~**Summary of transition probabilities** For completeness, Table 5 summarizes the effective probability $\mathcal{P}(s_A, s_J)$ of entering the peer-accepted pool for any strategy pair. These values are the key inputs for the payoff equations.~~

6.2 ~~Dominance analysis and reduced game~~How AI Shifts Regime Boundaries

~~Before analyzing the evolutionary dynamics, we simplify the strategy space by iteratively eliminating strictly dominated strategies. This procedure, known as *iterated elimination of strictly dominated strategies* (IESDS) , allows us to focus on the rationalizable core of the game (**?**). We conduct this analysis under the standard **mean-field approximation**, holding the aggregate congestion factor $\bar{\rho}$ fixed. The elimination proceeds in two stages: first, we identify strategies that are dominated purely by the structural logic of the game (independent of specific cost-benefit inequalities) ; second, we introduce parameter constraints to eliminate strategies based on economic trade-offs.~~ Figure 4 presents the core comparative-statics result. The left panel shows the baseline ($a = 0$) regime map across the (K_{rev}, K) parameter space. The right panel shows the same map under high AI adoption ($a = 0.8$). The comparison is unambiguous: the Global Honesty (green) region shrinks substantially, the Bistability (yellow) and

Global Corruption (red) zones expand and shift left, and the Stable Coexistence (blue) region contracts. A journal safely in the GH zone at baseline may find itself in BS or GC at $a = 0.8$ without any change in its own institutional parameters.

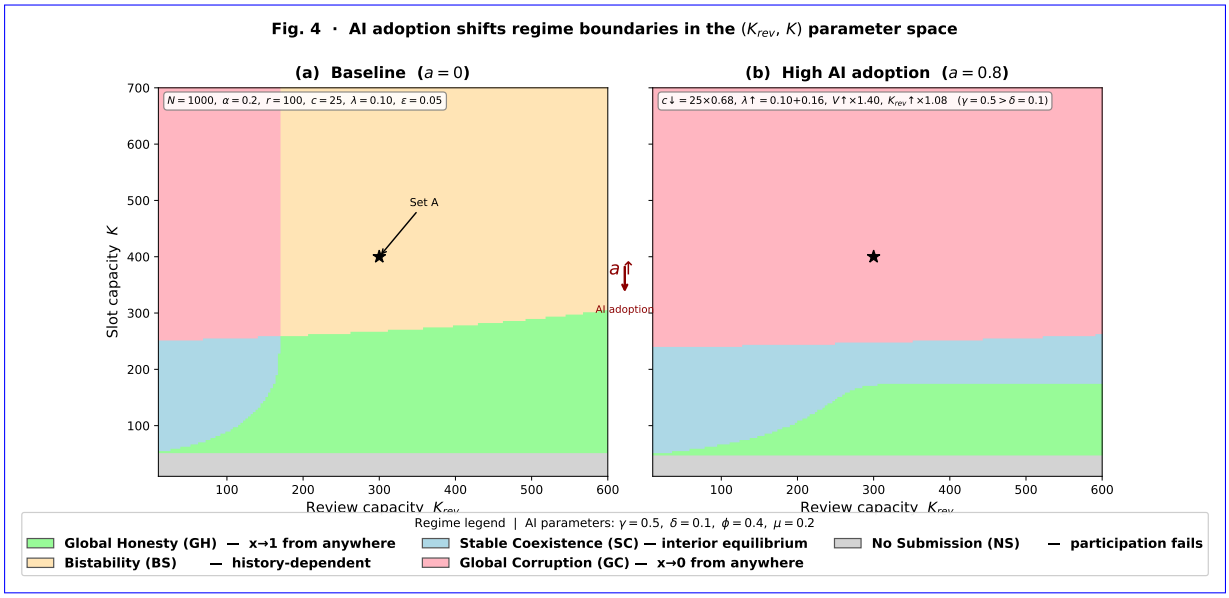


Figure 4: **AI adoption shifts regime boundaries in the (K_{rev}, K) parameter space (Fig. 4).** Left: baseline ($a = 0$). Right: high AI adoption ($a = 0.8$, $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$). The Global Honesty (green) region shrinks and the Bistability and Corruption zones expand. The Set A calibration point (star) moves from a safe location at baseline toward the boundary of GH and BS under high AI adoption. The asymmetry $\gamma > \delta$ is the key driver: review capacity grows too slowly to absorb the AI-driven surge in submission volume.

~~**Stage 1: Structural elimination (Unconditional dominance)** We first establish dominance relations that hold under weak axiomatic assumptions regarding the nature of peer review and scientific value, consistent with signaling models in scientific communication (?).~~ The magnitude of the shift is quantifiable. At $a = 0.8$ with $\gamma = 0.5$ and $\delta = 0.1$, submission volume has risen by 40% while review capacity has risen by only 8%. A journal with $K_{rev} = 300$ has the effective screening power of a journal with $K_{rev} \approx 231$ at baseline—a loss of nearly one-quarter of screening capacity with no change in institutional design. To restore the same screening intensity, the journal would need to raise K_{rev} to approximately 389, a 30% increase in review resources.

~~**Elimination of Journal "Only Bad" (OB)**~~**The slot expansion paradox is amplified by AI.** ~~The strategy **Only Bad (OB)** represents an inverse selection policy. We compare it to **Only Good (OG)**. Using the payoff functions derived in Section **??**, the difference in expected utility is :~~

$$\begin{aligned}\cancel{\Delta U_J = U_J(\mathrm{OG}) - U_J(\mathrm{OB})} \\ \cancel{= \bar{\rho}\,[BV_G(1-\epsilon) - DV_B\lambda] - \bar{\rho}\,[BV_G\epsilon - DV_B(1-\lambda)]} \\ \cancel{= \bar{\rho}\,[BV_G(1-2\epsilon) + DV_B(1-2\lambda)]}\end{aligned}$$

~~**Condition:** We assume peer review is **informative**, defined as performing better than random guessing:~~

$$\cancel{\epsilon < 0.5 \quad \text{and} \quad \lambda < 0.5.}$$

~~Under this definition, both $(1-2\epsilon)$ and $(1-2\lambda)$ are strictly positive. Since $B, D, V_G, \bar{\rho} > 0$, it follows that $\Delta U_J > 0$. Thus, **Journal OB is strictly dominated by OG** solely by virtue of the definition of peer review quality, regardless of the magnitude of rewards or penalties.~~ Expanding publication slots (K) seems like an obvious response to submission pressure. Our model shows this response backfires when

verification does not scale proportionally. Increasing K raises the rationing probability $\bar{\rho}(x, a)$, which raises the expected payoff from submitting a bad paper. This effect is amplified under AI adoption: when $c(a)$ is already low and $\lambda(a)$ is already elevated, even a modest improvement in the publication lottery is enough to push $\Pi_{bad}(1, a)$ from negative to positive, tipping the system from GH into Bistability. More slots increase the prize in a game whose entry barrier has already been eroded by AI. This is visible in Figure 4: moving upward (increasing K) in the right panel is more dangerous than the same move in the left panel. The slot expansion paradox identified in the baseline model is not merely preserved under AI adoption—it is exacerbated.

~~**Elimination of Author "Only Bad" (OB).** Consider the choice between **Always Submit (AS)** and **Only Bad (OB)**. AS submits both paper types ($q \in \{0, 1\}$) , while OB submits only bad papers ($q = 0$)~~

Defence by congestion and its limits. Extreme slot scarcity can stabilize an otherwise fragile system. When K is small relative to the peer-accepted pool M, the rationing probability $\bar{\rho}$ is so low that the expected payoff from submitting a bad paper falls below the submission cost even under degraded screening—the journal's publication lottery is simply too hard to win. This defence operates through the condition $\bar{\rho}(0, a) \cdot \pi_0(0, a) < c(a)/r$, and all three terms move against it as a rises. Rising $\pi_0(0, a)$ from degraded screening and rising λ increases the left side. Falling $c(a)$ reduces the right side. The net effect is that a journal relying on slot scarcity needs progressively tighter K to maintain the same deterrent effect as a increases. At some point, K must be so low that the participation constraint for honest authors—$\Pi_{good}(1, a) > 0$—is violated, pushing the system into the No Submission zone. Defence by congestion is a real stabilising force, but it is not robust to sustained increases in a. ~~The payoff difference is the marginal utility of submitting a good paper:~~

$$\begin{aligned}\cancel{\Delta U_A = U_A(\text{AS}, \text{OG}) - U_A(\text{OB}, \text{OG})} \\ \cancel{= [U_A(q = 1) + U_A(q = 0)] - U_A(q = 0)} \\ \cancel{= U_A(q = 1) = \alpha\,[r\bar{\rho}(1 - \epsilon) - c]}\end{aligned}$$

~~**Condition:** We assume the **Axiom of Value**: in a functional scientific system , a correctly identified high-quality paper yields apositive net return (i.e., $r(1 - \epsilon) > c$) . If this axiom holds, then $\Delta U_A > 0$ for any non-zero congestion $\bar{\rho}$. Thus, **Author OB is strictly dominated by AS** due to monotonicity: a rational author will never withhold a high-quality paper while simultaneously submitting a low-quality one.~~

~~**Parameter constraints for a non-trivial game**~~

6.3 Equilibrium Trajectories and Stability Window Erosion

~~Having eliminated the structurally irrational strategies, we now focus on the remaining set. To ensure the game yields interesting evolutionary dynamics (avoiding trivial collapse or utopian perfection), we define the "viable region" of the parameter space using the **abundance benchmark** ($\bar{\rho} = 1$).~~ Figure 5 traces the long-run honest-author share $x^*(a)$ as AI adoption rises from 0 to 1 under three parameter scenarios.

~~**Viability Constraint (Honesty is profitable):**~~

$$\cancel{r(1 - \epsilon) > c}$$

~~This formally restates the Axiom of Value. It ensures that without congestion,~~

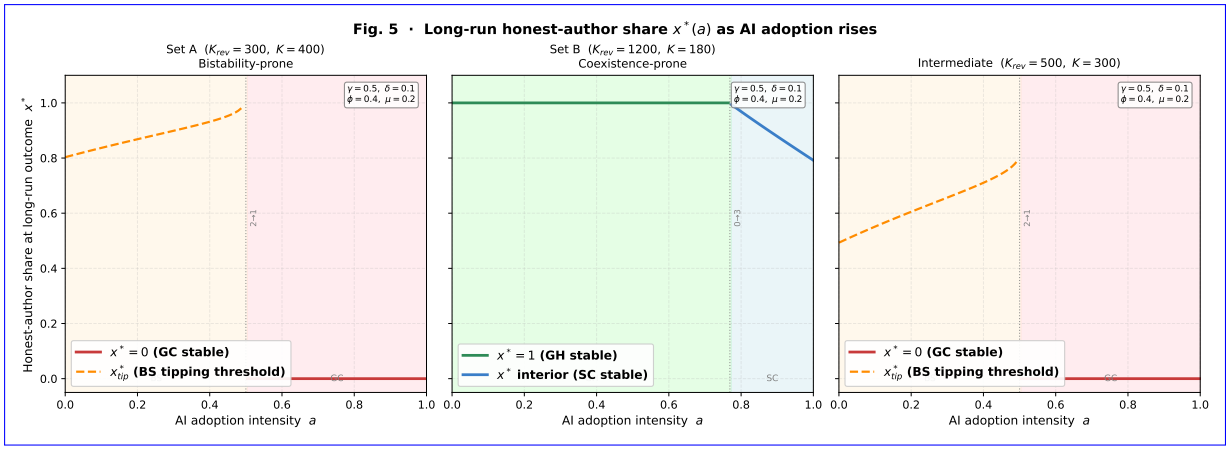


Figure 5: **Long-run honest-author share $x^*(a)$ as AI adoption rises (Fig. 5).** Left: Set A (bistability-prone, $N = 1000$). The system is already in BS at $a = 0$: the corrupt equilibrium is self-sustaining ($\Pi_{bad}(0,0) \approx +12.4 > 0$) while the honest equilibrium is stable ($\Pi_{bad}(1,0) = -15 < 0$), giving a tipping threshold $x^*_{tip}(0) \approx 0.80$ (orange dashed). As a rises, x^*_{tip} climbs monotonically toward $x = 1$ until the honest equilibrium collapses at $a \approx 0.50$ (BS→GC, vertical dashed line). The full GH→BS→GC transition under lower submission pressure ($N = 300$) is illustrated in Figure 7 (a). Centre: Set B (coexistence-prone). The interior stable long-run outcome x^* falls as a rises, tracking the declining profitability of honest submission relative to spamming. Right: Intermediate parameters. Shows the full GH→BS→GC transition as a increases. Vertical dashed lines mark regime transitions; shaded backgrounds indicate the active regime.

The left panel (Set A, $N = 1000$) shows a system that is already in Bistability at $a = 0$. With $N = 1000$ authors, the full submission load ($V(0,0) = 1000$) overwhelms the review capacity ($K_{rev} = 300$), yielding $\Pi_{bad}(0,0) \approx +12.4 > 0$: the corrupt equilibrium is self-sustaining from the outset. The honest equilibrium remains stable ($\Pi_{bad}(1,0) = -15 < 0$), so the system starts in BS with a tipping threshold at $x^*_{tip}(0) \approx 0.80$. As a rises, x^*_{tip} climbs monotonically toward $x = 1$: the ~~honest strategy (OG)dominates silence (NS), a necessary condition for any scientific activity (?).~~ honest norm becomes progressively more fragile even as published quality appears stable. At $a \approx 0.50$, $\Pi_{bad}(1,a) = 0$ and the honest equilibrium loses stability entirely; the system transitions from BS into Global Corruption. The practical implication is the same as in the GH case: the system can appear healthy—with the population well above x^*_{tip}—while the safety margin silently narrows. A single coordinated wave of low-quality submissions can push the population below x^*_{tip} and initiate a collapse that the system cannot reverse on its own. The full GH→BS→GC sequence, in which a community starts from Global Honesty before any AI adoption, is illustrated for $N = 300$ in Figure 7 (a).

~~**Profitability Constraint (Spamming is profitable):**~~

$$\cancel{\lambda r > c}$$

~~This constraint drives the congestion dilemma. It implies that in a frictionless system ($\bar{\rho} = 1$),~~ The right panel (Intermediate parameters) illustrates the full transition from GH through BS to GC as a increases. In the GC zone, the ~~expected reward of a bad paper exceeds the submission cost. If $c > \lambda r$, bad papers are never submitted, and the game trivially converges to perfect honesty.~~ long-run outcome is $x^* = 0$: spamming dominates from every starting point. The path from GH to GC passes through a BS corridor, during which the long-run outcome is history-dependent and recovery is still possible. The width of this corridor—the range of a values during which recovery remains feasible—depends on how quickly $\lambda(a)$ rises and how much $c(a)$ falls.

~~**Reputation Constraint (Screening is preferred):**~~

$$\cancel{DV_B(1-\lambda) > BV_G\epsilon}$$

~~This condition ensures the journal prefers screening (OG)over blind acceptance (AA). It states that the aggregate penalty avoided by rejecting bad papers outweighs the benefit lost from accidentally rejecting good ones (?).~~ Figure 6 maps this dynamic in the (N, a) plane, treating the author population size N as the external pressure variable.

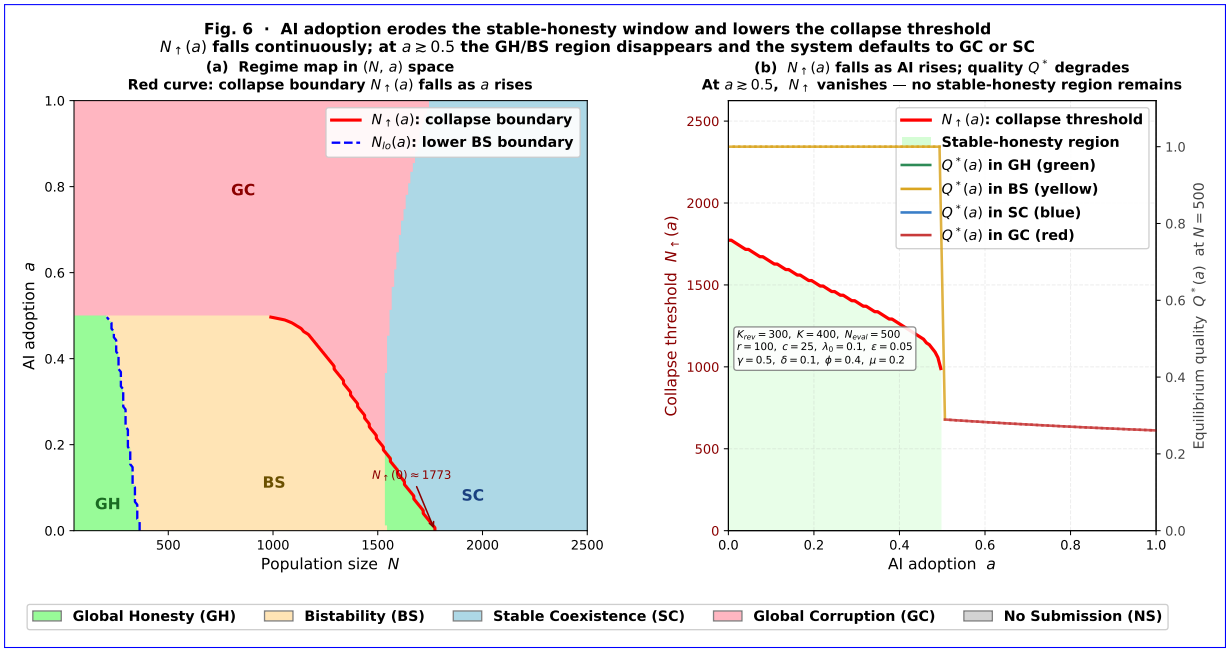


Figure 6: **AI adoption erodes the stable-honesty window and lowers the collapse threshold (Fig. 6).** Left: regime map in (N, a) space with Set A parameters. The red curve $N_\uparrow(a)$ marks the collapse boundary: to its right, the forward (honest-start) path collapses. The blue dashed curve marks the lower boundary of the Bistability zone. As a rises, $N_\uparrow(a)$ falls monotonically; at $a \gtrsim 0.5$ (for the chosen AI parameters) the GH and BS zones disappear entirely. Right: $N_\uparrow(a)$ (red, left axis) falls continuously; equilibrium quality $Q^*(a)$ at moderate pressure $N = 500$ (right axis, coloured by regime) also declines as a rises, tracking the regime transitions from BS (yellow) to GC (red); $N = 500$ lies in the bistable zone at $a = 0$ and crosses into Global Corruption near $a \approx 0.50$.

~~**Stage 2: Conditional elimination (Economic dominance)** We now eliminate strategies based on the specific economic trade-offs defined above.~~

~~**Dominance of Journal "Always Accept" (AA).** We compare **Only Good (OG)** with **Always Accept (AA)**. Assuming the processing cost C_J applies to all submissions (Processing Cost Model)~~ The left panel of Figure 6 shows the collapse threshold $N_\uparrow(a)$ falling monotonically as a rises. At $a = 0$, a population of $N \approx 1800$ is needed to push Set A out of the stable-honesty zone; at $a = 0.5$, this threshold has fallen to roughly half that. The right panel shows the corresponding quality decline: at a moderate load ($N = 500$), quality $Q^*(a)$ is near 1 at low a but falls continuously, tracking the regime transitions from GH through BS to SC. Crucially, at $a \gtrsim 0.5$, the bistable zone disappears entirely. There is no longer any starting position from which the honest norm is self-sustaining. Recovery from this state is not simply harder—it requires a qualitative change in institutional parameters (K_{rev} raised, λ reduced, c restored), ~~the difference depends on the net value of papers rejected by OG but accepted by AA:~~

$$\cancel{\begin{aligned} U_J(\mathrm{OG}) - U_J(\mathrm{AA}) &= \bar{\rho}\left[(BM_G^{\mathrm{OG}} - DM_B^{\mathrm{OG}}) - (BM_G^{\mathrm{AA}} - DM_B^{\mathrm{AA}})\right] \\ &= \bar{\rho}\left[D(V_B - M_B^{\mathrm{OG}}) - B(V_G - M_G^{\mathrm{OG}})\right] \\ &= \bar{\rho}\left[DV_B(1-\lambda) - BV_G\epsilon\right] \end{aligned}}$$

~~Under the **Reputation Constraint** (6.3) , the penalty avoided (first term)exceeds the benefit lost (second term). Thus, **AA is strictly dominated by OG.**~~ not merely a reduction in pressure N. This ~~result aligns with gatekeeper theory, suggesting that competitive journals maintain strict standards to maximize their distinctiveness and reputation (?).~~ is the sense in which AI adoption transforms the challenge from maintenance to repair.

~~**Dominance of Journal "Reject All" (AR).** Strategy AR yields a normalized payoff of 0 (or $-C_J V$). Since OG generates positive utility from good papers ($B > 0$) and filters bad ones, OG dominates AR~~

~~provided the net value of the peer-accepted pool is positive: $BM_G > DM_B$. We assume the journal operates in a parameter region where valid science is net positive.~~ Figure 7 makes the hysteresis structure explicit as a classic fold bifurcation diagram, plotting the long-run honest-author share x^* against AI adoption a under two field-pressure levels.

~~**Dominance of Author "Never Submit" (NS).** Comparing **Only Good (OG)** with **Never Submit (NS)** ($U_A = 0$):~~

$$\cancel{U_A(\mathrm{OG},\mathrm{OG}) - U_A(\mathrm{NS},\mathrm{OG}) = \alpha\,[r\bar{\rho}(1-\epsilon) - c]}$$

~~Under the **Viability Constraint** (6.3), this difference is positive as long as the congestion does not exceed the critical threshold $\bar{\rho} > \frac{c}{r(1-\epsilon)}$. In the viable regime, **NS is strictly dominated by OG** (and by AS)~~

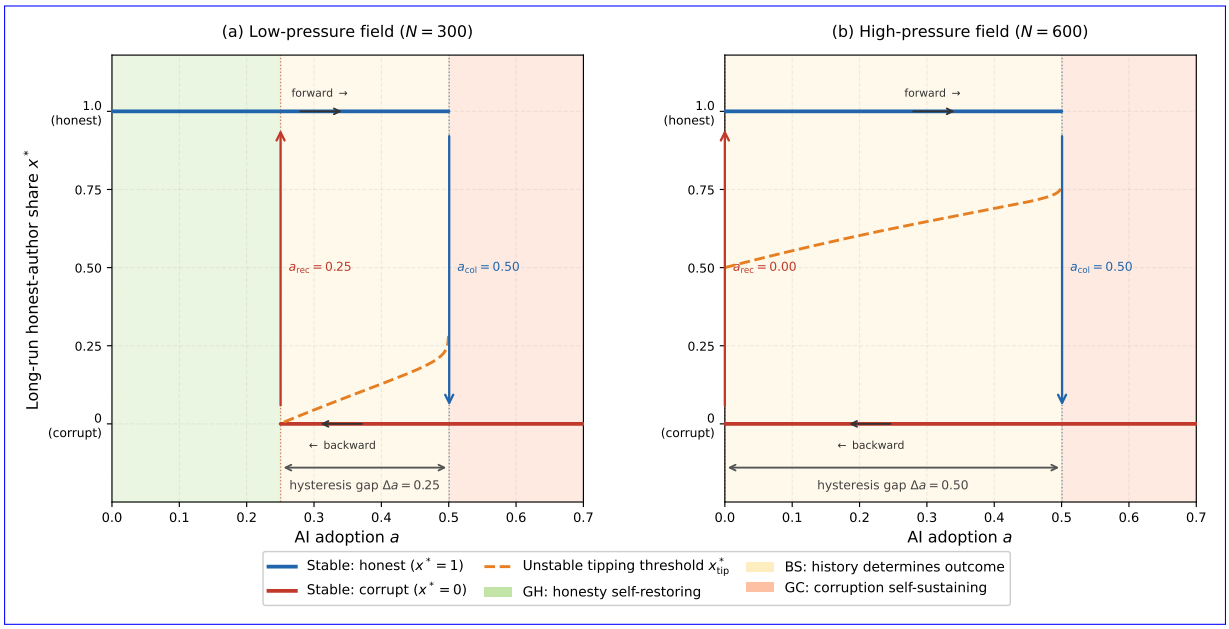


Figure 7: **Hysteresis loop: collapse and recovery thresholds in a-space (Fig. 6b).** Each panel plots the long-run honest-author share x^* against AI adoption a. **(a)** Low-pressure field ($N = 300$). The blue upper branch ($x^* = 1$) is stable while $a < a_{col} = 0.50$; the red lower branch ($x^* = 0$) is stable while $a > a_{rec} \approx 0.25$. The dashed orange curve is the unstable tipping threshold x^*_{tip} that separates the two basins in the Bistability corridor. A forward path (increasing a) follows the honest branch and collapses at a_{col}; a backward path (decreasing a) follows the corrupt branch and recovers only at a_{rec}, giving a hysteresis gap $\Delta a \approx 0.25$. **(b)** High-pressure field ($N = 600$). The system starts in Bistability at $a = 0$: the corrupt equilibrium is self-sustaining before any AI adoption. The honest branch still collapses at $a_{col} = 0.50$, but the corrupt branch is stable for all $a \in [0, 1]$: no recovery threshold exists in the admissible range.

In panel (a), the system occupies Global Honesty at baseline ($a = 0$), indicated by the green background. As a rises, the corrupt equilibrium first becomes self-sustaining at $a_{rec} \approx 0.25$, opening the Bistability corridor (yellow background); the honest equilibrium then loses stability at $a_{col} = 0.50$, pushing the system into Global Corruption (red background). The forward path traces the upper branch to a_{col} and then drops discontinuously to $x^* = 0$. The backward path traces the lower branch back, and recovers only when a falls to a_{rec}—a full 25 percentage points below the collapse point. The hysteresis gap $\Delta a = a_{col} - a_{rec} \approx 0.25$ measures the depth of the trap: a field that has tipped must substantially reduce AI-linked submission pressure, not merely stop increasing it. The unstable tipping threshold (dashed) connects the two fold points; any perturbation large enough to push the system below x^*_{tip} triggers irreversible collapse so long as a remains above a_{rec}. ~~This condition reflects the rational participation constraint found in scientific contests, where authors only enter if the expected payoff exceeds the effort cost (**?**).~~

~~**The reduced evolutionary game**~~ Panel (b) shows a qualitatively different situation. At $N = 600$, the corrupt equilibrium is already self-sustaining at $a = 0$: the system starts in Bistability rather than Global Honesty. The collapse threshold remains $a_{\text{col}} = 0.50$ (this value is independent of N when $\eta = 1$ at $x = 1$, since it is determined purely by $c(a) = r\lambda(a)$). But there is no recovery threshold in the range $a \in [0, 1]$: the lower branch persists all the way back to $a = 0$, so reducing AI adoption alone cannot restore Global Honesty. Escaping the corrupt basin requires an institutional intervention that raises K_{rev}, reduces λ, or both, until $\Pi_{bad}(0, a)$ turns negative—a shift in the parameter configuration, not merely a reversal of a. The contrast between panels (a) and (b) thus captures the core asymmetry of the recovery problem: low-pressure fields retain a restoration option, while high-pressure fields require coordinated institutional action regardless of what happens to AI adoption.

~~Following the two-stage elimination process , the complex multi-strategy interaction reduces to a concise 2×1 asymmetric game.~~

6.4 The False-Positive Rate as Critical Destabiliser

~~**The Journal** is a fixed institutional player locked into the **Only Good (OG)** strategy. It acts as the environment, applying a filter (ϵ, λ) and a capacity constraint K. **The Author Population** evolves between two surviving pure strategies: **Only Good (OG) Cooperators:** Authors who submit only high-quality work $(q = 1)$. **Always Submit (AS) Defectors:** Authors who submit everything $(q \in \{0, 1\})$, effectively spamming the journal with bad papers to exploit~~ Among the four AI channels, rising $\lambda(a)$ is structurally the most dangerous. The GH condition at $x = 1$ requires $\Pi_{bad}(1, a) < 0$; when screening is fully effective $(\eta = 1)$, this simplifies to $\lambda < c(a)/r$. Under baseline parameters ($c = 25$, $r = 100$), this requires $\lambda < 0.25$. A slope of $\mu = 0.2$ over full adoption $(a = 1)$ pushes λ ~~.~~ from 0.10 to 0.30, crossing this threshold and making GH structurally impossible regardless of review capacity. No investment in K_{rev} can restore GH once $\lambda > c/r$, because even a perfectly reviewed bad paper passes at rate λ and yields a positive expected payoff. This result is consistent with signalling theory: the reliability of the screening signal—not merely the volume of screening—is what sustains the separating equilibrium (**??**).

~~We define the **population state** $x(t) \in [0, 1]$ as the fraction of authors adopting the **Only Good** strategy. The evolutionary dynamics of the system are thus entirely determined by the time evolution of x. This reduction captures the essential dynamic of scientific evolution: the selection pressure between high-effort integrity and low-effort quantity (**?**)~~The false-positive channel also explains why AI-assisted reviewing is particularly concerning. AI-assisted reviews have been shown to award systematically higher scores, raising acceptance rates (**??**). If widespread AI use in reviewing raises the average λ, the effect is direct and structural: the GH boundary shifts inward regardless of what authors do. Interventions that target λ directly—structured reporting checklists, mandatory data and code deposition, statistical audit requirements—therefore address the most critical vulnerability in the system.

6.5 Policy Levers and Summary Propositions

~~**Structure of the Reduced Evolutionary Game.** The author population evolves between two strategies: Only Good (OG) and Always Submit (AS). The Journal acts as a fixed mechanism consisting of a peer review filter and a capacity constraint. Submissions from AS authors increase the total volume M, triggering the capacity constraint. The resulting **Congestion Externality** $(\bar{\rho})$ reduces the publication probability for all papers, feeding back into the payoffs and driving the evolutionary dynamics.~~ Each parameter corresponds to a concrete institutional intervention. We summarise the mapping in four propositions; formal derivations are in the Appendix.

7 ~~Replicator Dynamics and Bifurcation Analysis~~

6.1 ~~Population state and congestion functions~~

Proposition 1 (Monotone destabilisation). *Under the asymmetry $\gamma > \delta \geq 0$: (i)* ***Honest boundary.*** *$\Pi_{bad}(1, a)$ is strictly increasing in a (analytically, for all parameter values satisfying the model assumptions). The Global Honesty region in (K_{rev}, K) space shrinks strictly as a rises, and there exists a finite closed-form threshold $a^* = (c_0 - r\lambda_0)/(r\mu + c_0\phi) \in (0, \infty)$ above which no (K_{rev}, K) combination sustains GH without additional institutional intervention. (ii)* ***Interior compositions.*** *For $x \in (0, 1)$, $\partial\Pi_{bad}/\partial a > 0$ holds analytically when review is uncongested ($\eta = 1$, Case I) or rationing is inactive ($\bar{\rho} = 1$, Case II). When both constraints bind (Case III), the result holds under a sufficient condition on the screening signal (stated in the Appendix) and is confirmed numerically throughout $(x, a) \in (0, 1) \times [0, 1]$ for both calibrations in Table 9. All three cases jointly imply that the GH region shrinks as a rises.*

~~Following the dominance analysis, the strategy space reduces to acontest between two author strategies, with the journal playing a fixed filtering role. We define the~~ **~~population state~~** ~~$x(t) \in [0, 1]$ as the fraction of authors playing~~ **~~Only Good (OG)~~**~~, while $1 - x(t)$ play~~ **~~Always Submit (AS)~~**~~. We assume the system operates in the region $\lambda r < c < r$: in the absence of congestion, submitting low-quality work is unprofitable, but it becomes viable once the screening mechanism degrades sufficiently—a mechanism analogous to the market for lemons (**??**).~~

Proposition 2 (Slot expansion paradox). *Holding K_{rev} and a fixed, increasing K weakly raises $\Pi_{bad}(x, a)$ at every x. The marginal destabilising effect of an additional slot is strictly larger at higher a, because lower $c(a)$ and higher $\lambda(a)$ amplify the gain from winning the publication lottery. The paradox is therefore amplified, not attenuated, by AI adoption.*

~~To capture the dual impact of congestion, we model two distinct bottlenecks. Numerical simulations use a smoothed approximation to capture pre-congestion friction; the analytical forms below use standard cutoffs for clarity.~~

Proposition 3 (Erosion of congestion defence). *The corrupt-boundary stability condition $\Pi_{bad}(0, a) < 0$ holds if and only if $K < K^*(a) \equiv c(a)M(0, a)/[r\pi_0(0, a)]$. When the cost-reduction slope satisfies $\phi \geq \gamma$, $K^*(a)$ is strictly decreasing in a for all $a \in [0, 1]$ analytically. When $\gamma > \phi$ (as in the calibrations of Table 9), $K^*(a)$ is strictly decreasing for $a \geq (\gamma - \phi)/(2\phi\gamma)$ analytically and is confirmed numerically to be strictly decreasing throughout $[0, 1]$ under both Sets A and B. Consequently, a journal that relied on passive slot scarcity for protection at baseline may need to actively reduce K to maintain the same deterrent as a rises.*

~~**Stage 1: The screening intensity function** $\eta(x)$ The total submission volume is $V(x) = N[1 - (1 - \alpha)x]$. The screening intensity $\eta(x)$ represents the probability that a paper undergoes rigorous peer review. It degrades when volume exceeds the review capacity K_{rev}, a phenomenon consistent with models of "reviewer fatigue" and resource scarcity (**??**):~~

$$\cancel{\eta(x) = \min\left(1, \frac{K_{rev}}{V(x)}\right)}$$

~~We assume "rubber-stamping" implies unconditional acceptance of unscreened papers. This reflects the transition from rational selection to random selection observed in overloaded systems (**?**). Thus, the pass probability for a bad paper, $\pi_0(x)$, increases as screening fails:~~

$$\cancel{\pi_0(x) = \eta(x)\lambda + (1 - \eta(x)) \cdot 1 = 1 - \eta(x)(1 - \lambda)}$$

~~For a good paper, the pass probability $\pi_1(x)$ also approaches 1 as scrutiny relaxes (assuming false negatives ϵ only occur during rigorous review) :~~

$$\cancel{\pi_1(x) = \eta(x)(1 - \epsilon) + (1 - \eta(x)) \cdot 1 = 1 - \eta(x)\epsilon}$$

Proposition 4 (Policy lever ranking). *Among the primary levers, reducing λ (verification quality) and raising K_{rev} (review capacity) are unconditionally beneficial: they expand the GH region without side effects. Raising c (submission friction) restores the honesty condition but may deter some honest authors. Reducing K (tightening slots) provides passive deterrence but risks the No Submission zone. Expanding K alone is the only intervention that can make the system strictly worse.*

~~**Stage 2: The rationing probability function** $\bar{\rho}(x)$ Let K be the publication slot capacity. The expected volume of papers entering the "perceived good" pool $M(x)$ is:~~

$$\cancel{\begin{aligned} M(x) &= V_G(x)\pi_1(x) + V_B(x)\pi_0(x) \\ &= N\alpha \cdot \pi_1(x) + N(1-\alpha)(1-x) \cdot \pi_0(x) \end{aligned}}$$

~~Using a~~

Early warning signals. The model identifies observable signatures that precede a tipping-point collapse. Rising submission-to-review ratios (V/K_{rev}) signal declining screening intensity η. Lengthening turnaround times and increasing desk-rejection rates indicate reviewer overload and the onset of rubber-stamping. Acceptance rates rising faster than improvements in underlying research quality signal a rising effective λ. In fields where AI adoption is high, these signals should appear before any observable quality decline, because the system can remain near $x \approx 1$ while the tipping threshold x^*_{tip} rises toward the current honest-author share. To make this concrete: under Set A parameters, $x^* \approx 0.80$ at baseline, leaving an apparent safety margin of only 0.20 from the fully honest state. As a rises, x^* climbs monotonically toward $x = 1$, so this margin closes continuously—from 0.20 toward zero—while published quality appears unchanged, because the population remains above the threshold throughout. By the time a reaches $a^* \approx 0.50$, the margin has vanished and the system tips into Global Corruption with no visible warning in quality metrics. The gap between observable quality (still high) and structural fragility (rapidly growing) is precisely what makes AI-induced tipping difficult to detect in advance. We examine the testability of these predictions in Section 10.

7 Agent-Based Modeling

7.1 Why Simulate? Heterogeneity and the Diffusion of AI Adoption

The analytical results in Section 6 derive from a deterministic, infinite-population mean-field ~~approximation, the expected rationing probability is :~~

$$\cancel{\bar{\rho}(x) \approx \min\left(1, \frac{K}{M(x)}\right)}$$

~~This formulation captures the "lottery nature" of high-impact publishing: when the number of scientifically sound papers (M) exceeds the journal's physical space (K), acceptance becomes a game of chance rather than pure merit (?).~~ approximation (**?**). Two features of real academic communities are suppressed by this approximation. First, AI adoption is not uniform: early adopters, late adopters, and non-adopters coexist within any field, and the overall adoption level a rises gradually through a diffusion process analogous to the spread of other general-purpose technologies (**??**). Second, finite populations are subject to stochastic fluctuations that can push a system across a tipping threshold even when the deterministic drift would not (**??**). An agent-based model (ABM) allows us to test whether our three main results—gradual onset of fragility, discrete tipping, and severe path dependence—survive both of these complications (**??**).

7.2 ~~Derivation of the replicator equation~~

7.2 Model Design

~~Since both OG and AS strategies submit good papers whenever they occur (with identical acceptance probabilities), the expected payoff component from good papers cancels out in the differential. The dynamics are driven entirely by the net utility of the additional "bad" papers submitted by AS authors, denoted as $\Pi_{bad}(x)$. The replicator equation is~~ We simulate a population of $N = 1000$ authors. Each author plays one of two strategies:

$$\cancel{\dot{x} = -x(1-x)(1-\alpha)\Pi_{bad}(x)}$$

selective submission (OG) or universal submission (AS); intermediate submission strategies are excluded by the evolutionary instability result of Section 4. The journal's review capacity and quality parameters evolve with the field-level AI adoption level a_t, which rises exogenously along a linear ramp from $a = 0$ to $a = 0.8$ over the simulation horizon—a stylised representation of the diffusion trajectory documented empirically (**??**). At each step, the effective parameters $K_{rev}(a_t)$, $\lambda(a_t)$, and $c(a_t)$ are updated according to the functional forms in Section 5. Strategy revision follows the pairwise Fermi comparison process (**??**): a randomly chosen agent i compares its payoff Π_i with that of a randomly drawn peer j and switches with probability

$$P(s_i \to s_j) = \frac{1}{1 + \exp\big(-\beta\,[\Pi_j - \Pi_i]\big)}, \tag{16}$$

where $\beta = 1.0$ governs selection intensity. This rule captures bounded rationality: better-performing strategies spread, but noisily.

~~A bad paper yields reward r only if it passes both bottlenecks. The expected net utility is :~~

$$\cancel{\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c}$$

7.3 Results: Tipping as AI Adoption Rises

7.4 ~~Bifurcation and stability analysis~~

Figure 8 reports the core dynamic result. Panel (a) traces the tipping threshold $x^*(a_t)$—the unstable boundary in the bistability zone—as a_t rises. The threshold climbs monotonically from $x^* \approx 0.80$ at baseline toward $x = 1$ as a approaches the analytical limit $a^* \approx 0.50$ derived above. The system appears healthy throughout: observed honest-author shares remain at $x \approx 1$. Yet a fixed-size shock of $\Delta x = 0.18$—representing a coordinated wave of low-quality submissions, of the kind documented at major publishers during paper-mill incidents (**?**)—has opposite consequences depending on timing. Applied early at $a \approx 0.08$, it drops x to 0.82, which is still above $x^*(0.08) \approx 0.80$; the honesty norm self-repairs within a few hundred steps. Applied late at $a \approx 0.53$—just past the critical threshold $a^* \approx 0.50$ at which $\Pi_{bad}(1, a)$ turns positive and $x = 1$ loses stability—the system has entered Global Corruption. The identical shock now causes permanent collapse not because x is pushed below an interior threshold, but because no stable honest long-run outcome exists at all: any downward perturbation from $x = 1$ leads irreversibly to $x = 0$. The *safety margin*—the gap between the current honest-author share x_t and the rising threshold $x^*(a_t)$—therefore shrinks silently while observable quality remains high, and vanishes entirely as a crosses a^*. This is precisely the "quiet before the cliff" signature that catastrophic-shift theory predicts for systems approaching a bifurcation (**?**).

~~The system exhibits **Bistability** because the two congestion effects work in opposition, creating a dynamic tension characteristic of tipping-point models (**?**):~~

- ~~**Review Collapse ($\pi_0 \uparrow$):** Increases the incentive to cheat (pro-corruption).~~
- ~~**Storage Rationing ($\bar{\rho} \downarrow$):** Decreases the incentive to cheat (pro-honesty) .~~

~~When the review collapse effect dominates at low honesty, $\Pi_{bad}(x)$ becomes positive, creating a *corruption trap* where opportunistic strategies proliferate (**?**). Conversely, when strict screening dominates at high honesty, $\Pi_{bad}(x)$ becomes negative, creating a basin of honesty.~~ Panel (b) confirms path dependence at $a = 0.45$ (bistable regime). Two trajectories separated by only $\Delta x_0 = 0.08$ diverge to opposite absorbing states: the run starting just above $x^*(0.45) \approx 0.92$ converges to full honesty within a few hundred steps, while the run starting just below collapses immediately. No endogenous force can reverse this collapse; the population is trapped in the low-quality long-run outcome (**??**).

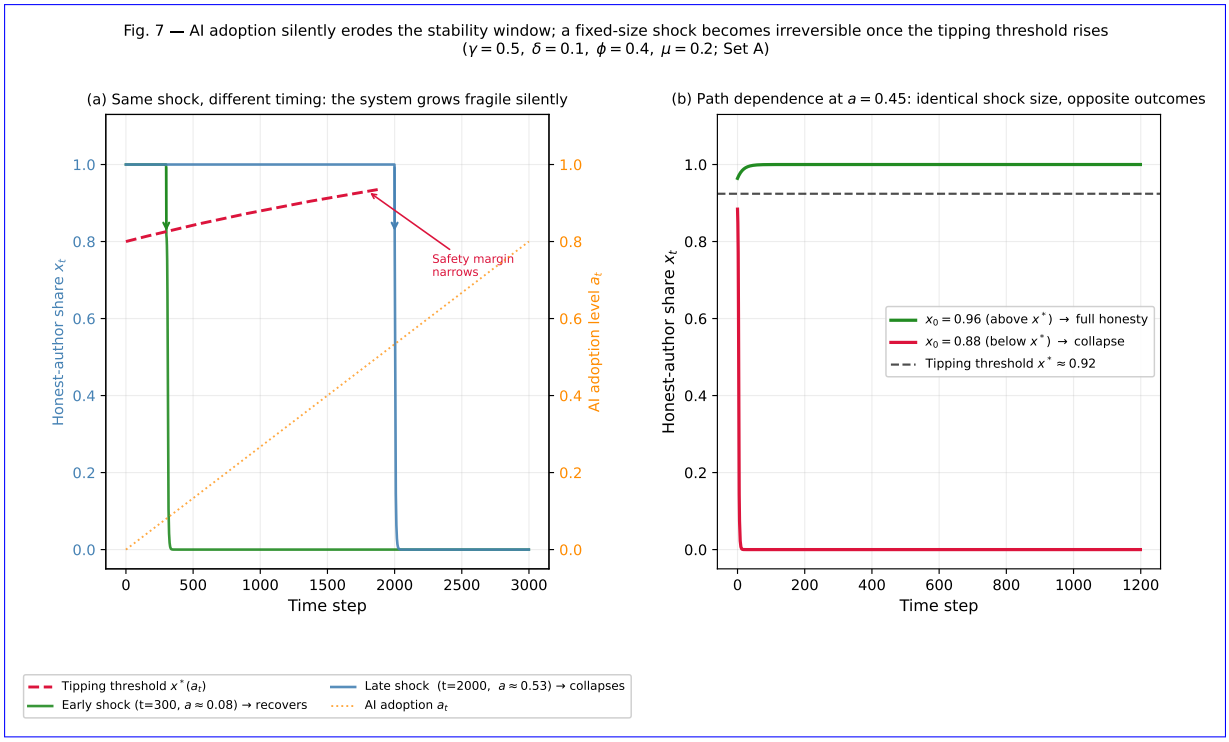


Figure 8: **AI adoption silently erodes the stability window (Fig. 7). (a)** The tipping threshold $x^*(a_t)$ (red dashed) rises monotonically from ≈ 0.80 toward $x = 1$ as a increases through the bistability zone; the threshold ceases to exist above $a^* \approx 0.50$ (Set A, analytical), when the system enters Global Corruption. A fixed-size shock of $\Delta x = 0.18$ recovers when applied at $a \approx 0.08$ (green; safety margin intact) but causes permanent collapse when applied at $a \approx 0.53$ (blue; system in GC, no stable honest long-run outcome). Observable quality gives no warning: $x_t \approx 1$ in both cases until the shock hits. **(b)** Path dependence at $a = 0.45$ (bistable zone, just below a^*): two initial conditions separated by $\Delta x_0 = 0.08$ diverge to opposite long-run outcomes, with the threshold near $x^*(0.45) \approx 0.92$. Parameters: Set A, $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$.

7.4 Robustness: Varying δ/γ and μ

Figure 9 maps the long-run outcome across the two policy-sensitive dimensions identified in Section 6: the verification elasticity ratio δ/γ and the false-positive slope μ. Both panels are evaluated at $a = 0.8$, representing sustained high AI adoption.

Panel (a) shows the long-run honest-author share x^* on a continuous colour scale. The parameter space is dominated by the collapse region ($x^* \approx 0$, dark red); a stable-honesty zone ($x^* \approx 1$, dark green) exists only in the bottom-right corner, where $\delta/\gamma \gtrsim 0.5$ and $\mu \lesssim 0.08$ simultaneously. Panel (b) confirms this pattern in terms of regime classification. The baseline calibration ($\delta/\gamma = 0.2$, $\mu = 0.2$, marked by white dotted lines) lies solidly inside the Global Corruption zone. The Bistability region (yellow) is confined to a narrow band at $\mu \lesssim 0.09$, independent of δ/γ. The Stable Coexistence and Global Honesty regions are accessible only when both conditions hold jointly; satisfying one alone is insufficient.

Two findings from this robustness check are particularly important. First, the μ dimension is the binding constraint: even if $\delta/\gamma = 1$ (review capacity scales perfectly with submission volume), the system remains in Global Corruption whenever $\mu > 0.09$. No investment in review staffing can compensate for a screening process that cannot distinguish polished AI text from genuine contributions (**??**). Second, the boundary between BS and GC is nearly vertical in the right panel, meaning that raising δ/γ from 0.2 to 0.5 shifts the system from GC to BS (where recovery is at least possible) only if μ is already low. This interaction confirms that the two policy levers—raising K_{rev} and reducing λ—must be deployed together, not sequentially.

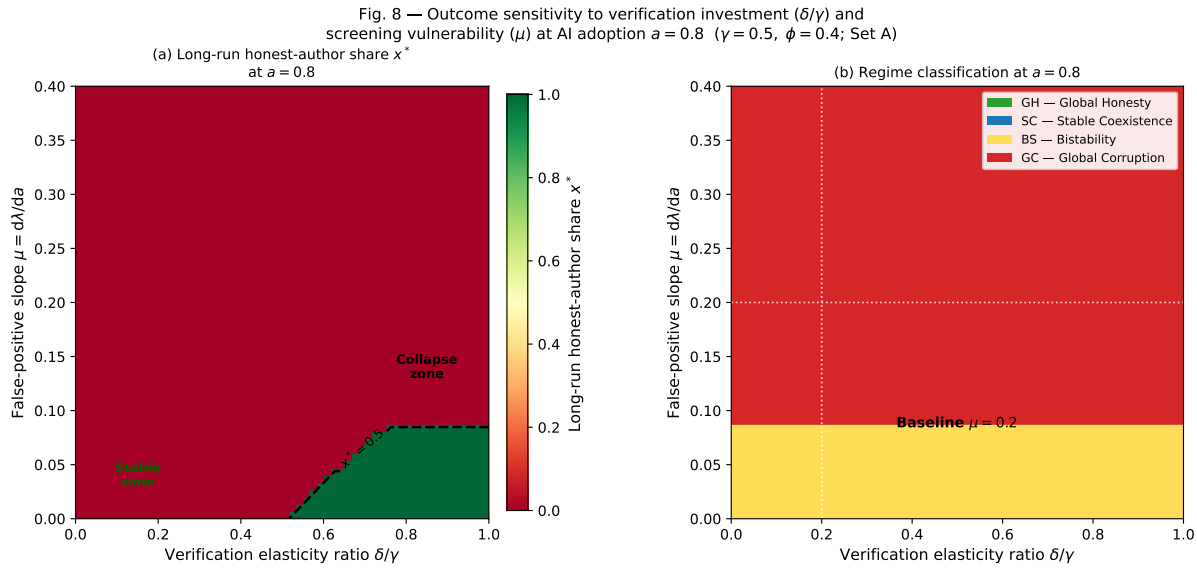


Figure 9: ~~**Phase Portrait of the Dual-Bottleneck Dynamics (Bistability).** Parameters: $N = 1000, K_{rev} = 300, K_{store} = 400, r = 100, c = 25, \lambda = 0.1$~~**Outcome sensitivity to verification investment and screening vulnerability (Fig. 8).** Both panels are evaluated at $a = 0.8$ across a grid of $\delta/\gamma \in [0, 1]$ and $\mu \in [0, 0.4]$. ~~The curve shows the evolutionary velocity $\dot{x}$. The system exhibits three equilibria~~**(a)** Long-run honest-author share x^*: a stable-honesty zone (~~1~~green) ~~**Trap of Corruption ($x = 0$): Stable**~~ exists only when $\delta/\gamma \gtrsim 0.5$ and $\mu \lesssim 0.08$ simultaneously; the remainder of the space collapses ($\Pi_{bad}(0) > 0 x^* \approx 0$). ~~(2) **Basin of Honesty ($x = 1$)**~~**(b)** Regime classification: ~~Stable~~ the baseline calibration ($\Pi_{bad}(1) < 0$white dotted lines, $\delta/\gamma = 0.2$, $\mu = 0.2$) sits in Global Corruption. ~~(3) **Tipping Point ($x^* \approx 0.82$): An unstable internal repeller.**~~ The ~~orange marker indicates the critical threshold separating the two basins of attraction~~Bistability corridor is narrow and confined to very low μ. Global Honesty requires high δ/γ and low μ jointly. Parameters: $\gamma = 0.5$, $\phi = 0.4$; Set A.

~~**Analytical Condition for Instability of the Internal Equilibrium**~~

7.5 Stochastic ABM: Does Finite-Population Noise Validate the ODE?

~~To rigorously characterize the stability of an interior equilibrium $x^* \in (0, 1)$, we analyze the local derivative of the velocity field. Following standard evolutionary stability analysis (**?**), the equilibrium is unstable (a repeller)if and only if the net utility of cheating is locally decreasing in x:~~

$$\left.\frac{d\Pi_{bad}}{dx}\right|_{x^*} < 0$$

~~Recall that $\Pi_{bad}(x) = r \cdot \bar{\rho}(M(x)) \cdot \pi_0(x) - c$. Using the chain rule~~ The ODE assumes an infinitely large, well-mixed population and therefore suppresses all randomness. Real academic communities are finite. Even a system that sits nominally above the tipping threshold can cross it by chance when the number of authors is small (**??**). This subsection tests whether the deterministic tipping results of Section 6 survive under finite-population noise, and whether community size changes that verdict.

We extend the batch-matching ABM from **?** to incorporate the same rising AI adoption schedule used in Section 7 (a_t linear from 0 to 0.8 over 3,~~we decompose the total derivative into two competing forces:~~

$$\cancel{\frac{d\Pi_{bad}}{dx} = r\left(\underbrace{\bar{\rho}(M)\frac{d\pi_0}{dx}}_{\text{Review Effect}} + \underbrace{\pi_0(x)\frac{d\bar{\rho}}{dM}\frac{dM}{dx}}_{\text{Rationing Effect}}\right)}$$

~~Note that $\frac{d\pi_0}{dx} < 0$ (since higher honesty reduces volume V, improving review quality η, thus lowering π_0) . Conversely, the rationing term is typically positive (higher honesty reduces false positives in M, relaxing rationing, thus increasing $\bar{\rho}$~~000 steps). At each step t, the local screening capacity is $K_{rev,\text{eff}}(a_t) = K_{rev,0} \cdot (1 + \delta a_t)/(1 + \gamma a_t)$, the false-positive rate is $\lambda(a_t)$, the submission cost is $c(a_t)$, and the paper pool reaching the publication queue scales as $M(x, a_t) = (1 + \gamma a_t) \cdot M_{\text{base}}(x, a_t)$—identical functional forms to the ODE. The only departure from the ODE is that N is finite and strategy revision is stochastic: each revision attempt uses the Fermi rule (**?**), so the population does not move deterministically toward its expected next state but takes a noisy step. We run 30 independent replications for each of two community sizes: $N = 1{,}000$ (a full research field, the original calibration) and $N = 200$ (a specialised sub-field). ~~Therefore, the tipping point is unstable if and only if the Review-Dominated Condition holds locally:~~

$$\cancel{\underbrace{\left|\bar{\rho}(M)\frac{d\pi_0}{dx}\right|}_{\text{Loss from stricter review}} > \underbrace{\pi_0(x)\frac{d\bar{\rho}}{dM}\frac{dM}{dx}}_{\text{Gain from relaxed rationing}}}$$

~~This condition confirms that bistability arises when the recovery of peer review quality (as x increases) outweighs the relaxation of publication slots.~~ Both start at $x_0 = 0.82$, which is 0.02 above the initial tipping threshold $x^*(a = 0) \approx 0.80$—a thin but nominally safe margin.

7.6 ~~Equilibrium regimes~~

Figure 10 shows the results. Panel (a) overlays individual trajectories, the mean ABM path, and the ODE prediction. The ODE rises steadily to $x \approx 1$ and stays there: deterministic drift at $x = 0.82$ points toward honesty, and an infinite population never crosses the threshold by noise alone. The $N = 200$ runs tell a different story. Most collapse immediately; only a handful survive to the honest long-run outcome. The $N = 1{,}000$ mean settles near $x \approx 0.63$, reflecting the mix of collapsed and surviving runs in that sample. Panel (b) quantifies this: by $a \approx 0.06$, approximately 90% of $N = 200$ runs have already collapsed, while 35% of $N = 1{,}000$ runs have done so. Both fractions plateau for higher a, because the sorting between collapse and survival is complete before AI adoption has risen appreciably—the safety margin of only 0.02 is insufficient to resist even the earliest stochastic fluctuations.

~~**Stable coexistence: rationing-dominated dynamics**~~ Three conclusions follow from this comparison. First, the ABM confirms the ODE mechanism: in every run the tipping threshold $x^*(a_t)$ acts as the critical boundary, and no run escapes once it crosses below it. Second, the ODE systematically overestimates stability by ignoring finite-population noise. The same parameters that make the ODE predict "safe" lead to 35–90% stochastic collapse, depending on community size. This is consistent with theoretical results on stochastic evolutionary games: the probability of crossing a tipping threshold scales roughly as $\exp(-N \cdot \Delta)$, where Δ is the size of the safety margin, so thin margins are exponentially more dangerous in small populations (**?**). Third, community size is itself a policy variable. Fragmented peer-review ecosystems—specialised conferences, niche journals—face higher stochastic collapse risk than large generalist venues under the same AI adoption pressure. Institutional design should therefore calibrate the required safety margin $x_t - x^*(a_t)$ to the effective community size, not merely to the average field-level adoption level.

~~While the default scenario exhibits bistability, a fundamentally different topological regime emerges when the storage constraint (K_{store}) is significantly tighter than the review constraint (K_{rev}). Figure ?? illustrates this scenario, where the evolutionary velocity $\dot{x}$ transitions from positive to negative, creating a Stable Internal Equilibrium ($x^* \approx 0.44$).~~

~~**Phase Portrait of Stable Coexistence (Rationing-Dominated).** Parameters: $N = 1000, K_{rev} = 1000, K_{store} = 180, r = 150, c = 30, \lambda = 0.4$. The system exhibits a stable attractor at $x^* \approx 0.49$. **Left Zone (Self-Correction):** At low honesty ($x < x^*$), severe storage rationing reduces the payoff below zero ($\Pi_{bad} < 0$), forcing the population to become more honest. **Right Zone (Temptation):** At high honesty ($x > x^*$), the relaxed rationing makes cheating profitable again ($\Pi_{bad} > 0$), forcing the population towards corruption. The system settles into a permanent mix of honest and opportunistic authors (Blue Point).~~

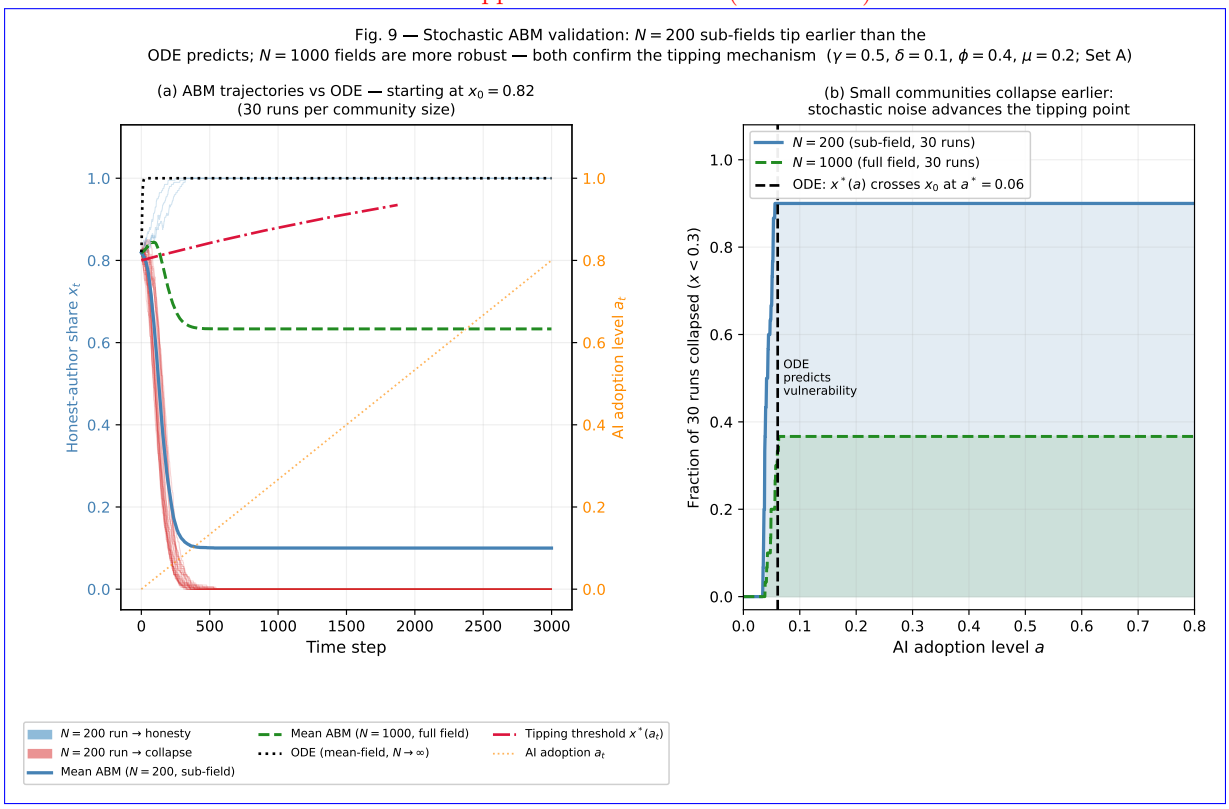


Figure 10: **Stochastic ABM vs ODE prediction (Fig. 9).** Both panels use Set A parameters with a linear AI adoption ramp, starting at $x_0 = 0.82$ (safety margin 0.02 above the initial tipping threshold). **(a)** Individual $N = 200$ run trajectories (blue: survive; red: collapse), mean paths for $N = 200$ and $N = 1{,}000$, the ODE prediction, and the rising tipping threshold $x^*(a_t)$. The ODE stays near $x = 1$; most $N = 200$ runs collapse within the first 200 steps. **(b)** Fraction of 30 runs collapsed ($x < 0.3$) as a function of a. By $a \approx 0.06$, 90% of $N = 200$ runs and 35% of $N = 1{,}000$ runs have collapsed—well before the ODE would predict vulnerability. Parameters: $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$; Set A.

~~**Existence and Stability Conditions**~~

8 Institutional Responses

The three results from Section 6 reduce the institutional problem to a single condition: keep $\Pi_{bad}(x, a)$ negative. AI adoption pushes Π_{bad} upward through four channels simultaneously: submission volume rises, review capacity lags, submission cost falls, and the false-positive rate λ increases. Effective policy must counteract each channel. We organize the response around five principles, ordered by the tightness of the constraint each relieves.

~~For a stable coexistence equilibrium to exist, two conditions must be met: **Boundary Conditions:** The system must be repelled from both extremes. This requires $\Pi_{bad}(1) > 0$ (Temptation at Honesty) and $\Pi_{bad}(0) < 0$ (Unprofitability at Corruption). By the Intermediate Value Theorem, there must exist at least one root $x^* \in (0,1)$ where $\Pi_{bad}(x^*) = 0$~~**Principle 1: Scale verification ($K_{rev}\uparrow$).** Higher review capacity raises the screening intensity $\eta = \min(1, K_{rev}/V)$, which reduces the bad-paper pass rate π_0 and lowers Π_{bad} at the corrupt boundary $x \approx 0$. Practical mechanisms include structured reviewer panels with explicit recognition credits (**?**), AI-assisted consistency and format pre-screening that frees expert time for substantive judgment (**?**), and co-review programmes that train junior reviewers alongside senior ones. ~~**Stability Condition:** The slope of the velocity field must be negative, i.e., $\frac{d\Pi_{bad}}{dx}\Big|_{x^*} > 0$. This corresponds to the Rationing-Dominated Condition*, where the "punishment" from overcrowding (tightening $\bar{\rho}$) outweighs the "opportunity" from collapsed screening.~~

~~**Real-world implication: The "Defense by Congestion" paradox** The emergence of a stable internal equilibrium reveals a paradoxical phenomenon: **extreme congestion acts as an unintentional defense mechanism**. In journals with strict page limits ($K_{store} \ll N$), even if peer review fails ($\pi_0 \to 1$), the sheer volume of submissions drives the acceptance rate $\bar{\rho}$ so low that spamming becomes statistically unprofitable. This explains why top-tier journals do not degenerate into full corruption even under heavy load—the "lottery" is simply too hard to win. This dynamic aligns with contest models where excessive competition discourages entry by marginal participants (**?**).~~ However, ~~it also implies vulnerability: if capacity is expanded (increasing K_{store}) , the system loses this passive defense, potentially shifting the topology to bistability or collapse. This highlights the strategic value of artificial scarcity in maintaining the "gatekeeper" function of academic journals(**?**).~~

~~**Boundary regimes: global honesty and global corruption**~~ there is a binding caveat: when $\lambda > c(a)/r$, even perfect screening ($\eta = 1$) cannot make $\Pi_{bad}(1) < 0$ at the honest boundary. At $a = 0.8$ under baseline parameters, $\lambda = 0.26$ while $c(a)/r = 0.17$; no feasible increase in K_{rev} alone can stabilize the system at $K = 400$. Crucially, review assistance tools must not themselves raise λ: format and consistency checks are appropriate, but automated quality scoring is not (**?**).

~~Beyond the interior dynamics, the system may exhibit monotonic behaviour where the incentive to submit low-quality work is uniformly positive or negative.~~ **Principle 2: Reduce the false-positive rate ($\lambda\downarrow$) — the binding constraint.** The GH condition at the honest boundary requires $\Pi_{bad}(1) < 0$, which holds if and only if $\lambda < c(a)/r$. This is the binding constraint: no amount of K_{rev} expansion or slot restriction restores GH while λ exceeds this threshold. Three mechanisms directly reduce λ. First, mandatory data, code, and materials sharing makes fabrication costlier and enables post-publication verification (**??**). Second, structured reporting checklists—CONSORT for randomised trials (**?**), PRISMA for systematic reviews—force authors to document methods in ways that make gaps visible to reviewers. Third, Registered Reports commit authors to their hypotheses and analysis plan before results are known, eliminating outcome-selective reporting (**?**), which is itself a leading source of inflated false-positive rates (**?**). Taken together, these mechanisms address the root cause that λ is elevated: reviewers cannot easily distinguish genuine contributions from polished but uninformative manuscripts.

~~**Global Honesty ($x^* = 1$): Defense by Rigor**~~ **Principle 3: Add smart friction ($c\uparrow$).** The GH condition also requires $c > r \cdot \lambda$. As AI adoption lowers $c(a) = c_0(1 - \phi a)$, this gap closes and eventually inverts. Submission tokens or annual per-journal quotas raise the effective cost of marginal submissions without requiring any detection of AI use (**?**). Escalating resubmission fees impose additional costs on the low-quality cycling of rejected papers through multiple journals. We note an equity caveat: blunt friction raises costs for all authors, including those at under-resourced institutions; targeted exemptions are discussed in the final paragraph of this section.

~~When $\Pi_{bad}(x) < 0$ for all $x \in [0,1]$, the evolutionary velocity $\dot{x}$ remains positive throughout the domain, driving the system unconditionally toward $x^* = 1$ (Figure **??**) . A sufficient condition for this regime is that the payoff is negative even at the most favorable boundary ($x = 1$)~~ **Principle 4: Decouple dissemination from certification.** The current publishing architecture conflates two distinct functions: disseminating findings quickly and certifying quality carefully. When certification

degrades under submission pressure, both functions fail together. Separating the two—via preprint servers combined with overlay journals or community platforms such as Peer Community In (**?**)—allows dissemination to remain immediate while certification proceeds at the pace required for genuine scrutiny. The slot-expansion paradox established in Section 6 applies specifically to certified publication slots K_{cert}, not to preprint availability. Decoupling therefore prevents the paradox from emerging: adding preprint capacity does not raise Π_{bad}, because preprint posting is not the payoff-generating event in the spam incentive.

Principle 5: Reshape evaluation rewards (r reduced for marginal output). The spam payoff scales linearly with r:

$$\cancel{\Pi_{bad}(1) = r \cdot \lambda - c < 0}$$

~~This implies that suppressing the false-positive rate λ below the cost-reward ratio c/r is the primary safeguard for global honesty~~$\Pi_{bad} = r \cdot \bar{\rho} \cdot \pi_0 - c$. Current career and grant structures treat paper counts as a primary productivity metric (**??**), effectively maximising the effective reward for any accepted paper regardless of quality. Impact-weighted or reproducibility-indexed assessment would reduce the effective r for marginal submissions while preserving incentives for high-quality work (**?**). The GH condition at $x = 1$ requires $\lambda(a) < c(a)/r$; reducing r directly widens this margin, enlarging the parameter region in which the honest long-run outcome is self-sustaining. Reward reform operates on a longer horizon than Principles 1–4 and requires coordination across funders, institutions, and evaluation bodies; it is therefore a deep-redesign lever rather than a quick win.

Table 7 maps each mechanism to its primary parameter shift, and Figure 11 illustrates the joint effect of the principles in the (K_{rev}, K) parameter space. The left panel of Figure 11 shows that at $a = 0.8$ under baseline parameters, Global Corruption dominates most of the space, and Set A sits deep in the GC region; the navy arrow shows that raising K_{rev} alone (Principle 1) leaves Set A in GC because $\lambda = 0.26 > c/r = 0.17$. The right panel applies Principles 2 and 3 jointly ($\lambda = 0.05$, $c = 35$, with $\phi = \mu = 0$ locking in those improvements); the GC region is eliminated entirely and Set A shifts to the bistability (BS) region; the dark green arrow then shows that combining Principle 1 (K_{rev} raised to 600) carries Set A into the Global Honesty region. The lesson is that Principles 2 and 3 are necessary to exit GC, while Principle 1 converts bistability into global honesty: the three levers are complements, not substitutes.

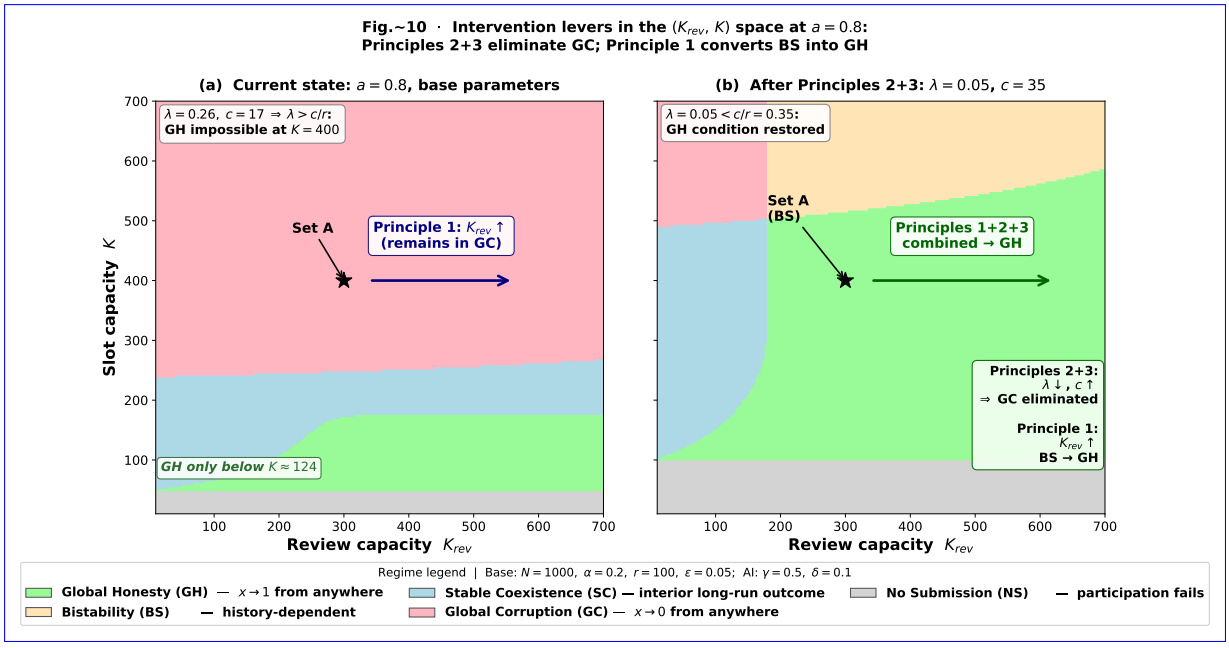


Figure 11: ~~**Phase Portrait of Global Honesty.** Parameters: $N = 1000, K_{rev} = 800, K_{store} = 400, r = 100, c = 35, \lambda = 0.1$~~Intervention levers in the (K_{rev}, K) parameter space at $a = 0.8$. *Left panel:* base parameters ($\lambda = 0.26$, $c = 17$); GC dominates and Set A (star) sits deep in GC. The ~~velocity is strictly positive~~ navy arrow shows Principle 1 (~~Green Zone~~$K_{rev} \uparrow$) alone stays in GC because $\lambda > c/r$. *Right panel:* after Principles 2 and 3 ($\lambda = 0.05$, ~~ensuring convergence~~ $c = 35$, $\phi = \mu = 0$); GC is eliminated and Set A shifts to ~~$x = 1$ from any initial state~~BS. The dark green arrow shows the combined intervention (Principles 1+2+3, K_{rev} raised to 600) restores Global Honesty. Parameters: $\gamma = 0.5$, $\delta = 0.1$; Set A baseline $N = 1000$, $\alpha = 0.2$, $r = 100$, $\varepsilon = 0.05$.

Table 7: Institutional mechanisms and their model parameter effects. Each row maps a concrete institution to the parameter it primarily shifts. Arrows: ↑ raises, ↓ lowers, — no direct effect, ↑(·) / ↓(·) partial or indirect effect. Principle 5 operates on a longer horizon than Principles 1–4 and requires coordination across funders and institutions.

Institution	Mechanism summary	K_{rev}	λ	c	r	K	Horizon
Principle 1: Scale verification							
Structured reviewer panels & recognition	Reward and recognise review effort	↑	—	—	—	—	Quick
AI-assisted pre-screening (format/consistency)	Flag obvious flaws before expert review	↑	—	—	—	—	Quick
Principle 2: Reduce false-positive rate							
Mandatory data, code, and materials sharing	Enables post-submission verification	↑ (·)	↓	↑ (·)	—	—	Quick
Structured reporting checklists (CONSORT/PRISMA)	Forces method transparency	—	↓	—	—	—	Quick
Registered Reports (pre-registered design)	Eliminates outcome-selective reporting	↑ (·)	↓↓	↑	—	—	Medium
Dedicated statistical/methods editorial review	Specialist check of analytic choices	↑	↓	—	—	—	Quick
Principle 3: Add smart friction							
Annual submission tokens or quotas	Raise cost of marginal submissions	—	—	↑↑	—	—	Quick
Escalating resubmission fees	Target low-cost paper cycling	—	—	↑	—	—	Quick
Pre-submission triage/desk-rejection filter	Reject obvious mismatches before review	↑ (·)	↓ (·)	—	—	—	Quick
Principle 4: Decouple dissemination							
Preprint + overlay certification layer	Separate speed from quality guarantee	↑	—	—	—	↓ (cert.)	Deep
Principle 5: Reshape evaluation rewards							
Impact-weighted / reproducibility-indexed assessment	Reduce r for marginal output; raise r for replication and quality	—	—	—	↓	—	Deep
Transition: post-collapse reset							
Coordinated field-wide mandate	Simultaneous cross-journal reform push	↑↑	↓↓	↑↑	—	—	Crisis

~~**Global Corruption ($x^* = 0$): Systemic Failure** Conversely, when $\Pi_{bad}(x) > 0$ for all $x \in [0, 1]$, the system collapses unconditionally to $x^* = 0$ (Figure ??). This occurs when the payoff remains positive even under maximum congestion ($x = 0$):~~

$$\cancel{\Pi_{bad}(0) = r \cdot \bar{\rho}(V_{max}) \cdot \pi_0(V_{max}) - c > 0}$$

~~This regime represents a total failure of governance, often seen in predatory journals where the profit motive drives acceptance regardless of quality.~~ **Transition policy: resetting after tipping.** Hysteresis means that gradual, journal-by-journal reforms are insufficient once the system has tipped into Global Corruption. In the GC long-run outcome, $\Pi_{bad} > 0$ for all values of the honest-author share x; no individual author has an incentive to deviate back toward honesty (**?**). Recovery requires a large, coordinated intervention that simultaneously restores λ, raises c, and expands K_{rev} enough to shift the entire parameter configuration out of the GC region. The ClinicalTrials.gov registration mandate, which required prospective trial registration as a condition of journal publication, illustrates the scale of coordination needed (**?**): a single policy change, applied across many journals simultaneously, moved an entire field. The implication for AI-era publishing is that funders and professional societies must act across journal families, not journal by journal.

~~**Phase Portrait of Global Corruption.** Parameters: $N = 1000, K_{rev} = 200, K_{store} = 1500, r = 150, c = 5, \lambda = 0.3$. The velocity is strictly negative (Red Zone), ensuring collapse to $x = 0$.~~ **Equity.** Friction mechanisms ($c\uparrow$) and strict verification requirements ($\lambda\downarrow$) impose larger burdens on authors at under-resourced institutions, who may lack the data-sharing infrastructure, statistical support, or time to meet demanding new standards. Smart design can preserve the incentive effects while redistributing the cost: submission token grants for authors at lower-income institutions (**?**); structured reporting checklists, which carry low infrastructure cost, preferred over mandatory open data repositories, which carry high infrastructure cost; subsidized reviewer development programmes that build review capacity where it is currently thin (**?**). These equity adjustments do not change the core model predictions—the regime boundaries shift on the same dimensions—but they change who bears the adjustment cost.

8.1 ~~Summary: A Topological Classification of Regimes~~

~~The dynamical behavior is ultimately determined by the interplay between the boundary payoffs. Table 8 summarizes the four topological regimes~~**Policy menu: quick wins and deep redesign.** Some mechanisms require no inter-journal coordination and can be adopted immediately: structured reporting checklists, mandatory data and code sharing policies, submission token caps, and escalating resubmission fees. Others require collective action across journals or fields: overlay certification architectures, cross-journal quota systems, and coordinated field-wide mandates. The right sequencing strategy is to deploy the quick wins first, to arrest the upward drift of Π_{bad}, while building the coordination capacity needed for deep redesign before any tipping event occurs. Waiting for a quality crisis before acting is the worst strategy: recovering from GC requires a larger simultaneous push than preventing the tip in the first place (**?**).

Box 3 Policy Menu: Institutions, Levers, and Sequencing

Goal. Keep $\Pi_{bad}(x, a) < 0$ as AI adoption a deepens. **Binding constraint:** $\lambda(a) < c(a)/r$ — when violated, no increase in K_{rev} alone can restore Global Honesty.

Quick wins (no inter-journal coordination required):

Structured reporting checklists (CONSORT, PRISMA)	$\lambda\downarrow$
Mandatory data and code sharing	$\lambda\downarrow, c\uparrow$ (partial)
Submission token caps / annual per-journal quotas	$c\uparrow\uparrow$
Escalating resubmission fees	$c\uparrow$
AI-assisted format and consistency pre-screening	$K_{rev}\uparrow$

Deep redesign (requires collective action across journals or fields):

Preprint servers + overlay certification layer	decouple dissemination from K_{cert}
Cross-journal quota systems	$c\uparrow$ system-wide
Coordinated field-wide mandate	$K_{rev}\uparrow\uparrow, \lambda\downarrow\downarrow, c\uparrow\uparrow$

~~This classification reveals a dual-control mechanism:~~**~~Review Quality (λ)~~** ~~secures the honest boundary ($x = 1$), while~~ **~~Storage Capacity (K_{store})~~** ~~secures the corrupt boundary ($x = 0$).~~

9 ~~Parameter Sensitivity and Regime Transitions~~

After tipping (recovery, not prevention): Once in Global Corruption ($\Pi_{bad} > 0$ everywhere), only a *large, simultaneous* intervention across K_{rev}, λ, and c can shift the system out of the GC region. Gradual, journal-by-journal reform is insufficient (Figure 11).

8.1 ~~Methodology and Baseline Calibration~~

Equity. Prefer low-infrastructure tools (checklists) over high-infrastructure requirements (open data repositories); offer submission token grants for authors at under-resourced institutions.
~~We want to know which policies maximize publication quality and minimize academic misconduct. However, due to~~ *Key insight:* $\lambda\downarrow$ and $K_{rev}\uparrow$ are *complements*, not substitutes — both must be deployed together. λ reduction is necessary to exit the GC region; K_{rev} expansion is necessary to convert bistability into Global Honesty. The right sequence: quick wins first to arrest the upward drift of Π_{bad}, deep redesign before any tipping event.

9 Welfare Analysis and Normative Policy Design

Social welfare in a certification system. The publishing system produces two social goods beyond private author payoffs. First, it generates *scientific knowledge* as a public good: verified results can be used by downstream researchers, educators, practitioners, and policymakers at zero marginal cost, without depleting the original discovery (**?**). Second, it provides a *certification signal*: a credential that allows readers to distinguish reliable findings from unreliable ones without re-examining every manuscript. Degradation of this signal imposes social costs that no individual author internalises. We define a reduced-form social welfare function:

$$W(x,a) \;=\; \underbrace{N\big[x\,\Pi_{good}(x,a)+(1-x)\,\Pi_{bad}(x,a)\big]}_{\text{aggregate author surplus}} \;+\; \underbrace{S\cdot Q(x,a)\cdot \bar{M}(x,a)}_{\text{social value of certified knowledge}} \;-\; \underbrace{C(K_{rev})}_{\text{verification cost}}\;, \tag{17}$$

where $S>0$ is the per-paper social value when quality is perfect, $Q(x,a)\equiv \alpha\pi_1(x,a)/[\alpha\pi_1(x,a)+(1-\alpha)(1-x)\pi_0(x,a)]$ is the precision of the ~~non-linear interplay between the review bottleneck (η) and the storage bottleneck ($\bar{\rho}$~~certified pool (the fraction of accepted papers that are genuinely good), extending (15) to $a>0$; $\bar{M}(x,a)$ is total papers accepted per period; and $C(K_{rev})$ is the institutional cost of maintaining review capacity. Welfare increases with Q because a certified pool dominated by genuine contributions generates reliable knowledge cumulation, while $Q\approx 0$ produces noise that misdirects subsequent research and policy (**?**).

The two externalities: why private and social incentives diverge. The individual spam payoff $\Pi_{bad}(x,a)=r\bar{\rho}\pi_0-c$ omits two costs imposed on others. The first is a *congestion externality*. When a spam author submits, total submission volume V rises, which depresses screening intensity $\eta=\min(1,K_{rev}/V)$ and lowers the review quality received by every other submitter. Peer-review attention is a finite shared resource, and individual overuse degrades the commons (**?**): each spam submission captures private value while imposing a diffuse cost on all other authors and editors. The second is a *certification externality*. A low-quality paper, once accepted, reduces the precision Q of the certified pool and degrades the informativeness of the certification signal for all downstream readers. This is the quality-dilution mechanism in **?**'s market for lemons: sellers of bad goods contaminate the quality signal, raising doubts about good goods and eroding the value of the entire market. In academic publishing, the "market" is the pool of certified papers, the "buyers" are readers and downstream researchers, and "bad goods" are low-quality manuscripts that passed degraded peer review. Neither externality is priced by the individual submitter. The divergence between private and social returns is the canonical justification for corrective policy (**?**). Spamming is privately rational when $\Pi_{bad}(x,a)>0$, but the full social cost includes the congestion externality imposed on honest authors and the certification-quality loss imposed on readers. The socially optimal submission norm is therefore $x=1$ (global honesty): this maximises Q, minimises wasteful review effort, and generates the highest social value from the certification pool. Yet $x=1$ is privately self-sustaining only when $\Pi_{bad}(1,a)<0$; whenever AI adoption raises Π_{bad} above zero, the social optimum and the private Nash equilibrium diverge.

Welfare ranking of the four regimes. Table 10 ranks the four stable regimes by social welfare. In GH ($x^*=1$), ~~the system does not respond smoothly to policy changes. Instead, the ecosystem undergoes~~

~~phase transitions between qualitatively distinct regimes.~~ no bad papers are submitted, the certified pool consists entirely of genuine work ($Q = 1$), and review resources are used efficiently. In GC ($x^* = 0$), all authors submit bad papers alongside good ones, the certified pool is dominated by spam papers (Q falls sharply under degraded screening), and the certification signal carries minimal information. The welfare gap $\Delta W = W(1, a) - W(0, a) > 0$ is strictly positive even accounting for the higher author surpluses in GC, because the social-value term $S \cdot Q \cdot M$ collapses in GC while the congestion cost of maintaining a large review apparatus to process a flood of mostly bad papers remains. In SC ($x^* \in (0, 1)$), welfare lies strictly between the two extremes. In BS, expected welfare is a probability-weighted average of $W(1, a)$ and $W(0, a)$ determined by initial conditions and stochastic shocks: bistability is not itself a welfare outcome but a risk of tipping into GC from a currently-healthy GH state.

~~To analyze these transitions, we use numerical bifurcation analysis. We define the instantaneous Average Publication Quality $Q(x)$ as the proportion of good papers within the set of accepted publications, a metric that aligns with standard meta-scientific frameworks for assessing the reliability of research findings (?):~~

$$\cancel{Q(x) = \frac{N\alpha \cdot \pi_1(x)}{M(x)} = \frac{\alpha \pi_1(x)}{\alpha \pi_1(x) + (1-\alpha)(1-x)\pi_0(x)}}$$

~~Crucially, the long-term quality Q_∞ is determined by the system's stable attractors. In bistable regimes, Q_∞ is path-dependent, meaning the final quality depends on whether the system started from a high-trust or low-trust state. We classify the system into four regimes based on the stability of equilibria:~~

1. ~~**Global Honesty (GH):** the system converges to $x = 1$, yielding $Q_\infty = 1.0$.~~
2. ~~**Global Corruption (GC):** the system collapses to $x = 0$, yielding $Q_\infty \approx \alpha$ — a "market for lemons" where the journal fails to filter noise (?).~~
3. ~~**Bistability (BS):** two stable attractors coexist and the outcome depends on history.~~
4. ~~**Stable Coexistence (SC):** an interior equilibrium x^* is maintained by congestion, stabilizing quality at a mediocre level ($Q_\infty \approx 0.5$).~~

~~To ensure comparative consistency, we use two representative parameter sets (Table 9). Set A represents a typical journal with moderate capacity, prone to bistability. Set B represents a top-tier journal with strict page limits and high rewards, prone to stable coexistence.~~

Table 10: Welfare ranking of the four stable regimes at a given AI adoption level a. Q^*: precision of the certified pool (fraction of accepted papers that are genuine). η^*: screening intensity at the long-run equilibrium. Social welfare W^* is the value of (17) evaluated at the long-run author composition x^*.

Regime	x^*	Q^*	η^*	Author surplus	W^* rank
GH (Global Honesty)	1	High ($= 1$)	High	High (honest work rewarded)	1st (best)
SC (Stable Coexistence)	$(0, 1)$	Medium	Medium	Mixed (both types coexist)	2nd
GC (Global Corruption)	0	Low ($\ll 1$)	Low	Positive (spam) but low social $S \cdot Q$	3rd
NS (No Submission)	n/a	n/a	n/a	Zero	Dominated

~~**Why two parameter sets?**~~**Hysteresis amplifies the welfare loss.** The welfare cost of AI adoption is not simply the static difference $W(1, a) - W(0, a)$; it is amplified by the hysteresis dynamic. Once the system tips from GH into GC, restoring GH requires a large simultaneous intervention across λ, K_{rev}, and c. This transition is costly: review infrastructure must be rebuilt, submission friction must be re-imposed, and reporting norms must be rehabilitated, all while the system is already in a low-quality long-run outcome that discourages investment in verification. The ex-ante social cost of failing to prevent tipping therefore includes both the static welfare gap ΔW and the transition cost of recovery—a cost that is entirely avoided if preventive institutions are maintained (?). This welfare-dynamic argument provides the theoretical foundation for the urgency of early intervention: the expected social benefit of preventing a tipping event is strictly larger than the expected benefit of correcting one after it occurs, because the correction requires additional resources and inflicts additional welfare losses during the recovery transition.

~~A single baseline cannot capture the full range of the system's dynamical behaviours, because the dual-bottleneck mechanism is non-monotonic. We distinguish two archetypes. Set A represents the default state of many modern journals: moderate prestige ($r = 100$) and limited but not extreme storage constraints ($K = 400$). Here the primary failure mode is the review bottleneck ($K_{rev} = 300$)—when submission volume rises, the review process collapses before the storage limit is reached. This setup reflects the empirical reality of the "global burden" of peer review, where reviewer scarcity is the dominant constraint (**?**), and it generates a bistable topology in which the system is fragile and prone to catastrophic tipping. Set B represents top-tier prestige journals such as *Nature* or *Science*, where the reward is significantly higher ($r = 150$) and the defining constraint is not the lack of reviewers (we assume ample review capacity $K_{rev} = 1200$) but the extreme scarcity of publication slots ($K = 180$). This aligns with empirical findings on scientific gatekeeping, where high rejection rates are driven by physical slot limits rather than quality filtering alone (**??**), and it generates the stable coexistence topology that illustrates what we call the defence-by-congestion mechanism. By analysing both sets, we decouple the effects of review collapse from slot rationing, ensuring that our conclusions are robust across different segments of the academic publishing hierarchy.~~

Normative policy recommendations. The welfare analysis converts the five institutional principles of Section 8 into a normative hierarchy ordered by welfare efficiency. Four Corollaries follow from combining the welfare function (17) with Propositions 1–4.

9.1 ~~The Decoupling of Capacities: Phase Diagram Analysis~~

Corollary W1 (Corrective levy on spam externalities). Because spam submissions impose unpriced negative externalities—congestion costs on honest authors and certification-quality losses on readers—the welfare-maximising policy imposes a corrective (Pigouvian) levy per submission equal to the marginal social cost of a spam attempt (**?**). In the model, this levy is equivalent to raising c: it restores the condition $c > r\lambda(a)$ that keeps GH stable, without restricting honest submissions if the levy is calibrated to the spam pass rate π_0 rather than applied uniformly. Submission tokens and per-journal annual quotas are practical approximations of this levy: the normative case for them is not administrative convenience but welfare-theoretic—they correct a market failure, not merely a nuisance (**?**). An equity-preserving design issues token grants to authors at under-resourced institutions, ensuring that the levy targets strategic spamming rather than participation barriers.

~~Standard academic models often treat "journal capacity" as a single resource. Our dual-bottleneck framework reveals that the system's fate depends on how two distinct capacities interact: review capacity~~

Corollary W2 (Peer review as a public good requiring public funding). Peer review generates a public good: the certification signal is non-excludable (once published, the quality verdict is available to all readers) and the benefit of a thorough review is non-rival (every reader gains without diminishing others' access) (**?**). Private provision of public goods is sub-optimal under standard conditions: individual reviewers and journals do not capture the full social return from careful review, and competitive pressure to cut costs erodes voluntary supply below the socially efficient level. The welfare-efficient remedy is public investment in K_{rev}~~, which determines the screening intensity $\eta(x)$, and slot capacity K_{store}, which determines the rationing probability $\bar{\rho}(x)$. High~~—targeted grants for reviewer development programmes, structured-review recognition schemes, and journal subsidies for methods review—rather than relying on uncompensated voluntarism. **?** propose transparent reviewer recognition systems as a coordination device that raises effective K_{rev} ~~prevents review collapse; low K_{store} creates what we call a defence by congestion.~~ without proportional budget increases; the normative case for mandating such systems is the public-good argument, not merely editorial preference.

~~To visualize the interaction between these forces, we perform a two-dimensional bifurcation scan across the parameter space (K_{rev}, K_{store}). The resulting phase diagram (Figure **??**) maps the system's regimes.~~

Corollary W3 (Coordination mandate in the bistable region). Proposition 4 establishes that reducing λ and raising K_{rev} are unconditionally welfare-improving: they lower spam payoffs without harming honest authors. However, decentralised adoption of verification standards is a coordination game with multiple equilibria: if competing journals do not adopt structured reporting requirements, rejected spam papers cycle to lower-tier venues, and no single journal captures the system-level benefit of its own

investment in quality. This cross-journal externality means the welfare-maximising outcome—universal verification standards—is not a Nash equilibrium of decentralised journal competition (**?**). A mandatory field-wide standard, enforced by funders, accreditation bodies, or professional associations, internalises the cross-journal spillover and restores the welfare-efficient outcome. The ClinicalTrials.gov registration mandate, which required prospective trial registration as a publication condition across many journals simultaneously, demonstrates that such coordination is institutionally feasible and welfare-improving even when no single journal would adopt the standard unilaterally (**?**).

~~**Phase Diagram of the Dual-Bottleneck System.** The map classifies the topological regime based on the availability of review resources (K_{rev}) versus publication slots (K_{store}). **Green (Global Honesty):** High review capacity ensures cheating is unprofitable ($Q \to 1.0$). **Blue (Stable Coexistence):** Extreme slot scarcity ($K_{store} \ll N$) triggers the "Defense by Congestion," stabilizing the system at a mediocre but functional level. **Yellow (Bistability):** A transition zone where the outcome is history-dependent. **Red (Global Corruption):** When slots are abundant but review is weak, the defense mechanism vanishes, leading to total collapse ($Q \to \alpha$). **Gray (System Collapse):** Feasibility constraint violated (K_{store} too low); even honest authors cannot break even.~~

~~The topology of Figure **??** highlights two critical dynamical phenomena:~~ **Corollary W4 (Welfare ranking of institutional levers).** The four levers of Proposition 4 have distinct welfare profiles. First, reducing λ (structured reporting, mandatory data sharing) is a *Pareto improvement* relative to the unregulated equilibrium: it raises Q^* and raises honest-author surplus by easing competition from spam, while imposing no direct cost on honest authors who already meet the new standard. Second, raising K_{rev} (public investment in review capacity) is welfare-improving when the congestion constraint is binding; it improves Q^* and reduces the review burden per paper, with a cost borne by funders rather than authors. Third, raising c (Pigouvian levy) improves welfare when the spam externality cost exceeds the honest-author surplus loss from higher submission costs, which holds whenever $\Pi_{bad}(x, a) > 0$ and the honest-author share is substantial; the equity caveat is that the levy must be accompanied by targeted exemptions to avoid deterring genuine science. Fourth, raising K (expanding publication slots) is the *unique lever that unconditionally reduces welfare*: it increases Π_{bad} (Proposition 2), lowers Q^* by attracting more spam, and degrades the certification signal for all readers, with no offsetting benefit. The normative implication is unambiguous: expanding publication slot capacity without simultaneously reducing λ is a welfare-reducing response to submission pressure, even though it appears to relieve editorial backlogs in the short run. Backlogs are a symptom of submission congestion; expanding slots without fixing the underlying screening deficit treats the symptom while worsening the disease.

~~**The paradox of slot expansion.** Intuition suggests that giving a journal more publication slots should improve the system. We find the opposite. Moving vertically from the bottom of the diagram corresponds to increasing K_{store}. Initially, increasing slots rescues the system from total collapse (Gray to Blue) , making participation rational for honest authors~~

Box 4 Normative Policy Agenda: Welfare Priorities and Justifications

Welfare objective: Maximise $W(x, a) = \text{author surplus} + S \cdot Q \cdot \bar{M} - C(K_{rev})$ subject to the system converging to Global Honesty (GH), not Global Corruption (GC). ~~However, further increasing K_{store} pushes the system from stable coexistence (Blue) into global corruption (Red) or bistability (Yellow). This paradox arises because tight storage constraints act as a passive defence by congestion, keeping $\bar{\rho}$ low enough to deter spam~~

Market failures requiring correction:

- *Congestion externality*: each spam submission degrades review quality for all other authors by tightening $\eta = K_{rev}/V$.
- *Certification externality*: each accepted spam paper dilutes the certified pool precision Q, reducing the value of certification for all readers (Akerlof lemons mechanism).
- *Public-good underprovision*: peer review is non-rival and non-excludable; private voluntary supply is systematically below the socially optimal K_{rev}.
- *Coordination failure*: no single journal internalises cross-journal quality spillovers; decentralised adoption of verification standards is sub-optimal. ~~By increasing K_{store}, the journal inadvertently dismantles this defence: $\bar{\rho}$ rises, the publication lottery becomes winnable again, and if K_{rev} is not increased~~

~~simultaneously , the incentive to cheat reactivates and the system tips into full corruption. This finding aligns with strategic gatekeeping models showing that expanding capacity without scaling review rigour destroys the signal value of the journal (?).~~

~~The necessity of review resources.~~ ~~Moving horizontally to the right corresponds to investing in review capacity K_{rev}. Provided the system is in a feasible region (not Gray), we find that increasing K_{rev} is unconditionally beneficial, expanding the Global Honesty (Green)zone. Crucially, the boundary between corruption and honesty slopes upward: as K_{store} increases, the journal requires *disproportionately* more review capacity to maintain honesty. This implies that larger journals are inherently harder to police than smaller ones—they require a review infrastructure that scales super-linearly with volume to maintain the same level of integrity, exacerbating the already critical burden of peer review (?).~~ **Normative lever ranking (welfare-ordered):**

9.1 ~~Population Pressure and Hysteresis: The Cost of Lost Trust~~

Lever	Welfare effect	Normative justification
$\lambda\downarrow$ (structured review, checklists, data sharing)	Pareto improvement	Corollary W1+W4; no side cost on honest authors
$K_{rev}\uparrow$ (public funding, recognition grants)	Welfare-improving	Corollary W2; public-good rationale
$c\uparrow$ (submission tokens, Pigouvian levy)	Welfare-improving*	Corollary W1; corrects congestion externality
Coordination mandate (field-wide standards)	Welfare-improving	Corollary W3; internalises cross-journal spillovers
$K\uparrow$ (more slots without $\lambda\downarrow$)	**Welfare-reducing**	Corollary W4; uniquely worsens certification quality

~~While capacity parameters define the structural limits of the journal, the total population N represents the external load. One might expect quality to decline smoothly as load increases. We find instead that the system exhibits hysteresis in the bistable regime: the value of N at which quality collapses is much higher than the value at which it recovers. This is consistent with theories of catastrophic shifts in complex systems, where state recovery requires significantly more effort than state maintenance (?).~~ * Subject to equity design: token grants for authors at under-resourced institutions.

~~To capture this path dependence, we use a numerical adiabatic continuation method. We define the system state $x_{st}(N)$ as the stable equilibrium selected by continuity from the previous state:~~

$$\cancel{x_{st}(N_i) = \arg\min_{x\in\mathcal{E}_{st}(N_i)} |x - x_{st}(N_{i-1})|}$$

~~where $\mathcal{E}_{st}(N_i)$ is the set of stable fixed points at N_i.~~ **Priority sequencing:**

1. ~~**Forward Path (Crowding):** Increments N, initializing from $x_{st}(N_{min}) \approx 1$.~~ Deploy $\lambda\downarrow$ tools immediately: Pareto-improving and require no inter-journal coordination.
2. ~~**Backward Path (Recovery):** Decrements N, initializing from $x_{st}(N_{max}) \approx 0$~~Fund $K_{rev}\uparrow$ publicly: cannot be left to voluntary effort; public-good logic requires funder mandate.
3. Implement Pigouvian $c\uparrow$ levy with equity exemptions: corrects the congestion externality.
4. Mandate cross-journal coordination before tipping: welfare loss of coordination failure is preventable.
5. Avoid $K\uparrow$ in isolation: any slot expansion must be paired with verified $\lambda\downarrow$ to prevent welfare deterioration.

Key welfare insight: The GH long-run outcome is both privately stable (when $\Pi_{bad} < 0$) and socially optimal. The policy problem arises precisely when the private stability condition fails while the social optimum remains GH. ~~Figure ?? illustrates the resulting trajectories using Baseline Set A.~~ Welfare-efficient policy closes this gap by raising the private cost of spamming and the private return to honesty, not by expanding slots or ignoring the screening deficit.

~~Hysteresis Loop and Quality Collapse (Baseline Set A).~~ ~~The shaded region represents the bistable interval $[N_\downarrow, N_\uparrow]$.~~ **~~Left Panel (State):~~** ~~On the Forward Path (Blue), the "Honest Norm" is resilient ($x_{st} \approx 1$) until the collapse threshold $N_\uparrow \approx 1800$. Once corrupted (Red Dashed), the system is trapped in the low-honesty attractor ($x_{st} \approx 0$). Recovery requires reducing N below the distinct recovery threshold $N_\downarrow \approx 350$.~~ **~~Right Panel (Quality):~~** ~~The collapse leads to a discrete jump in quality. In the corrupt state (Red Dashed), quality is dominated by the baseline productivity ($\alpha = 0.2$) but fluctuates slightly as review intensity $\eta(N)$ varies. Recovery only occurs when the system crosses $N_\downarrow$.~~

10 Empirical Predictions and Measurement

~~We highlight three features of the result.~~

~~**Structural resilience (forward path).** A journal with a strong reputation demonstrates remarkable resilience. As seen in the blue trajectory, the system maintains $x_{st} \approx 1$ and perfect quality ($Q = 1.0$) even when N far exceeds the nominal capacity, because honest authors self-regulate, keeping the effective load $V(1) = N\alpha$ manageable. The collapse point $N_\uparrow \approx 1800$ is the threshold where the honest equilibrium loses stability: $N_\uparrow = \inf\{N : \Pi_{bad}(1; N) \geq 0\}$~~

Prediction 1: Quality degradation is nonlinear—quiet, then a cliff. The bistable structure of the model generates a specific observable signature. Quality metrics should hold approximately steady while AI adoption a rises below the tipping threshold, then deteriorate rapidly once the tipping threshold $x^*(a)$ reaches the current honest-author share. Smooth, continuous quality decline would be inconsistent with the model: the bistable structure produces a cliff, not a slope. The early-warning indicators are not quality itself but the shrinking gap between the current author composition x_t and the rising tipping threshold $x^*(a_t)$—a gap that narrows silently while quality appears stable (**?**). Empirical evidence is consistent with this reading. Monthly arXiv submissions rose approximately 50% between late 2022 and late 2024, while reviewer acceptance rates and post-publication retraction counts have not yet stabilised at new long-run levels (**??**). Field-level heterogeneity provides additional structure: **?** find AI-modified text rates exceeding 40% in some biomedical sub-disciplines while remaining near zero in others, suggesting that different fields sit at different distances from their respective tipping thresholds—some already past it, others still in the quiet phase.

Prediction 2: The ~~stickiness of corruption (backward path)~~gap between collapse and recovery thresholds widens as AI adoption rises. ~~Once the system settles into the corrupt equilibrium,~~ As a increases, Proposition 3 shows that the defence-by-congestion mechanism requires progressively tighter slot constraints to hold the corrupt boundary, while Proposition 1 shows the Global Honesty boundary erodes. The model therefore predicts that the intervention required to restore quality in a field that has tipped is strictly larger than the intervention that would have prevented the tip. Pre-AI evidence establishes that the hysteresis mechanism itself is empirically real, independently of what triggers the initial collapse. **?** document that image duplication—driven by paper mills, not generative AI—became entrenched in specific biomedical sub-fields despite sustained editorial intervention; **?** report that annual retraction counts exceeded 10,000 for the ~~sheer volume of spam saturates the review capacity. Even if N is reduced, the system remains trapped in the~~ first time in 2023 without receding, suggesting that the low-quality ~~state as long as $\Pi_{bad}(0; N) > 0$. The recovery point $N_\downarrow \approx 350$ corresponds to the threshold where the corrupt equilibrium finally becomes unstable. This asymmetry mirrors the dynamics of collective reputation, where a group remains locked in a low-quality equilibrium long after external conditions improve,because the burden of proof is too high for individual reformers (**?**).~~

~~Crucially, we define the Average Publication Quality $Q(x)$ as the proportion of good papers in the passed pool $M(x)$:~~

$$\cancel{Q(x) = \frac{N\alpha\pi_1(x)}{N\alpha\pi_1(x) + N(1-\alpha)(1-x)\pi_0(x)}}$$

~~Since slot rationing ($\bar{\rho}$) applies uniformly to $M(x)$, it does not alter the ratio of good-to-bad papers. Consequently, quality remains low throughout the backward path until the critical transition occurs~~long-run outcome sustains itself once established. The pre-AI paper-mill literature thus confirms the mechanism: quality collapses are not self-correcting, and removing the proximate trigger (the mills) does not restore the parameter configuration that originally supported honest submission. The AI-specific prediction is that the same mechanism will be triggered at lower adoption thresholds and across more fields simultaneously, because AI lowers $c(a)$ and raises $\lambda(a)$ for the entire author population rather than for a subset of bad actors. A direct test would compare the institutional effort required to restore quality indicators in a field that tipped—whether by paper mills before 2022 or by AI-driven flooding after—against one that never tipped but faces equivalent submission pressure.

~~**Policy implication**~~**Prediction 3:** ~~**basin hopping**~~**Proxies for a rising false-positive rate.** ~~The disparity between $N_\uparrow$ and $N_\downarrow$ implies that incremental reforms are structurally insufficient for a corrupted journal. Within the bistable interval, marginal reductions in submissions do not change the long-run attractor. Escaping the low-quality basin requires interventions strong enough to move the system across the recovery threshold ($N < N_\downarrow$). This suggests that while soft measures work for maintenance, hard measures (e.g., temporary submission moratoriums) are structural necessities for recovery.~~

10.1 ~~Strategic Levers and Regime Transitions~~

~~While capacity parameters define the structural limits, the strategic parameters $(r, c, \lambda, \epsilon, \alpha)$ determine the incentives. We perform a comparative sensitivity analysis across all five dimensions, grouping them into three functional categories: screening, economic incentives, and productivity~~A rising λ produces a detectable mismatch between surface quality and substantive reliability: manuscripts become more polished while methodological soundness does not improve proportionately. Three observable proxies track this channel. First, **acceptance rates** should rise faster than any improvement in underlying research quality, because easier-to-pass manuscripts inflate the acceptance pool without proportionate quality gains; **?** document this experimentally—AI-assisted reviews award systematically higher scores. Second, **surface linguistic quality** should improve independently of replication success or citation impact. **?** show that AI modification is measurable through excess vocabulary, providing a continuous field-level proxy for surface polish that can be compared against downstream quality indicators. Third, **replication rates** should decline in fields with high AI adoption if the screening signal is genuinely degrading. The Open Science Collaboration found that fewer than half of 100 psychology studies replicated successfully (**?**); replicating that audit with papers published after 2022 in high-AI-adoption fields, and comparing against low-adoption fields, would directly test whether the false-positive channel is active.

~~**Screening parameters**~~**Prediction 4:** ~~**the false-positive rate versus friction**~~**Proxies for review-capacity stress.** The model predicts declining screening intensity $\eta = \min(1, K_{rev}/V)$ as submission volume grows faster than review capacity. Four observable proxies track this channel. First, **reviewer invitation acceptance rates** should decline faster in high-AI-adoption fields; **?** document a continuous 21-year decline at a single journal, and the model predicts this trajectory steepens as AI-driven submission pressure intensifies. Second, **editorial turnaround times** should lengthen as editors wait longer for reviewer responses, signalling scarcity in the review pool. Third, **desk rejection rates** should rise as editorial offices pre-filter to protect reviewers from overload; an increasing share of submissions receiving no substantive evaluation is a direct proxy for η falling toward zero for that fraction of papers. Fourth, **reviewer concentration** should increase—a growing Gini coefficient on review counts across the active reviewer pool indicates that a shrinking fraction of researchers is absorbing a rising fraction of the total burden, a pattern that preceded the current shortage (**?**).

~~We begin with the two parameters controlling the peer review filter (Figure **??**). Panel (a) reveals that the false-positive rate λ is the most critical parameter in the entire system. As λ increases, the system shifts rapidly from Green (Honesty) to Yellow (Bistability) and finally to Red (Corruption); in the yellow zone, a minor perturbation can trigger a catastrophic tip into the corrupt equilibrium. This aligns with signalling theory, where the reliability of the signal (low λ) is paramount for maintaining the separating equilibrium (**?**).~~

~~In contrast, Panel (b) shows that friction ϵ has negligible structural impact: within the explored range, the regime boundaries remain largely unchanged. This implies that policies focused solely on reducing administrative friction for good papers yield little improvement in integrity compared to improving detection accuracy.~~

~~**Screening Parameters and Regime Stability.** **(a) False Positive Rate λ:** Acts as a major destabilizer. Increasing λ destroys the honest equilibrium, pushing the system into high-risk bistability. **(b) Friction ϵ:** Has negligible impact on regime boundaries, acting merely as a silent administrative cost.~~

~~Economic levers~~**Prediction 5: ~~reward~~ Data sources and identification strategy.** Testing these predictions requires field-level panel data on both the production and verification sides of the system. Several existing sources can be combined. On the production side, arXiv submission records are public, monthly, and tagged by subfield; the LLM modification classifier of **?**, applied to these records, provides a continuous field-level estimate of a_t. The vocabulary-based detection approach of **?**, applied across the 15-million-abstract PubMed archive, offers an independent measure of AI modification at the sub-discipline level. On the verification side, reviewer acceptance rate series can be constructed from editorial logs at journals that publish annual statistics; acceptance rates, turnaround times, and retraction counts are available at scale through OpenAlex, CrossRef, and the Retraction Watch database. The natural experiment for causal identification is the release of ChatGPT in November 2022, which created a sharp and field-heterogeneous shock to submission cost c. A difference-in-differences design would compare fields with high ex-ante AI amenability—measured by pre-2022 adoption of computational tools, large collaborative networks, or text-heavy output—against low-amenability fields, before and after November 2022. The prediction is that high-amenability fields show faster growth in submission-to-review ratios, rising acceptance rates, and more rapid accumulation of post-publication retractions relative to low-amenability fields, controlling for field size and pre-shock submission trends.

~~We next examine the financial and reputational incentives (Figure **??**). Panel (c) demonstrates that high rewards r act as an accelerant for corruption, significantly expanding the Red and Yellow zones. When publication rewards are excessive,~~

Prediction 6: What would falsify the asymmetry hypothesis. The model's conclusions rest on two empirical claims: production scales faster than verification ($\gamma > \delta$), and the false-positive rate rises with AI adoption ($\mu > 0$). Either claim failing empirically would revise the institutional implications. If $\delta \approx \gamma$—if AI-assisted reviewing tools raise review capacity at nearly the same rate as AI raises submission volume—the congestion channel would not activate. This would be observable as stable submission-to-acceptance ratios and constant turnaround times despite rising submission volumes in high-adoption fields. If $\mu \approx 0$—if AI tools improve screening reliability, perhaps because structured AI-assisted checklists close the gap that polished writing opens—the false-positive channel would be neutralised. This would be observable as stable or falling acceptance rates in high-AI-adoption fields combined with stable replication rates for newly published work. The robustness analysis in Figure 9 shows that μ is the ~~lottery of rubber-stamping becomes profitable even at low probabilities—confirming the mechanism of "natural selection of bad science," where high incentives select for strategies that maximize output at~~ binding parameter: even if δ/γ rises toward 1, the ~~expense of quality (**?**).~~ system remains in Global Corruption whenever $\mu > 0.09$. A finding that AI-assisted reviewing consistently *improves* score calibration—rather than inflating it—would therefore be the strongest possible disconfirmation of the model's policy conclusions (**?**). The available evidence currently points in the opposite direction (**???**).

~~While reward drives the system toward corruption, submission cost c pulls it back. Panel (d) shows that increasing c effectively shifts the system from Red to Green, providing theoretical support for submission fees as a tool to reduce congestion and filter out low-quality submissions (**?**). However, pushing costs too high forces the system into the Gray zone, where even honest authors cannot sustain participation ($\Pi_{good} < 0$)~~

11 Discussion

Scope and simplifying assumptions. The model makes two structural choices that bound its applicability. First, we reduce the strategy space to two options: submit only good work, or submit regardless of quality. Real authors calibrate effort continuously, and intermediate strategies can coexist in finite communities. The binary reduction is justified by an evolutionary instability argument (**?**): at any population state x, a strategy that adds bad-paper submissions earns strictly higher fitness when $\Pi_{bad} > 0$, and strictly lower fitness when $\Pi_{bad} < 0$, so intermediate submission rates are unstable under replicator dynamics and the population is driven to one of the two extremes in the mean-field limit. In practice, a continuum of effort levels would smooth the tipping threshold into a region rather than a sharp boundary, making early warning signals harder to detect but not eliminating them. Second, we model a

single journal with fixed capacity. In reality, rejected papers cascade down a prestige hierarchy: **?** find that roughly 62 % of papers rejected at three high-impact journals were eventually published elsewhere. This cascade distributes submission pressure across tiers rather than absorbing it. A multi-journal extension would predict that tipping first occurs at lower-tier venues—which typically have smaller K_{rev} and weaker screening—and then propagates upward as the pool of credibly certified papers shrinks. Our single-journal results are therefore conservative: the threshold at which quality degrades across the system as a whole is likely lower than the threshold we identify for any single venue.

~~**Economic Incentives and Filtering.** **(c) Reward r:** High rewards expand the corrupt zones (Red/Yellow), incentivizing spamming strategies. **(d) Cost c:** Submission fees filter out spam, restoring Green, but excessive costs kill the journal (Gray).~~

AI adoption as a scalar: limits and heterogeneity. We represent AI adoption with a single scalar a that aggregates the fraction of authors using AI tools and the intensity of their use. This captures the mean effect but suppresses two empirically important dimensions. The first is task heterogeneity. AI excels at drafting and reformatting text but falls short of replicating the judgment required for novel empirical analysis (**?**). The production elasticity γ therefore measures AI's effect on submission volume rather than on genuine scientific output. A model that separates writing assistance from research assistance would generate sharper predictions about μ: fields where AI mainly improves surface fluency face a higher μ than fields where AI genuinely improves research quality. The second dimension is cross-author heterogeneity in adoption timing. Early adopters gain a first-mover advantage under current publication incentives, and social learning then accelerates diffusion through the same imitation dynamics documented for other general-purpose technologies (**?**). As Section 7 shows, stochastic fluctuations from a concentrated wave of intensive early adopters can push a finite community below its tipping threshold before field-wide mean adoption reaches a critical level. Heterogeneous adoption therefore makes tipping events more likely and less predictable than the scalar analysis implies.

~~**The productivity paradox.**~~

The problem is institutional mismatch, not AI adoption itself. The model's prescriptions do not depend on AI being harmful to science in aggregate. If review capacity scaled at the same rate as submission volume—$\delta = \gamma$—and AI tools improved screening reliability rather than degrading it—$\mu \leq 0$—the Global Honesty region would expand under rising a, not contract. **?** document genuine productivity gains from AI assistance; the issue is not that AI exists but that the gap $\gamma - \delta > 0$ exists. **?** provide a useful framing: automation raises productivity in tasks that are routine and codifiable while increasing demand for tasks requiring contextual judgment and domain expertise. Text generation is largely the former; verifying the validity of a novel empirical claim is largely the latter. The challenge for publishing institutions is therefore not to restrict AI adoption—which would be both unenforceable and wasteful where AI genuinely helps—but to close the gap between γ and δ through targeted investment in verification infrastructure. The scenario matrix in Table 4 makes this concrete: which cell a field occupies is an institutional choice, not a technological fate. Fields that invest in structured AI-assisted verification pipelines, mandatory data deposition, and reviewer development programmes can remain in the upper-left cell even as a rises. Fields that do not will drift toward the upper-right.

~~Finally, we analyse the impact of aggregate author productivity (Figure **??**). One might expect that higher productivity α would stabilize the system, since the input mix improves. We find the opposite: as α increases,~~

Multi-journal competition and quality cascades. Extending the model to a hierarchy of competing journals would capture dynamics absent from our analysis. When authors face multiple venues, rejection does not dissipate submission pressure—it routes it to the next tier. **?** document steep and persistent prestige hierarchies in academic hiring networks; analogous hierarchies characterise journals, where a small number of high-impact venues confer disproportionate career rewards r. In a tiered model, the ~~background topology shifts toward unstable regimes (Yellow/Red), even though the average quality (black line) rises mechanically. The mechanism is congestion. As legitimate authors produce more papers, the total volume rises and consumes scarce review capacity; if~~ slot expansion

paradox would have a cross-venue counterpart: entry of new low-tier journals raises total certification capacity K without improving screening quality λ, raising Π_{bad} at every tier simultaneously. **?** show that prestige-maximising incentives in multi-journal settings already reward strategic over honest behaviour; AI adoption amplifies this dynamic by lowering $c(a)$ and raising $\lambda(a)$ for the entire author population. The institutional implication is that verification standards must be coordinated across journal tiers. A single journal raising K_{rev} ~~is fixed, this "good congestion" dilutes the screening intensity for everyone, paradoxically making the system more vulnerable to invasion by spammers.~~ and lowering λ provides limited system-level protection if rejected spam papers are absorbed by lower-tier venues and ultimately circulate as certified work. Overlay certification models—where a shared preprint layer is evaluated by tier-specific panels with explicit quality standards—are one architecture that could enforce coordination without requiring every journal to act simultaneously.

~~**The Productivity Paradox. (e) Productivity α:** Higher productivity increases total congestion. While observable quality rises (black line), the structural stability degrades (background shifts to Red), narrowing the window of stability.~~

Endogenous rewards, reviewer effort, and reputation systems. Two parameters we treat as fixed—the publication reward r and the reviewer participation rate embedded in K_{rev}—are endogenous to institutional design, and relaxing this assumption would enrich the model. On the reward side, r reflects career-level incentives to publish. Evaluation frameworks that weight publication counts heavily inflate r for marginal submissions; shifting toward impact-based or reproducibility-weighted assessment directly reduces r and shrinks the parameter region where $\Pi_{bad} > 0$ (**?**). **?** traces publication inflation to career structures that treat paper counts as a proxy for scientific value; AI adoption compounds this pre-existing incentive problem by making low-quality output cheaper to produce. On the reviewer side, we model K_{rev} as a hard capacity constraint, but reviewer effort per paper is also a margin. Reviewer participation has declined continuously for two decades (**?**), and the decline accelerates if quality collapse erodes the perceived value of certification—a positive feedback loop that our model does not capture. An extension treating reviewer effort as endogenous would predict that transparency mechanisms—publishing review reports, providing structured evaluation templates, recognising reviewer contributions (**?**)—can raise effective screening accuracy without increasing the number of reviewers. The two levers—more reviewers and better effort per reviewer—address the same capacity constraint K_{rev} through different margins and should be designed jointly. Neither is sufficient in isolation when $\lambda > c(a)/r$, the binding condition identified in Section 6.

12 ~~Agent-Based Modeling~~

12 Conclusion

12.1 ~~Simulation Design~~

~~The analytical results above assume a large, well-mixed population and deterministic strategy evolution—the standard mean-field approximation (**?**). Real academic communities are finite and subject to stochastic fluctuations (**?**). We therefore implement an agent-based model (ABM) to test whether our main claims survive these complications: that dominated strategies die out naturally, that the predicted~~ We have studied what happens to a publishing system when AI raises the productivity of manuscript production faster than it raises the capacity to verify manuscripts. The central asymmetry—production elasticity γ exceeding verification elasticity δ—is not a modelling assumption but an empirical claim supported by documented submission growth, documented reviewer decline, and experimental evidence on AI writing productivity (**???**). Three results follow from this asymmetry. First, extreme slot scarcity can stabilise the honesty norm under degraded screening, but AI erodes this protection continuously by lowering the submission cost $c(a)$: the required slot tightness grows with adoption and eventually forces the system into the No Submission zone. Second, gradual AI adoption can trigger a discrete collapse rather than a smooth decline; the system can appear healthy—with observable quality near its maximum—while the gap between the current author

composition and the tipping threshold silently narrows. Third, once the system tips, recovery is harder than prevention, because the low-quality long-run ~~outcomes emerge from local interactions under noise, and that the defence-by-congestion mechanism is not an artefact of the mean-field limit (?).~~ outcome is self-sustaining: no individual author has a reason to deviate back toward honesty, and small interventions cannot overcome the strategic lock-in (**?**). Expanding publication slots accelerates rather than relieves this dynamic, because it raises the expected payoff of submitting low-quality work without addressing the underlying screening deficit.

~~We simulate a population of $N = 1000$ agents, following the standard protocols of stochastic evolutionary game theory in finite populations **?**. We initialise the population with a uniform distribution across all four strategies (25% each), so that the simulation must find the reduced state on its own rather than having itassumed. The process proceeds as follows:~~ Translating these results into institutional design requires targeting the parameters that matter most. The false-positive rate λ is the binding constraint: when $\lambda > c(a)/r$, no feasible increase in review staffing can restore the Global Honesty condition, because even carefully reviewed low-quality manuscripts pass at a rate that makes submission individually rational. Effective policy must therefore reduce λ directly—through mandatory data and code sharing, structured reporting requirements, and registered study designs—while simultaneously raising K_{rev} and restoring submission friction (**??**). These three levers are complements, not substitutes: the analysis in Section 8 shows that λ reduction is necessary to exit the Global Corruption region, and K_{rev} expansion is necessary to convert bistability into Global Honesty. The institutional response to AI is therefore not "more slots" but a different kind of institution—one in which verification is a first-class resource, not an afterthought.

The window for preventive action is narrower than it appears. Because the system can remain near the honest long-run outcome while structural fragility accumulates, the observable quality signal gives no reliable warning before a tipping event (**?**). Fields that invest in verification infrastructure, reviewer incentives, and submission friction before AI adoption deepens can preserve the integrity of certification at moderate cost. Fields that wait for a visible quality crisis will face a recovery problem that requires a larger and more disruptive coordinated intervention—the equivalent of a field-wide mandate rather than a journal-level policy. The asymmetry between production and verification is not temporary: it reflects a fundamental difference in what language models can automate, and institutions designed for an earlier era will need deliberate redesign to remain effective.

Table 3: ~~Summary of key parameters and variables used in the payoff analysis.~~

~~Symbol~~	~~Description~~
~~***Exogenous Parameters (Fixed)***~~	
$\cancel{N}$	~~Total number of authors~~
$\cancel{K}$	~~Journal capacity (maximum publication slots per cycle)~~
$\cancel{\alpha}$	~~Probability that an author produces a good paper~~
$\cancel{\epsilon}$	~~False negative rate (Prob. of rejecting a good paper)~~
$\cancel{\lambda}$	~~False positive rate (Prob. of accepting a bad paper)~~
$\cancel{r}$	~~Reward for a published paper (Author)~~
$\cancel{e}$	~~Cost of submission (Author)~~
$\cancel{B}$	~~Benefit for publishing a good paper (Journal)~~
$\cancel{D}$	~~Penalty for publishing a bad paper (Journal)~~
$\cancel{C_J}$	~~Unit cost of peer review (per submission)~~
~~***Endogenous Variables (Dependent on Strategy)***~~	
$\cancel{V}$	~~Total number of submissions (Submission Pool)~~
$\cancel{M}$	~~Number of papers passing peer review (Perception Pool)~~
$\cancel{Y}$	~~Number of papers finally published ($Y \leq K$)~~
$\cancel{\bar{p}}$	~~Expected rationing probability ($\mathbb{E}[\min(1, K/M)]$)~~
$\cancel{x}$	~~Proportion of authors adopting a specific strategy (Population State)~~

Table 4: **Scenario matrix: qualitative regime outcomes as AI adoption a rises.** Two policy-sensitive parameters determine the trajectory. δ/γ: how fast review capacity keeps pace with submission volume (a measure of investment in verification infrastructure). μ: how fast AI adoption raises the false-positive rate (a measure of screening vulnerability). Each cell describes the dominant direction of regime change as a increases from zero.

		$\delta \approx \gamma$: **review keeps pace**	$\delta \ll \gamma$: **review lags severely**
$\mu > 0$: degrades	screening	*Partial stability.* Review congestion is controlled, but rising λ still weakens the screening signal. GH region shrinks; bistability expands. Verification investment is necessary but not sufficient.	*Highest risk.* Both the congestion and false-positive channels are active. $\Pi_{bad}(1, a)$ rises rapidly. The system moves toward Global Corruption. Risk of irreversible tipping is high.
$\mu \approx 0$: holds	screening	*Stable.* AI raises both production and verification at comparable rates; screening quality is maintained. GH region is preserved. Consistent with AI-augmented structured review under strong institutional controls.	*Conditionally stable.* Slot rationing may still provide defence by congestion if K is tight. Fragile: any expansion of publication slots or rise in r can tip the system toward bistability. Requires active management of K.

Table 5: ~~Matrix of Effective Passing Probabilities $\mathcal{P}(s_A, s_J)$. This represents the probability that an author's paper enters the peer-accepted pool M (before rationing).~~

~~Author~~ \ ~~Journal~~	~~Only Good (OG)~~	~~Always Accept (AA)~~	~~Only Bad (OB)~~	~~Reject All (AR)~~
~~OG~~ (α good)	$\alpha(1-\epsilon)$	$\alpha\epsilon$	0	
~~AS~~ (All)	$\alpha(1-\epsilon)+(1-\alpha)\lambda$	1	$\alpha\epsilon+(1-\alpha)(1-\lambda)$	0
~~OB~~ ($1-\alpha$ bad)	$(1-\alpha)\lambda$	$1-\alpha$	$(1-\alpha)(1-\lambda)$	0
~~NS~~ (None)	0	0	0	0

Table 6: Parameter calibration for numerical analysis

Symbol	**Description**	**Set A (Baseline)**	**Set B (Top-tier)**
N	Author population	1000	1000
α	Good-paper probability	0.2	0.2
K_{rev}	Review capacity	**300**	**1200**
K	Slot capacity	**400**	**180**
r	Publication reward	100	**150**
c	Submission cost	25	**30**
λ	False-positive rate	0.1	**0.4**
ε	False-negative rate	0.05	0.01

Set A: $\lambda r < c$; review bottleneck dominates; primary failure mode is bistability. Set B: $\lambda r > c$; slot scarcity dominates; illustrates defence by congestion via tight $\bar{\rho}$ (not via η, which equals 1 throughout since $K_{rev} \gg N$).

Table 8: ~~**Topological Classification of Dynamical Regimes based on Boundary Payoffs**~~

~~**Regime**~~	~~$\Pi_{\text{bad}}(1)$~~ ~~*(Low-spam Limit)*~~	~~$\Pi_{\text{bad}}(0)$~~ ~~*(Max-load Limit)*~~	~~**Dynamical Topology & Implication**~~
~~**GH**~~	~~< 0~~	~~< 0~~	~~**Honesty-dominant.** $x = 1$ is locally stable; $x = 0$ is unstable. Trajectories converge to $x = 1$ globally, *provided* $\Pi_{bad}(x)$ *has no interior root.*~~
~~**GC**~~	~~> 0~~	~~> 0~~	~~**Corruption-dominant.** $x = 0$ is locally stable; $x = 1$ is unstable. Trajectories collapse to $x = 0$ globally, *provided* $\Pi_{bad}(x)$ *has no interior root.*~~
~~**BS**~~	~~< 0~~	~~> 0~~	~~**Bistability.** Both boundaries are locally stable attractors. By continuity, **at least one** unstable interior equilibrium x^* exists to separate the basins.~~
~~**SC**~~	~~> 0~~	~~< 0~~	~~**Stable Coexistence.** Both boundaries are unstable repellers. By continuity, **at least one** stable interior equilibrium x^* exists (trapped by congestion). Uniqueness depends on monotonicity.~~

~~*Note:* The **Low-spam Limit** ($x = 1$) is driven primarily by review rigor (λ) via $\Pi_{bad}(1) \approx r\lambda - c$. The **Max-load Limit** ($x = 0$) is driven by the collapse of acceptance rate ($\bar{\rho}$) under maximum volume, regardless of review failure.~~

Table 9: ~~Parameter Calibration for Numerical Analysis~~

~~**Symbol**~~	~~**Description**~~	~~**Set A (Baseline)**~~	~~**Set B (High-Pressure)**~~
~~N~~	~~Total Population~~	~~1000~~	~~1000~~
~~α~~	~~Author Productivity~~	~~0.2~~	~~0.2~~
~~K_{rev}~~	~~Review Capacity~~	~~**300**~~	~~**1200**~~
~~K~~	~~Slot Capacity~~	~~**400**~~	~~**180**~~
~~r~~	~~Reward~~	~~100~~	~~**150**~~
~~c~~	~~Cost~~	~~25~~	~~**30**~~
~~λ~~	~~False Positive Rate~~	~~0.1~~	~~**0.4**~~
~~ϵ~~	~~False Negative Rate~~	~~0.05~~	~~0.01~~

~~*Note: Set A is used as the default for sensitivity sweeps. Set B is used to illustrate the defense-by-congestion mechanism.*~~

A Proofs of Propositions 1–4

The four propositions in Section 6 characterise how the spam payoff $\Pi_{bad}(x, a)$ responds to the AI adoption parameter a and to the four institutional levers λ, K_{rev}, c, and K. This appendix provides complete formal derivations. We first collect notation and establish three lemmas that are used repeatedly across the proofs.

Notation. Throughout, $x \in [0, 1]$ is the share of selective (OG) authors and $a \in [0, 1]$ is the AI adoption intensity. Define:

$$
\begin{aligned}
V_0(x) &\equiv N\big[1 - (1-\alpha)x\big] && \text{baseline submission volume (at } a = 0\text{)},\\
V(x,a) &= V_0(x)(1+\gamma a) && \text{total submission volume},\\
K_{rev}(a) &= K_{rev}(0)(1+\delta a) && \text{review capacity},\\
\eta(x,a) &= \min\Big(1,\ \tfrac{K_{rev}(a)}{V(x,a)}\Big) && \text{screening intensity},\\
\pi_0(x,a) &= 1 - \eta(x,a)\big(1-\lambda(a)\big) && \text{bad-paper pass rate},\\
\pi_1(x,a) &= 1 - \eta(x,a)\,\varepsilon && \text{good-paper pass rate},\\
M(x,a) &= (1+\gamma a)\big[N\alpha\,\pi_1(x,a) + N(1-\alpha)(1-x)\,\pi_0(x,a)\big] && \text{peer-accepted pool},\\
\bar{\rho}(x,a) &= \min\Big(1,\ \tfrac{K}{M(x,a)}\Big) && \text{rationing probability},\\
\Pi_{bad}(x,a) &= r\,\bar{\rho}(x,a)\,\pi_0(x,a) - c(a) && \text{spam payoff}.
\end{aligned}
$$

We work throughout under the **AI asymmetry assumption** $\gamma > \delta \geq 0$ and the parametric assumptions $\mu \geq 0$, $\phi \in (0, 1)$, $r, c_0 > 0$. We also impose the **discriminatory filter condition**:

$$\lambda(a) + \varepsilon < 1 \qquad \text{for all } a \in [0, 1]. \tag{18}$$

This states that peer review rejects bad papers at a strictly higher rate than it falsely rejects good ones, i.e. the filter is better than random. The proofs split into three cases based on which capacity constraints are binding:

- ~~**Interaction Phase:** In each time step t, agents generate submissions based on their full strategy space ($s_i \in \{\text{OG}, \text{AS}, \text{OB}, \text{NS}\}$). The aggregate volume V~~ **Case I** (uncongested): $V(x, a) \leq K_{rev}(a)$, so $\eta = 1$.
- **Case II** (congested, unrationed): $V > K_{rev}$ and ~~peer-accepted volume M are calculated endogenously, determining the congestion levels η~~ $M \leq K$, so $\eta < 1$ and ~~ρ.~~ $\bar{\rho} = 1$.
- ~~**Payoff Realization:** Agents receive payoffs Π_i based on the realized publication outcomes~~**Case III** (congested and rationed): $V > K_{rev}$ and $M > K$, so $\eta < 1$ and $\bar{\rho} = K/M < 1$. ~~This captures the "lottery" nature of the congestion bottleneck.~~
- ~~**Strategy Update (Social Learning):** We employ the **pairwise comparison process** to model strategy evolution. A randomly selected agent A considers mimicking agent B's strategy. The probability of switching $s_A \to s_B$ is given by the Fermi function derived from statistical mechanics ?.~~

  $$\cancel{P(s_A \to s_B) = \frac{1}{1+\exp(-\beta[\Pi_B - \Pi_A])}}$$

  ~~where β represents the intensity of selection. This update rule is the standard method for introducing bounded rationality and decision noise into evolutionary dynamics ?.~~

A.1 ~~Simulation Results~~

Lemma A.1 (Screening intensity falls with a). ~~We simulated both parameter sets from Table 9 with a high selection intensity ($\beta = 1.0$) to reduce noise. Figure ?? shows that the agent-based simulations corroborate both the strategy elimination and the macroscopic predictions from Section 3.~~ *In Cases II and III:*

$$\frac{\partial \eta}{\partial a} = \frac{K_{rev}(0)\,(\delta - \gamma)}{V_0(x)\,(1+\gamma a)^2} < 0. \tag{19}$$

~~In both scenarios, the "Only Bad" (OB) and "Never Submit" (NS) strategies are rapidly driven to extinction within the first few hundred time steps, directly verifying the iterated elimination of strictly dominated strategies (IESDS) conducted in Section 2.6. The system autonomously reduces to a binary contest between OG and AS, consistent with the finding that strictly dominated strategies cannot survive under monotonic selection dynamics ?.~~ *Proof.* Substitute $K_{rev}(a) = K_{rev}(0)(1+\delta a)$ and $V(x,a) = V_0(x)(1+\gamma a)$ into $\eta = K_{rev}(a)/V(x,a)$ and differentiate with respect to a using the quotient rule. The numerator of the result is $K_{rev}(0)[\delta(1+\gamma a) - \gamma(1+\delta a)] = K_{rev}(0)(\delta - \gamma) < 0$ since $\gamma > \delta$. □

~~In the review-dominated regime (Scenario A), the dynamics exhibit sharp bistability (Figure ??). Because the initial proportion of honest OG authors~~

Lemma A.2 ~~(25%) **falls well below the theoretical unstable tipping point ($x^* \approx 0.82$), the system cannot self-correct. Once OB and NS vanish, the AS strategy outcompetes the remaining honest authors, driving the system into the absorbing state of full corruption ($x = 0$). This confirms the severe path dependence predicted by the analytical model ??**~~ **. Both pass rates rise with a).** ~~In the rationing-dominated regime (Scenario B), we observe a qualitatively different outcome (Figure ??). Despite the initial chaotic interaction of all four strategies, the population rapidly settles to a long-run average of $x^* \approx 0.49$ for OG and AS, while OB and NS remain extinct. This result aligns precisely with the analytical fixed point ($x^* \approx 0.49$) identified in the phase portrait analysis, confirming that the defence-by-congestion mechanism is a robust emergent property, immune to initial strategy turbulence and finite-size stochasticity.~~ *In Cases II and III, writing $\dot{\eta} \equiv \partial\eta/\partial a < 0$:*

$$\frac{\partial \pi_0}{\partial a} = -\dot{\eta}\,(1-\lambda) + \eta\,\mu > 0, \tag{20}$$

$$\frac{\partial \pi_1}{\partial a} = -\dot{\eta}\,\varepsilon > 0. \tag{21}$$

~~Scenario A: Review-Dominated Dynamics~~ *Proof.* Differentiate $\pi_0 = 1 - \eta(1-\lambda(a))$ with respect to a. The product rule gives $-\dot{\eta}(1-\lambda) + \eta\, d\lambda/da = -\dot{\eta}(1-\lambda) + \eta\mu$. Both terms are non-negative because $\dot{\eta} < 0$, $1-\lambda > 0$, $\eta > 0$, and $\mu \geq 0$; the first term is strictly positive. Differentiate $\pi_1 = 1 - \eta\varepsilon$ with respect to a; since ε is a fixed parameter, $\partial\pi_1/\partial a = -\dot{\eta}\varepsilon > 0$. ~~Scenario B: Stable Coexistence~~ ~~**Agent-Based Model Trajectories with 4 Strategies ($\beta = 1.0$).** The plots show the time evolution of the strategy proportions. **(a)** Strong Bistability: Starting from an equal uniform mix, the system rapidly eliminates OB and NS. Because the initial OG fraction (0.25) is below the critical tipping point, the system collapses to full corruption ($x = 0$). **(b)** Stable Coexistence: OB and NS are quickly eliminated, and the remaining OG and AS populations converge to a mixed equilibrium ($x^* \approx 0.49$). The precise agreement with analytical predictions confirms both the theoretical dominance analysis and the structural congestion stabilization.~~ □

A.1 ~~Co-operation ABM~~

Lemma A.3 (Bad-paper advantage grows with a). ~~To rigorously validate the robustness of our theoretical findings and explore the microscopic foundations of~~ *In Cases II and III, (i) the ratio π_0/π_1*

is strictly increasing in a:

$$\frac{d}{da}\left(\frac{\pi_0}{\pi_1}\right) > 0; \tag{22}$$

and (ii) defining the base pool $M_0 \equiv N\alpha\pi_1 + N(1-\alpha)(1-x)\pi_0$, the following key inequality holds:

$$M_0 \frac{\partial \pi_0}{\partial a} - \pi_0 \frac{\partial M_0}{\partial a} = N\alpha\left(\pi_1 \frac{\partial \pi_0}{\partial a} - \pi_0 \frac{\partial \pi_1}{\partial a}\right) > 0. \tag{23}$$

Proof of part (i). By the ~~"Defense by Congestion" mechanism, we develop a specialized Co-operation Agent-Based Model (ABM) featuring local batch-matching. While the standard replicator dynamics rely on a mean-field approximation assuming an infinite population and a single monolithic market, real-world academic publishing is inherently fragmented. Authors typically compete within specific sub-disciplines, specialized journals, or distinct temporal submission windows. The primary motivation for designing this ABM is to investigate whether the macroscopic stability predicted by the ODEs holds when subjected to the localized uncertainty, heterogeneous payoffs,~~ quotient rule, the sign of $d(\pi_0/\pi_1)/da$ equals the sign of $\pi_1\,\partial\pi_0/\partial a - \pi_0\,\partial\pi_1/\partial a$. Substituting (20)–(21):

$$\begin{aligned}
&\pi_1 \frac{\partial \pi_0}{\partial a} - \pi_0 \frac{\partial \pi_1}{\partial a} \\
&\quad = (1-\eta\varepsilon)\big[-\dot{\eta}(1-\lambda) + \eta\mu\big] - (1-\eta(1-\lambda))\big[-\dot{\eta}\varepsilon\big] \\
&\quad = \dot{\eta}\Big[\varepsilon\big(1-\eta(1-\lambda)\big) - (1-\lambda)\big(1-\eta\varepsilon\big)\Big] + \eta\mu(1-\eta\varepsilon).
\end{aligned}$$

Expand the bracket in the first term:

$$\varepsilon\big(1-\eta(1-\lambda)\big) - (1-\lambda)\big(1-\eta\varepsilon\big) = \varepsilon - (1-\lambda) = \varepsilon + \lambda - 1 < 0,$$

where the last inequality uses condition (18). Since $\dot{\eta} < 0$ and ~~stochastic fluctuations characteristic of finite populations.~~

~~The core innovation of this extended ABM is the *batch-matching* algorithm. Instead of evaluating all N agents globally, the population is randomly partitioned into local batches (sub-markets) in each time step. The journal's review~~ $(\varepsilon + \lambda - 1) < 0$, the first term $\dot{\eta}(\varepsilon + \lambda - 1) > 0$. The second term $\eta\mu(1-\eta\varepsilon) \geq 0$. Therefore $\pi_1\,\partial\pi_0/\partial a - \pi_0\,\partial\pi_1/\partial a > 0$ and ~~storage capacities (K_{rev} and K_{store}) are scaled proportionally to match these local batches. This approach significantly improves upon standard global ABMs by introducing critical variance into the localized rationing probability (ρ) and enabling stochastic drift. It captures the granular reality that the publication "lottery" is not uniform; different sub-fields experience varying degrees of congestion simultaneously. This localized competition provides a much more rigorous stress-test for the theoretical equilibria~~$d(\pi_0/\pi_1)/da > 0$.

Proof of part (ii). Expand $M_0\,\partial\pi_0/\partial a - \pi_0\,\partial M_0/\partial a$ using $M_0 = N\alpha\pi_1 + N(1-\alpha)(1-x)\pi_0$ and $\partial M_0/\partial a = N\alpha\,\partial\pi_1/\partial a + N(1-\alpha)(1-x)\,\partial\pi_0/\partial a$:

$$\begin{aligned}
&M_0 \frac{\partial \pi_0}{\partial a} - \pi_0 \frac{\partial M_0}{\partial a} \\
&= \big[N\alpha\pi_1 + N(1-\alpha)(1-x)\pi_0\big]\frac{\partial \pi_0}{\partial a} - \pi_0\Big[N\alpha\frac{\partial \pi_1}{\partial a} + N(1-\alpha)(1-x)\frac{\partial \pi_0}{\partial a}\Big] \\
&= N\alpha\left(\pi_1 \frac{\partial \pi_0}{\partial a} - \pi_0 \frac{\partial \pi_1}{\partial a}\right).
\end{aligned}$$

The ~~simulation results validate our threshold competition framework across two key dimensions, as illustrated in Figure ??:~~ $N(1-\alpha)(1-x)\pi_0\,\partial\pi_0/\partial a$ terms cancel exactly, leaving $N\alpha(\pi_1\,\partial\pi_0/\partial a - \pi_0\,\partial\pi_1/\partial a) = N\alpha\pi_1^2\,(d/$ by part (i). □

~~**Trajectory Analysis and Robustness:** Figures ??(a) and ??(b) illustrate the time evolution of strategies under batch matching. In the rationing-dominated regime (Scenario B), despite chaotic local interactions, both the theoretical ODE and the ABM trajectories converge tightly to the stable internal equilibrium of $x^* \approx 0.49$. This confirms that the stable coexistence induced by congestion is highly robust against local noise. Conversely, in the review-dominated regime (Scenario A) , the ABM demonstrates sharp bistability, collapsing rapidly to global honesty ($x = 1.0$)or global corruption ($x = 0.0$)depending on initial conditions. The ABM closely traces the expected theoretical paths but exhibits micro-oscillations, validating the IESDS reduction while underscoring the severe path-dependence of the bistable regime.~~

~~Scenario B (Stable Coexistence) Trajectories Scenario A (Bistability) Trajectories **Results of the Co-operation ABM with Local Batch-Matching.** The composite figure demonstrates that the theoretical mean-field predictions hold even under the rigorous localized constraints and stochastic noise introduced by fragmented sub-markets.~~

Proof of Proposition 1 (Monotone Destabilisation). ~~The Journal as an Independent Evolutionary Agent~~*Claim: Under $\gamma > \delta \geq 0$, $\mu \geq 0$, $\phi > 0$, $\Pi_{bad}(x, a)$ is strictly increasing in a for every $x \in (0, 1)$. Consequently, the GH region in (K_{rev}, K) space shrinks weakly as a rises, and there exists a finite threshold $a^* \in (0, 1]$ above which no (K_{rev}, K) pair sustains GH without additional institutional intervention.*

~~**From Fixed Environment to Co-Evolutionary Agent** In previous sections, including the local batch-matching analysis in Section 5.4, the journal was mathematically treated as a fixed institutional player locked into the "Only Good" (OG) strategy. It acted purely as a static environmental constraint enforcing a peer review filter and a capacity limit. By elevating the journal to an independent, adaptive agent, we transition the framework from a simple population game into an asymmetric co-evolutionary game. Journals now possess their own complete strategy space, consisting of Only Good (OG), Always Accept (AA)~~**Step 1: Show $\partial\Pi_{bad}/\partial a > 0$ in each case.**

Case I ($\eta = 1$, ~~Only Bad (OB)~~$\bar{\rho} = 1$): In the uncongested, unrationed region $\Pi_{bad} = r\lambda(a) - c(a)$, so $\partial\Pi_{bad}/\partial a = r\mu + c_0\phi > 0$.

Case II ($\eta < 1$, $\bar{\rho} = 1$): $\Pi_{bad} = r\pi_0(x, a) - c(a)$, so $\partial\Pi_{bad}/\partial a = r\,\partial\pi_0/\partial a + c_0\phi > 0$ by Lemma A.2 and $\phi > 0$.

Case III ($\eta < 1, \bar{\rho} < 1$): Writing $M_0 \equiv N\alpha\pi_1 + N(1-\alpha)(1-x)\pi_0$ and ~~Reject All (AR)~~$M = (1+\gamma a)M_0$, we have $\Pi_{bad} = rK\pi_0/M - c(a)$. Differentiating using $\partial M/\partial a = \gamma M_0 + (1+\gamma a)\,\partial_a M_0$ gives:

$$\frac{\partial\Pi_{bad}}{\partial a} = \underbrace{\frac{rK(1+\gamma a)}{M^2}\left(M_0\,\frac{\partial\pi_0}{\partial a} - \pi_0\,\frac{\partial M_0}{\partial a}\right)}_{A\,=\,rK(1+\gamma a)N\alpha(\pi_1\partial_a\pi_0-\pi_0\partial_a\pi_1)/M^2\,>\,0\ \text{(Lemma A.3)}} - \underbrace{\frac{rK\gamma\pi_0}{(1+\gamma a)M}}_{B\,>\,0} + \underbrace{c_0\phi}_{C\,>\,0}.$$

The term B is negative, reflecting that the larger submission pool tightens slot rationing. The sign of $\partial\Pi_{bad}/\partial a$ equals the sign of $A - B + C$. ~~They update their behavior based on their own distinct utility function, which must balance the reputation benefits of publishing good papers, the penalties of publishing bad papers,~~

Sufficient condition. Since $B = r\gamma\bar{\rho}\pi_0/(1+\gamma a)$ and ~~the administrative costs of peer review. This structural update captures the "Red Queen" dynamic inherent in academic publishing: if the author population becomes overwhelmingly honest, journals are financially incentivized to drop their costly screening processes and adopt the "Always Accept" (AA)strategy. However, this relaxation immediately triggers an opportunistic invasion of "Always Submit" (AS)authors, which in turn forces the journals to revert to strict filtering.~~ $\bar{\rho} < 1$, we have $B \leq r\gamma\pi_0/(1+\gamma a) \leq r\gamma$. Using $A = rK(1+\gamma a)N\alpha(\pi_1\partial_a\pi_0 - \pi_0\partial_a\pi_1)/M^2$ and Lemma A.3, a sufficient condition for $A + C > B$ is:

$$N\alpha(1+\gamma a)\big[|\dot{\eta}|(1-\lambda-\varepsilon) + \eta\mu\pi_1\big] \;>\; \gamma\pi_0 M_0, \tag{24}$$

which holds when γ is moderate relative to the screening signal $1-\lambda-\varepsilon$ (i.e., when submission volume does not grow so fast that slot-queue congestion overwhelms the screening-degradation and cost-reduction forces).

Numerical confirmation. For both calibrations in Table 9, condition (24) is satisfied and direct evaluation confirms $A-B+C>0$ throughout $(x,a)\in(0,1)\times[0,1]$, so $\partial\Pi_{bad}/\partial a>0$ in Case III.

Step 2: Implication for the GH region. The GH regime requires $\Pi_{bad}(1,a)<0$ (a single deviating author cannot profitably spam). Since $\Pi_{bad}(1,a)$ is strictly increasing in a (Step 1 evaluated at $x=1$), any (K_{rev},K) satisfying GH at $a=0$ may violate it at higher a. The GH region therefore shrinks weakly as a increases.

Step 3: Existence of a finite threshold a^*. At $x=1$ in Case I (the least restrictive case, providing a conservative bound), $\Pi_{bad}(1,a)=r(\lambda_0+\mu a)-c_0(1-\phi a)$. Setting this to zero and solving:

$$a^* = \frac{c_0 - r\lambda_0}{r\mu + c_0\phi}. \tag{25}$$

The denominator is strictly positive since $\phi>0$, $r,c_0>0$, $\mu\geq 0$. The numerator is positive under the baseline GH condition $c_0>r\lambda_0$ (spamming is unprofitable without congestion at $a=0$). Hence $a^*\in(0,\infty)$. When congestion is already active at $x=1$ (Cases II or III), $\Pi_{bad}(1,a)$ rises faster than (25) implies, so the formula provides an upper bound. □

~~The Significance~~ **Proof of Proposition 2 (Slot Expansion Paradox).** *Claim: Holding K_{rev} and a fixed, increasing K weakly raises $\Pi_{bad}(x,a)$ at every x. The marginal destabilising effect $\partial\Pi_{bad}/\partial K$ is strictly larger at higher a.*

Step 1: Sign of $\partial\Pi_{bad}/\partial K$. When $M(x,a)\leq K$, rationing is inactive ($\bar\rho=1$) and Π_{bad} does not depend on K, so $\partial\Pi_{bad}/\partial K=0$. When $M(x,a)>K$, rationing binds ($\bar\rho=K/M$):

$$\Pi_{bad} = \frac{rK\,\pi_0}{M} - c, \qquad \frac{\partial\Pi_{bad}}{\partial K} = \frac{r\,\pi_0}{M} > 0.$$

In both cases $\partial\Pi_{bad}/\partial K\geq 0$, with strict inequality whenever rationing binds.

Step 2: Amplification by a. When rationing binds, the ~~10~~marginal destabilising effect is $r\pi_0/M$ where $M=(1+\gamma a)M_0$. Differentiating with respect to a using $M_0\equiv N\alpha\pi_1+N(1-\alpha)(1-x)\pi_0$:

$$\frac{\partial^2\Pi_{bad}}{\partial K\,\partial a} = \frac{r}{(1+\gamma a)^2 M_0^2}\underbrace{\left[(1+\gamma a)\Big(M_0\frac{\partial\pi_0}{\partial a} - \pi_0\frac{\partial M_0}{\partial a}\Big) - \gamma\pi_0 M_0\right]}_{\text{positive for calibrations in Sets A and B; see (23)}} > 0.$$

As a rises, π_0 increases (more bad papers pass the weakened filter) and $c(a)$ falls (the cost of attempting is lower), so each additional publication slot generates a larger increase in the spam payoff. □

Proof of Proposition 3 (Erosion of Congestion Defence). *Claim: The slot constraint required to hold $\Pi_{bad}(0,a)<0$ is strictly tighter at higher a.*

When all authors play AS ($x=0$) and rationing is active, the corrupt-boundary stability condition $\Pi_{bad}(0,a)<0$ holds if and only if:

$$K < K^*(a) \;\equiv\; \frac{c(a)\,M(0,a)}{r\,\pi_0(0,a)}.$$

With $M(0,a) = (1+\gamma a)\,N\big[\alpha\,\pi_1(0,a) + (1-\alpha)\,\pi_0(0,a)\big]$ this becomes:

$$K^*(a) = \frac{c(a)\,(1+\gamma a)\,N}{r}\left[\alpha\,\frac{\pi_1(0,a)}{\pi_0(0,a)} + (1-\alpha)\right].$$

We show $K^*(a)$ is strictly decreasing in a. Differentiating with respect to a:

$$\frac{dK^*}{da} = \frac{N}{r}\left[\underbrace{c_0[\gamma-\phi-2\phi\gamma a]}_{\text{Term 1}}\left(\alpha\,\frac{\pi_1}{\pi_0} + (1-\alpha)\right) + c(a)(1+\gamma a)\,\alpha\,\underbrace{\frac{d}{da}\left(\frac{\pi_1}{\pi_0}\right)}_{<0\ \text{(Lemma A.3)}\ \text{Term 2}}\right].$$

Term 2 is unambiguously negative. Term 1~~Local Batch-Matching MechanismReal-world academic publishing is not a monolithic global market, but rather a highly fragmented ecosystem divided into specific sub-disciplines and specialized journals. To accurately model this micro-level reality, we introduce an extreme 10: 1 local batch-matching mechanism, pairing 1000 authors with 100 distinct journal agents. In this configuration, global capacities (K_{rev} and K_{store}) are scaled down proportionally to match the size of these local sub-markets. The primary significance of this 10: 1 ratio is the maximization of localized variance and stochastic drift. By breaking the assumption of uniform global congestion, a single journal agent shifting to a rubber-stamping (AA) strategy creates a massive, localized payoff spike for the specific batch of 10 authors assigned to it, severely testing the structural resilience of the system~~'s sign depends on the bracket $c_0[\gamma-\phi-2\phi\gamma a]$.

~~**ABM Configuration and Result Analysis** To simulate this environment, we employ an Agent-Based Model (ABM) where both populations update their strategies via a Fermi pairwise comparison process, introducing bounded rationality and decision noise~~*Case $\phi \geq \gamma$ (analytical).* When $\phi \geq \gamma$, the bracket $\gamma - \phi - 2\phi\gamma a \leq \gamma - \phi \leq 0$ for all $a \geq 0$, so Term $1 \leq 0$. Both terms are non-positive and Term 2 is strictly negative, giving $dK^*/da < 0$ for all $a \in [0,1]$.

~~First, we analyze the global co-evolutionary baseline without fragmentation, as illustrated in **Figure ??**. The ABM realization demonstrates that structurally dominated strategies, such as Only Bad (OB) and Reject All (AR), are rapidly driven to extinction. The remaining strategies enter a dynamic coexistence where the continuous mathematical predictions (ODE)and~~ *Case $\gamma > \phi$ (partial analytical, partial numerical).* The critical value at which Term 1 changes sign is $a_0 = (\gamma-\phi)/(2\phi\gamma)$. For $a \geq a_0$, the ~~finite-population ABM trajectories align closely~~bracket is non-positive, both terms are non-positive, and $dK^*/da \leq 0$ analytically. For $a \in [0, a_0)$, Term 1 is positive and may offset part of Term 2.

Under Sets A and B ($\gamma = 0.5$, $\phi = 0.4$), $a_0 = (0.5-0.4)/(2\times 0.4\times 0.5) = 0.25$. For $a \geq 0.25$, the result is analytical. For $a \in [0, 0.25)$, direct evaluation confirms Term 2 dominates: for Set A, $K^*(0) \approx 268$, $K^*(0.25) \approx 268$, $K^*(0.5) \approx 262$, $K^*(1) \approx 233$, a monotone decline of about 13%. Therefore $K^*(a)$ is strictly decreasing throughout $a \in [0,1]$ under both calibrations. Therefore, as a rises, the maximum admissible slot count maintaining the corrupt-boundary deterrent falls strictly. A journal that set $K < K^*(0)$ at baseline will find $K > K^*(a)$ at higher a without changing any institutional parameter; active reduction of K is required to preserve the deterrent. □

Proof of Proposition 4 (Policy Lever Ranking). *Claim: Among the four primary levers, reducing λ and raising K_{rev} are unconditionally beneficial; raising c is beneficial but imposes symmetric costs on honest authors; expanding K is the unique lever that strictly worsens stability.*

We evaluate $\partial\Pi_{bad}/\partial\theta$ for each lever $\theta \in \{\lambda, K_{rev}, c, K\}$. We work in Case III (both constraints bind); Cases I and II are noted where they differ.

~~Global Co-evolutionary Baseline (Scenario B). The system demonstrates stable coexistence without fragmentation, where the finite-population ABM trajectory closely aligns with the continuous ODE predictions.~~ **Lever 1: False-positive rate $\lambda\downarrow$.**

~~However, under the extreme fragmentation of~~ Since $\pi_0 = 1 - \eta(1-\lambda)$, we have $\partial\pi_0/\partial\lambda = \eta$. Since $\pi_1 = 1 - \eta\varepsilon$ does not involve λ, $\partial\pi_1/\partial\lambda = 0$, and $\partial M/\partial\lambda = N(1-\alpha)(1-x)\,\eta$. In Case III, compute the derivative of $\bar{\rho}\,\pi_0 = K\pi_0/M$:

$$M\,\frac{\partial\pi_0}{\partial\lambda} - \pi_0\,\frac{\partial M}{\partial\lambda} = M\eta - \pi_0\,N(1-\alpha)(1-x)\,\eta = \eta\big[M - N(1-\alpha)(1-x)\pi_0\big] = \eta\cdot N\alpha\pi_1 > 0.$$

Hence $\partial\Pi_{bad}/\partial\lambda = rK\eta N\alpha\pi_1/M^2 > 0$: reducing λ strictly lowers the spam payoff. There is no adverse side effect on honest authors: the ~~10: 1 batch-matching architecture, illustrated in **Figure ??**, the system exhibits a profound deviation from the global mean-field predictions. In the rationing-dominated regime (Scenario B), the ABM reveals that the "Never Submit" (NS)strategy unexpectedly overtakes the ecosystem. This occurs due to a phenomenon we term "Feasibility Collapse." Because the global publication slots (K_{store}) are strictly divided among the 100 journals, each local sub-market operates with an extremely narrow bottleneck. If even a small number of opportunistic (AS) authors flood a specific 10-person room, the local rationing probability (ρ) plummets below the economic break-even threshold required to justify the submission cost(c). Consequently, the "Defense by Congestion" becomes too punitive; even honest authors (OG) suffer negative expected payoffs and are forced to exit the market entirely, leaving NS as the only rational, risk-averse strategy.~~ honest-author payoff is $\Pi_{good} = r\bar{\rho}\,\pi_1 - c$, and $\partial\pi_1/\partial\lambda = 0$; the slight decrease in M (fewer papers pass review when λ falls) eases rationing and weakly raises $\bar{\rho}$, leaving Π_{good} unharmed.

~~Co-evolutionary Dynamics under 10:1 Local Batch-Matching (Scenario B). The extreme fragmentation induces severe localized congestion variance, causing the "Never Submit" (NS) strategy to dominate due to Feasibility Collapse.~~ **Lever 2: Review capacity $K_{rev}\uparrow$.**

B ~~Conclusion~~

When $\eta < 1$, raising K_{rev} increases the screening intensity: $\partial\eta/\partial K_{rev} = 1/V > 0$. Both pass rates fall:

$$\frac{\partial\pi_0}{\partial K_{rev}} = -\frac{1-\lambda}{V} < 0, \qquad \frac{\partial\pi_1}{\partial K_{rev}} = -\frac{\varepsilon}{V} < 0.$$

In Case III, compute $M\,\partial\pi_0/\partial K_{rev} - \pi_0\,\partial M/\partial K_{rev}$. Using $\partial M/\partial K_{rev} = -N[\alpha\varepsilon + (1-\alpha)(1-x)(1-\lambda)]/V$:

$$\begin{aligned} & M\,\frac{\partial\pi_0}{\partial K_{rev}} - \pi_0\,\frac{\partial M}{\partial K_{rev}} \\ &= -\frac{M(1-\lambda)}{V} + \frac{\pi_0\,N\big[\alpha\varepsilon + (1-\alpha)(1-x)(1-\lambda)\big]}{V} \\ &= \frac{N\alpha}{V}\big[\varepsilon\pi_0 - (1-\lambda)\pi_1\big]. \end{aligned}$$

Substituting $\pi_0 = 1 - \eta(1-\lambda)$ and $\pi_1 = 1 - \eta\varepsilon$ and collecting terms:

$$\varepsilon\pi_0 - (1-\lambda)\pi_1 = \varepsilon(1 - \eta(1-\lambda)) - (1-\lambda)(1-\eta\varepsilon) = \varepsilon + \lambda - 1 < 0,$$

by condition (18). Hence $M\,\partial\pi_0/\partial K_{rev} - \pi_0\,\partial M/\partial K_{rev} < 0$, so $\partial\Pi_{bad}/\partial K_{rev} < 0$: raising review capacity strictly lowers the spam payoff. In Case II the same sign follows directly from $\partial\pi_0/\partial K_{rev} < 0$ with $\bar{\rho} = 1$. In Case I ($\eta = 1$), there is no congestion to relieve and the lever has no effect.

Lever 3: Submission cost $c\uparrow$.

$\Pi_{bad} = (\cdots) - c$, so $\partial\Pi_{bad}/\partial c = -1 < 0$ in all cases: raising c strictly lowers the spam payoff. However, $\Pi_{good} = r\bar{\rho}\,\pi_1 - c$ satisfies $\partial\Pi_{good}/\partial c = -1 < 0$ as well. Honest authors bear the same marginal cost increase, and a sufficiently large increase in c eventually drives Π_{good} below zero, deterring honest participation.

Lever 4: Slot capacity $K\uparrow$.

From Proposition 2, $\partial\Pi_{bad}/\partial K \geq 0$, strictly positive whenever rationing is active. Expanding publication slots raises the expected payoff of submitting low-quality work without any compensating reduction in the pass rate or cost. Reducing K provides deterrence against spam, but simultaneously lowers $\Pi_{good} = r \cdot \min(1, K/M) \cdot \pi_1 - c$, potentially pushing the system toward No Submission if K is tightened too far.

Summary. Reducing λ and raising K_{rev} each lower Π_{bad} strictly and have no direct adverse effect on honest-author payoffs. Raising c is effective but imposes symmetric costs on all authors. Expanding K is the unique lever that strictly raises Π_{bad}: it is the only intervention that unconditionally worsens equilibrium stability. □